MINISTRY OF EDUCATION AND TRAINING

TRƯỜNG ĐẠI HỌC FPT

FPT UNIVERSITY

Capstone Project Document

iTranslator

GSP23SE29	
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Capstone Project code	SP23SE38

- Ho Chi Minh, April 2023 -

Table of Contents

Table of Contents 2

List of Tables 5

List of Figures 6

Acknowledgements 8

Definition and Acronyms 9

I. Project Introduction 10

1. Overview 10

1.1 Project Information 10

1.2 Project Team 10

2. Product Background 10

3.1 Fiverr 11

3.2 ProZ 11

4. Business Opportunity 11

5. Software Product Vision 12

6. Project Scope & Limitations 12

6.1 Major Features 12

6.2 Limitations 13

II. Project Management Plan 14

1. Overview 14

1.1 Scope & Estimation 14

1.2 Project Objectives 15

1.3 Project Risks 15

2. Management Approach 16

2.1 Project Process 16

2.2 Quality Management 16

2.3 Training Plan 16

3. Project Deliverables 17

4. Responsibility Assignments 17

5. Project Communications 18

6. Configuration Management 18
6.1 Document Management 18
6.2 Source Code Management 19
6.3 Tools & Infrastructures 20
III. Software Requirements Specification 21
1. Product Overview 21
2. User Requirements 21
2.1 Actors 21
2.2 Use Cases 22
3. Functional Requirements 26
3.1 System Functional Overview 26
3.2 Entity Relationship Diagram 43
4. Non-Functional Requirements 49
4.1 External Interfaces 49
4.2 Quality Attributes 49
5. Requirement Appendix 50
5.1 Business Rules 50
5.2 Application Messages List 53
IV. Software Design Description 54
1. System Design 54
1.1 System Architecture Overview 54
1.2 Package Diagram 55
2. Database Design 57
3. Detailed Design 64
3.1 Project Management Feature 64
3.2 Process Management Feature 71
3.3 Payment Management Feature 75
3.4 Activity Diagram Order 81
V. Software Testing Documentation 111
1. Overall Description 111

1.1 Test Scope 111
1.2 Test Model 111
1.3 Testing Levels 111
2. Test Plan 111
2.1 Test Stages 111
2.2 Resources 111
2.3 Test Milestones 111
3. Test Cases and Test Reports 112
VI. Release Package & User Guides 113
1. Deliverable Package 113
1.1 Source codes & documents 113
1.2 Known Issues, Limitations & Restrictions 113
2. Installation Guides 113
2.1 System Requirements 113
2.2 Setup Files 114
2.3 Installation Instruction 114
3 User Manual 118
3.1 Overview 118
3.2 Application Usage 118

List of Tables

Table 1 - *Definition and Acronyms* 9
Table 2 - *Supervisors* 10
Table 3 - *Team Members* 10
Table 4 - *Limitations* 13
Table 5 - *Scope & Estimation* 15
Table 6 - *Project Objectives* 15
Table 7 - *Project Risks* 15
Table 8 - *Training plan* 17
Table 9 - *Project deliverables* 17
Table 10 - *Responsibility Assignments* 18
Table 11 - *Project Communications* 18
Table 12 - *Tools & Infrastructures* 20
Table 13 - *Actors list and description* 21
Table 14 - *Use Case list* 25
Table 15 - *Account Management Function* 29
Table 16 - *Project Manage Function* 33
Table 17 - *Authenticate Function* 34
Table 18 - *Profile Management Function* 34
Table 19 - *Register Function* 35
Table 20 - *Categories Management Function* 36
Table 21 - *Topic Management Function* 38
Table 22 - *Languages Management Function* 39
Table 23 - *Achievement Management Function* 40
Table 24 - *Chat Function* 40
Table 25 - *Payment Management Function* 41
Table 26 - *Invitation Management Function* 41
Table 27 - *<Screen authorization> Web application* 42
Table 28 - *Non-screen functions* 43
Table 29 - *Logic entities and attributes description* 49
Table 30 - *Business rules* 52
Table 31 - *Application messages* 53
Table 32 - *Table Data Descriptions* 58
Table 33 - *Attribute Data Description* 63
Table 34 - *Project Service Class Specification* 65
Table 35 - *Process Service Class Specification* 72
Table 36 - *Process Service Class Specification* 75
Table 37 - *Test Stages* 111
Table 38 - *Resources* 111
Table 39 - *Test Milestones* 111
Table 40 - *Test Cases and Test Reports* 112
Table 41 - *Source codes & documents* 113
Table 42 - *<System Requirements> Web application* 113
Table 43 - *<System Requirements> Software requirement* 114

List of Figures

Figure 1 - *Scum process* 16
Figure 2 - *Document management* 19
Figure 3 - *Context diagram* 21
Figure 4 - *<Use case overview> iTranslator Web Application* 22
Figure 5 - *<Screen flow diagram> Admin web application* 26
Figure 6 - *<Screen flow diagram> Moderator web application* 26
Figure 7 - *<Screen flow diagram> Customer web application* 27
Figure 8 - *<Screen flow diagram> Translator web application* 28
Figure 9 - *Conceptual entity relationship diagram* 43
Figure 10 - *Logical entity relationship diagram* 44
Figure 11 - *System architecture* 54
Figure 12 - *Web API Package Diagram* 55
Figure 13 - *Web Application Package Diagram* 56
Figure 14 - *Database Design* 57
Figure 15 - *Project Management Feature Class Diagram* 64
Figure 16 - *Create Project Activity Diagram* 66
Figure 17 - *View Project Activity Diagram* 67
Figure 18 - *Edit Project Activity Diagram* 68
Figure 19 - *Delete Project Activity Diagram* 69
Figure 20 - *Create Project Sequence Diagram* 70
Figure 21 - *Create Project State Machine Diagram* 71
Figure 22 - *Process Management Feature Class Diagram* 71
Figure 23 - *View Process Activity Diagram* 73
Figure 24 - *Create Process Sequence Diagram* 74
Figure 25 - *Payment Management Feature Class Diagram* 75
Figure 26 - *View Payment History Activity Diagram* 76
Figure 27 - *Deposit Activity Diagram* 77
Figure 28 - *Withdraw Activity Diagram* 78
Figure 29 - *Deposit Sequence Diagram* 79
Figure 30 - *Withdraw Sequence Diagram* 80
Figure 31 - *<Activity Diagram> Admin view account* 81
Figure 32 - *<Activity Diagram> Admin Ban account* 82
Figure 33 - *<Activity Diagram> Mod create category* 83
Figure 34 - *<Activity Diagram> Mod create topic* 84
Figure 35 - *<Activity Diagram> Mod create language* 85
Figure 36 - *<Activity Diagram> Mod create achievement* 86
Figure 37 - *<Activity Diagram> Mod view project* 87
Figure 38 - *<Activity Diagram> Mod view topic* 88
Figure 39 - *<Activity Diagram> Mod category* 89
Figure 40 - *<Activity Diagram> Mod view language* 90
Figure 41 - *<Activity Diagram> Mod view achievement* 91
Figure 42 - *<Activity Diagram> Mod approve project* 92
Figure 43 - *<Activity Diagram> Mod reject project* 93
Figure 45 - *<Activity Diagram> Customer create project* 95
Figure 46 - *<Activity Diagram> Customer view project* 96
Figure 47 - *<Activity Diagram> Customer view translator* 97

Figure 48 - *<Activity Diagram> Customer view process* 98
Figure 49 - *<Activity Diagram> Customer view history rating* 99
Figure 50 - *<Activity Diagram> Customer accept translator into project* 100
Figure 51 - *<Activity Diagram> Customer reject customer into project* 101
Figure 52 - *<Activity Diagram> Customer delete project* 102
Figure 53 - *<Activity Diagram> Customer login* 103
Figure 54 - *<Activity Diagram> Customer logout* 104
Figure 55 - *<Activity Diagram> Translator view history rating* 105
Figure 56 - *<Activity Diagram> Translator view invitation* 106
Figure 57 - *<Activity Diagram> Translator view project* 107
Figure 58 - *<Activity Diagram> Translator approve invitation* 108
Figure 59 - *<Activity Diagram> Translator reject invitation* 109
Figure 60 - *<Activity Diagram> Translator request to join project* 110
Figure 61 - *Setup Backend Api* 114
Figure 62 - *Setup Backend Api* 115
Figure 63 - *Setup Backend Api* 115
Figure 64 - *Setup Backend Api* 116
Figure 65 - *Setup Backend Api* 116
Figure 66 - *Setup Management Web* 117
Figure 67 - *Setup Management Web* 117
Figure 68 - *Setup Management Web* 117

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We acknowledge that due to our lack of experience and unfamiliarity with certain technologies used in the project, mistakes were inevitable. We hope that people will overlook and forgive any errors or shortcomings. Thank you for your unwavering support. Regards

Definition and Acronyms

Acronym	Definition
BA	Business Analysis
BR	Business Rule
ERD	Entity Relationship Diagram
GUI	Graphical User Interface
PM	Project Manager
SDD	Software Design Description
SRS	Software Requirement Specification
UAT	User Acceptance Test
UC	Use Case
API	Application Programming Interface
BE	Backend
FE	Frontend

Table 1 - Definition and Acronyms

I. Project Introduction

1. Overview

1.1 Project Information

- Project name: **iTranslator**
- Project code: SP23SE38
- Group name: GSP23SE29
- Software type: Web Application

1.2 Project Team

a. Supervisors

Full Name	Role	Email	Mobile
Nguyễn Thế Hoàng	Lecturer	hoangnt20@fe.edu.vn	0986.628.525

Table 2 - Supervisors

b. Team Members

Full Name	Role	Email	Mobile
Võ Nhứt Thanh Long	Leader	longvntse140630@fpt.edu.vn	0865.416.142
Võ Nhị Thanh Long	Member	longvntse140143@fpt.edu.vn	0865.416.143
Phạm Đức Anh Tuấn	Member	tuanpdase141182@fpt.edu.vn	0932.178.403
Lê Đức Quang	Member	quangldse140637@fpt.edu.vn	0375.152.247

Table 3 - Team Members

2. Product Background

Reading across several languages appears to have grown challenging for readers in today's culture, when the requirement to do so is prevalent in many nations. This is because there aren't many platforms available to address the issue. Even for those who frequently work in and speak the language they have acquired, reading comprehension is difficult due to the variety of information.

There isn't currently a widely used, practical, or user-friendly translation tool for readers in Vietnam. Even when using well-known forehead late machines like google translate, translation still contains faults that might cause reading to be halted or cause readers to misinterpret the concepts being delivered in books and stories, which makes reading more monotonous. As a result, ITranslator was developed with the aim of resolving the aforementioned issue, facilitating communication between clients and translators, and simplifying the process of reading books and stories.

3. Existing Systems

3.1 Fiverr

- **Link:** https://www.fiverr.com/categories/writing-translation
- **Description:** The website fiverr.com is a platform that allows customers to rent any online services such as Graphics & Design, Digital Marketing, Writing & Translation, Video & Animation, Music & Audio, Programming & Tech, Business, Lifestyle, .. This application as a 3rd party helps connect customers and sellers quickly.
- Fiverr.com is trusted by Facebook, Google, Netflix, PayPal. The seller will work freelance waiting for customers to hire.
- **System actor:**
  - Seller: They will post articles to help translate, write, and advise their customers.
  - Customer: Go to see the services you need to use and hire a service seller.
- **Features:**
  - Seller: post a post about the capabilities and services that I can provide to customers.
  - Customers: search for services, view seller information, view reviews of service sellers.
- **Advantage and Disadvantage**
  - Advantages: Safe, diverse products, reputable suppliers, friendly interface.
  - Disadvantages: Only allow sellers to post, but not actively find customers waiting to use the service. Less choice of service packages.

3.2 ProZ

- **Link:** https://www.proz.com/
- **Description:** ProZ.com's mission is to empower language industry professionals to achieve their business objectives and realize their full potential. ProZ.com does this by: being committed to member success, providing access to state-of-the-art tools, educating and inspiring, and fostering collaboration among positive, like-minded professionals.
- **System actors:**
  - Seller: They will post articles to help translate, write, and advise their customers.
  - Customer: Go to see the services you need to use and hire a service seller.
- **Features:**
  - Seller: post a post about the capabilities and services that I can provide to customers.
  - Customers: search for services, view seller information, view reviews of service sellers.
- **Advantage and Disadvantage**
  - Advantages: Safe, diverse products, reputable suppliers, friendly interface.
  - Disadvantages: Only allow sellers to post, but not actively find customers waiting to use the service. Less choice of service packages.

4. Business Opportunity

The customers who need to translate the project such as film, manga,.. are increasing day by day. It is quite difficult to find the suitable platform. Meanwhile, translation systems that already exist in the market to connect customers and translators often have a scope of

activities that is quite wide and does not focus on serving one field, leading to no focus on advertising to the right customer. Our platform will focus on translation.

5. Software Product Vision

For customers, they can find suitable translators. Translators can find suitable projects for their demand through iTranslator which directly introduce and provide product information to the customer . Thanks to iTranslator, customers and translators can easily connect together.

6. Project Scope & Limitations

6.1 Major Features

- **Admin:**
  - FE-01: View, ban account
  - FE-02: Manage permission
  - FE-03: Manage revenue
- **Moderator:**
  - FE-04: View, reject, approve project
  - FE-05: View rating
  - FE-06: View, create, update category
  - FE-07: View, create, update topic
  - FE-08: Approve payment
  - FE-09: View, add, edit language
  - FE-10: View, add, edit achievement
  - FE-11: Chat
- **Customer:**
  - FE-12: View, create, delete, edit project
  - FE-13: Search, view, invite, accept, reject translator
  - FE-14: Rate project
  - FE-15: View process of project
  - FE-16: View payment
  - FE-17: Make a Payment
  - FE-18: View history rating
  - FE-19: Withdraw money
  - FE-20: Chat
- **Translator:**
  - FE-21: View, accept, reject invitation
  - FE-22: Search project
  - FE-23: View project
  - FE-24: Request to join project
  - FE-25: View translation experience
  - FE-26: Update translation

- FE-27: Send project completion notice
- FE-28: View history rating
- FE-29: Chat
- FE-30: Withdraw money

- **System:**
  - FE-31: Notify user
  - FE-32: Suggest suitable translator

6.2 Limitations

#	Limitations
LI-01	There is only one admin and one moderator in the system
LI-02	Online payment only
LI-03	Not yet automatically proposing prices for customers

Table 4 - Limitations

II. Project Management Plan

1. Overview

1.1 Scope & Estimation

#	WBS Item	Complexity	Est. Effort (man-days)
1	***Requirement analysis***		***12***
1.1	Define background and context of product	Complex	7
1.2	Find problems of users	Medium	2
1.3	Existed system analysis	Simple	1
1.4	Business opportunity analysis	Medium	2
2	***Technology training***		***18***
2.1	Learn .Net API framework	Complex	6
2.2	Learn Angular framework	Complex	6
2.3	Learn DevOps tool and third-party	Complex	6
3	Workspace setup		***7***
3.1	Setup resources storage workspace	Medium	3
3.2	Setup resources and version control	Medium	4
4	Document writing		***35***
4.1	Write project introduction	Medium	2
4.2	Write project management plan	Medium	3
4.3	Write software requirement specification	Complex	6
4.4	Write software design document	Complex	6
4.5	Write test document	Medium	4
4.6	Write user guide document	Medium	4
4.7	Write final document	Complex	10
5	Interface design		***45***
5.1	Design web application interface	Complex	25
5.2	Design web application interface	Complex	20
6	***Implementation***		**167**
6.1	Implement authentication/authorization features	Complex	7

6.2	Implement admin features	Complex	40
6.3	Implement moderator features	Complex	20
6.4	Implement customer features	Complex	50
6.5	Implement translator features	Complex	50
7	***Testing***		**24**
7.1	Unit Test	Complex	8
7.2	Integration Test	Complex	8
7.3	System Test	Complex	8
8	***Closing***		**8**
8.1	Maintenance	Medium	5
8.2	Prepare slides for thesis defence	Medium	3
		Total Estimated Effort (man-days)	***309***

Table 5 - Scope & Estimation

1.2 Project Objectives

#	Testing Stage	No. of Defects	% of Defect	Notes
1	Reviewing			
2	Unit Test			
3	Integration Test			
4	System Test			

Table 6 - Project Objectives

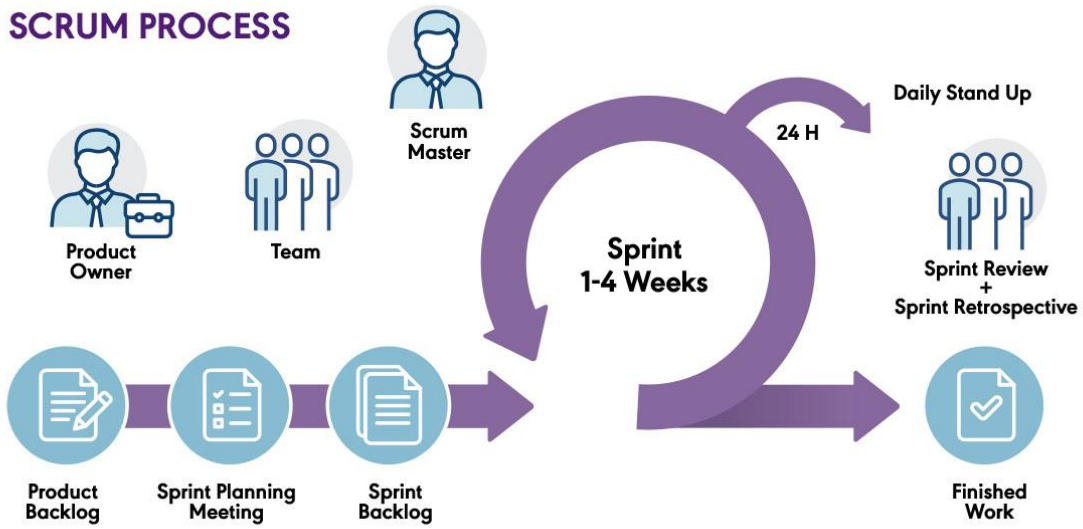
1.3 Project Risks

#	Risk Description	Impact	Possibility	Response Plans
1	High weight points are skewed to Frontend	Critical	Medium	Assign more members and spend more time doing frontend tasks.

Table 7 - Project Risks

2. Management Approach

2.1 Project Process



References: https://www.pm-partners.com.au/the-agile-journey-a-scrum-overview/

Figure 1 - Scum process

iTranslator project applied the Scrum model that is an Agile framework for software development with following reasons:

- We developed software with a small team of 4 people so that we have no effort to complete all stages in the old process like Waterfall. During a period in sprints, we will easily analyse what is next and refactor if needed.
- With little time in sprints, we will be pressured to do tasks quickly.

2.2 Quality Management

- Review code
- Coding convention
- Unit Testing
- Integration Testing
- System Testing

2.3 Training Plan

Training Area	Participants	When, Duration	Waiver Criteria
.Net API Framework	Võ Nhứt Thanh Long, Lê Đức Quang, Phạm Đức Anh Tuấn	Week 1 - 6 days	Mandatory

DevOps tools (Docker, Github, Jenkin…)	Võ Nhứt Thanh Long, Võ Nhị Thanh Long	Week 1 - 3 days	Mandatory
Angular 2.0	Võ Nhị Thanh Long, Lê Đức Quang	Week 1 - 6 days	Mandatory
Research Third-party	Phạm Đức Anh Tuấn, Lê Đức Quang	Week 1 - 3 days	Mandatory

Table 8 - Training plan

3. Project Deliverables

#	Deliverable	Due Date	Notes
1	Project Plan Document	1/1/2023	Almost completing the project plan
2	Use case specification	12/1/2023	Functional Requirement
3	Design documents	20/3/2023	UI design, Architecture design
4	Web admin & moderator modules	20/4/2023	Web for admin and moderator
5	Web translator & customer modules	20/04/2023	Web for customer and translator

Table 9 - Project deliverables

4. Responsibility Assignments

Responsibility	Vo Nhut Thanh Long	Vo Nhi Thanh Long	Pham Duc Anh Tuan	Le Duc Quang

Backend for system	x	x	x	x
Front end web		x		x
DevOps		x		
Others	x		x	x

Table 10 - Responsibility Assignments

5. Project Communications

Communication Item	Who/ Target	Purpose	When, Frequency	Type, Tool, Method(s)
Working with advisor	Supervisor (Nguyen The Hoang)	• Review documentation • Demonstrate features • Evaluate progress and result	1 time per week	Face to face
Working with team	Team members	• Raise problem • Review source code • Evaluate progress and result	Always	Messenger, Google meet, and Face to face

Table 11 - Project Communications

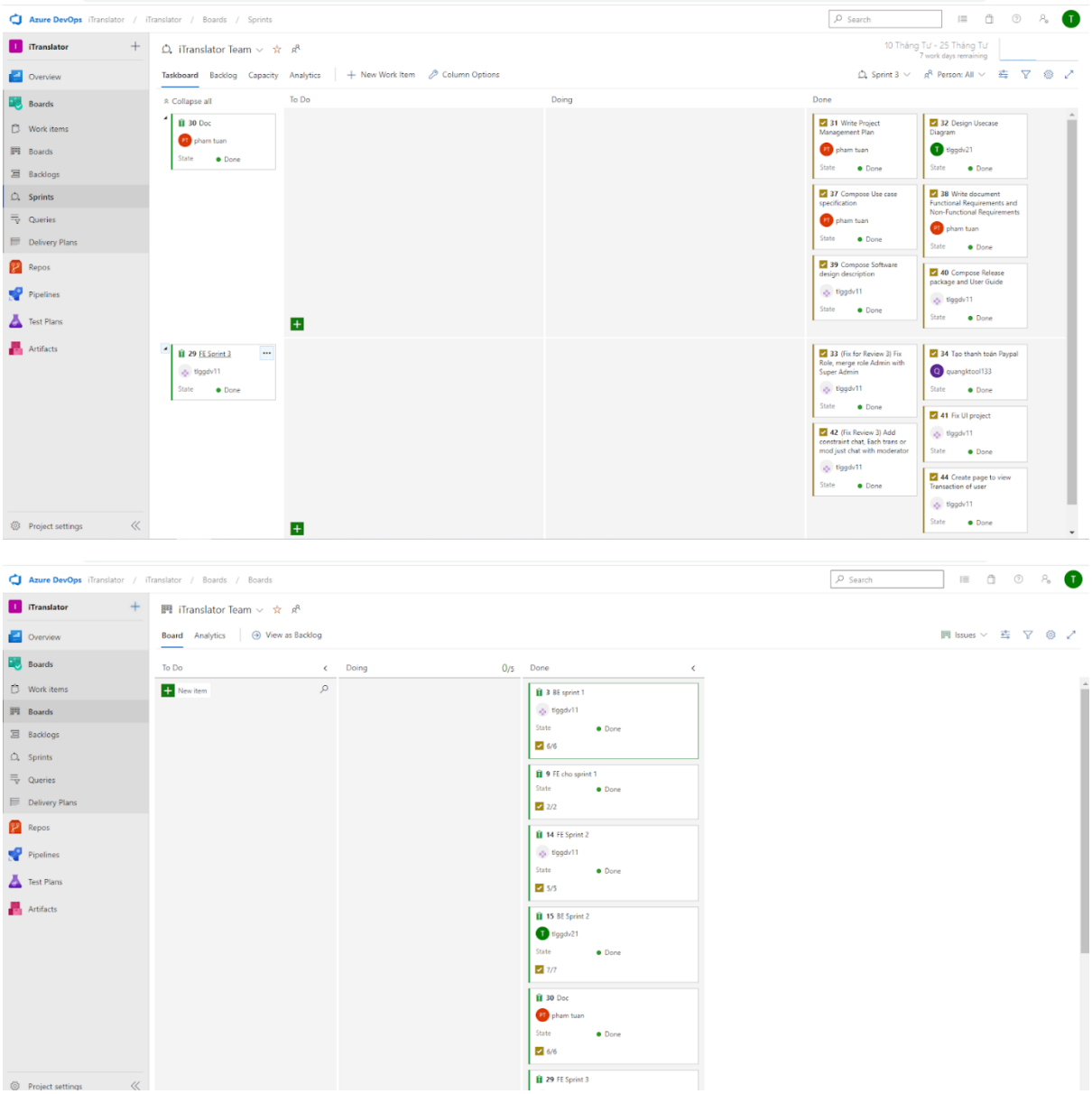
6. Configuration Management

6.1 Document Management

We used Azure and Google for storing document:

- Using Google to save images, diagrams, etc.
- Azure provide a Page Project that contains a topic, content, comment features to discuss problems so that we can review thereafter.
- Use Azure to create tasks, bugs, reports, etc., and then we apply SCRUM with dynamic time for the sprints.

Below is the screenshot of Azure:



Reference: https://dev.azure.com/iTranslator/iTranslator/_boards/board/t/iTranslator%20Team/Issues

Figure 2 - Document management

6.2 Source Code Management

Creating an organisation that covered 3 source projects for FE, BE. We can define many rules in the organisation that will apply to all projects.

- In each source project, we created a dev branch that will be cloned from other branches for building features. After that, we made a pull request from the dev to the master branch that is deployed into the server.
- During pull request, merged codes will be sent to the server for passing pipelines and accept merge thereafter.

6.3 Tools & Infrastructures

Category	Tools / Infrastructure
Technology	Angular (FrontEnd), C# .Net 5 (BackEnd)
Database	SQL Server 2019
IDEs/Editors	Visual Studio Code, Visual Studio 2019
Diagramming	Diagrams.net
Documentation	Ms Office, Google Docs/ Sheets/ Slides
Version Control	Azure(Source Code), Google Drive(Documents)
Deployment server	Azure
Project management	Azure(Task management), Google Drive(Media files management)

Table 12 - Tools & Infrastructures

III. Software Requirements Specification

1. Product Overview

The iTranslator Platform is a software system that connects the customers to the translator, provides fast connect solutions to save time for the customers and the translators. The context diagram below illustrates the external entities and system interfaces

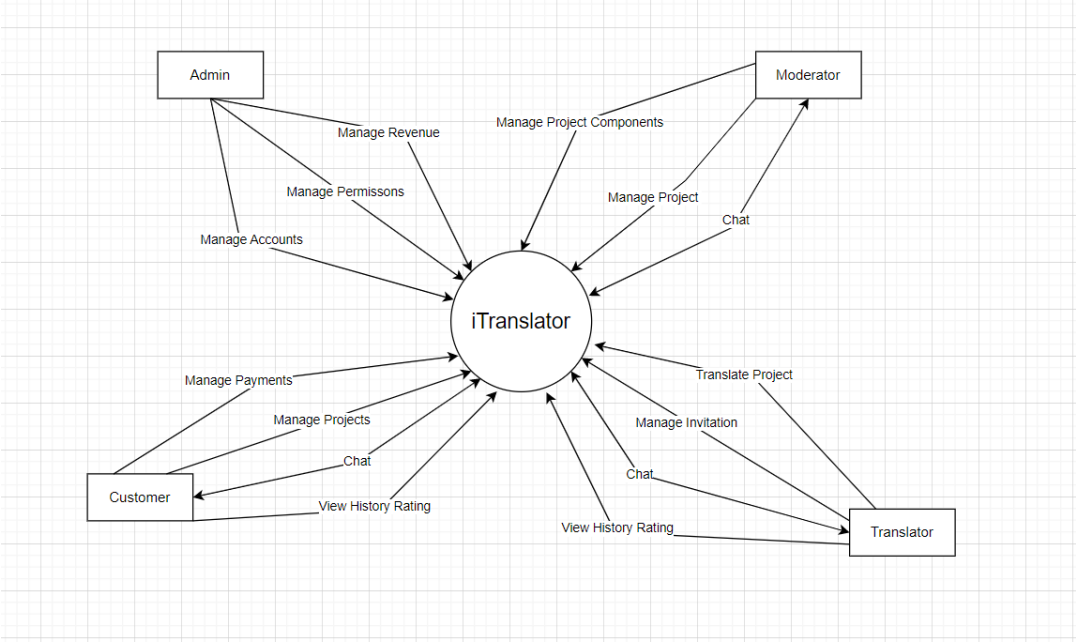


Figure 3 - Context diagram

2. User Requirements

2.1 Actors

#	Actor	Description
1	Administrator	The administrators who are responsible for managing the system. Concretely, they can manage permission, manage accounts, manage revenue.
2	Moderator	The moderators who are responsible for managing projects, managing topic, approving payment, chating and approving project. Moreover, they can view, create, update category.
3	Customer	The customers who are responsible for managing project and managing payment. Moreover, they can chat and view history rating.
4	Translator	The translators who are responsible for translating, managing invitation, looking for project and sending project completion notice.
5	Guest	Guests are customers who can visit the web pages.

Table 13 - Actors list and description

2.2 Use Cases

2.2.1 Diagram

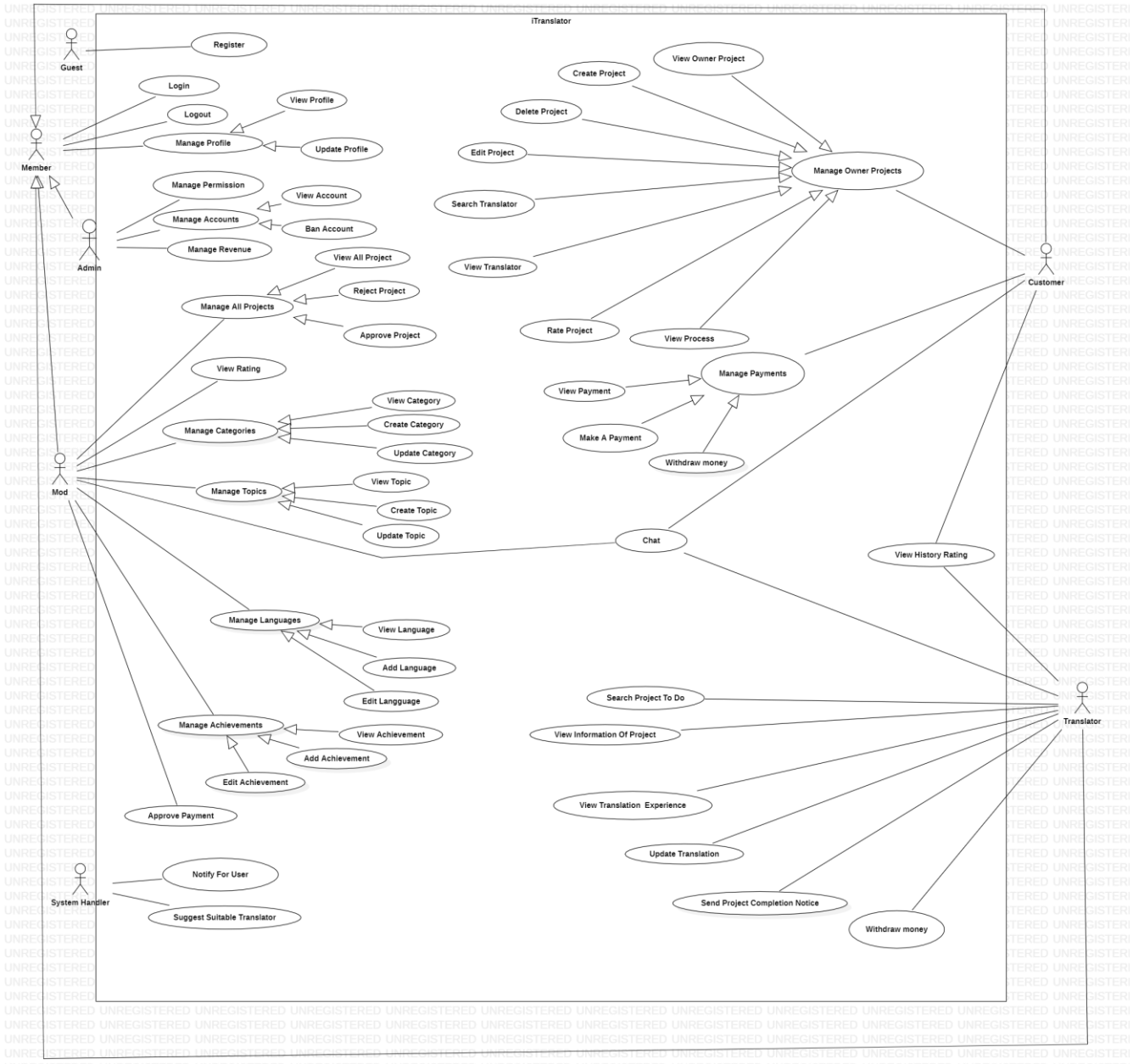


Figure 4 - <Use case overview> iTranslator Web Application

2.2.2 Use Case List with description

ID	Use Case	Actors	Description
UC01	Register	Guest	Guest register for account
UC02	Login	Admin, Moderator, Customer, Translator	Unauthenticated User login to iTranslator to verify role, and give them access to more features in the system.
UC03	Logout	Admin, Moderator, Customer, Translator	This feature allows all user logout the system
UC04	View profile	Admin, Moderator, Customer, Translator	This feature allows all user view their profile
UC05	Update profile	Admin, Moderator, Customer, Translator	This feature allows users update their profile
UC06	View account	Admin	This feature allows admin view all accounts
UC07	Ban account	Admin	This feature allows admin ban account
UC08	View project	Mod	This feature allows mod view project
UC09	Reject project	Mod	This feature allows mod reject project
UC10	Approve project	Mod	This feature allows mod approve project
UC11	View rating	Mod	This feature allows mod view rating
UC12	View category	Mod	This feature allows mod view category
UC13	Create category	Mod	This feature allows mod create a new category
UC14	Update category	Mod	This feature allows mod update category

UC15	View topic	Mod	This feature allows mod view topic
UC16	Create topic	Mod	This feature allows admin create a new topic
UC17	Update topic	Mod	This feature allows mod updates topic
UC18	Approve payment	Mod	This feature allows mod approve payment
UC19	Chat	Mod, Customer, Translator	This feature allows mod, customer, translator chat to the other.
UC20	View language	Mod	This feature allows mod view language
UC21	Add language	Mod	This feature allows mod add language
UC22	Edit language	Mod	This feature allows mod edit language
UC23	View achievement	Mod	This feature allows mod view achievement
UC24	Add achievement	Mod	This feature allows mod add achievement
UC25	Edit achievement	Mod	This feature allows mod edit achievement
UC26	View project	Customer	This feature allows customer view project
UC27	Create project	Customer	This feature allows customer create new project
UC28	Delete project	Customer	This feature allows customer delete project
UC29	Edit project	Customer	This feature allows customer edit project
UC30	Search translator	Customer	This feature allows customer search translator
UC31	View Translator	Customer	This feature allows customer view translator

UC32	Rating project	Customer	This feature allows customer rating process
UC33	View process	Customer	This feature allows customer view process
UC34	View payment	Customer	This feature allows customer view payment
UC35	Make a payment	Customer	This feature allows customer make a payment
UC36	Withdraw money	Customer, Translator	This is feature allows customer withdraw money
UC37	View history rating	Customer, Translator	This feature allows customer, translator view history rating
UC38	Search project	Translator	This feature allows translator search project
UC39	View project	Translator	This feature allows translator view project
UC40	View translation experience	Translator	This feature allows translators view their experience
UC41	Update translation	Translator	This feature allows translator update translation of project
UC42	Send project completion notice	Translator	This feature allows the translator to send project completion notice.

Table 14 - Use Case list

3. Functional Requirements

3.1 System Functional Overview

3.1.1 Screens Flow

3.1.1.1 <Admin> Web Application Screen Flow

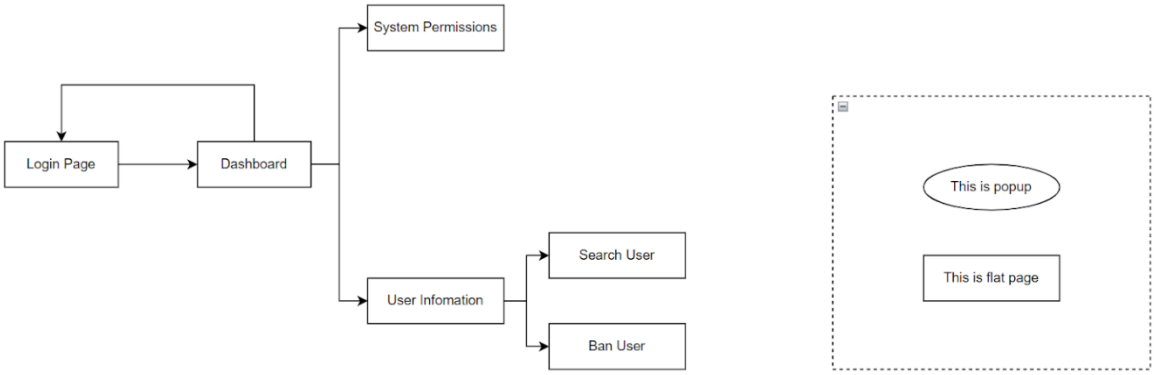


Figure 5 - <Screen flow diagram> Admin web application

3.1.1.2 <Moderator> Web Application Screen Flow

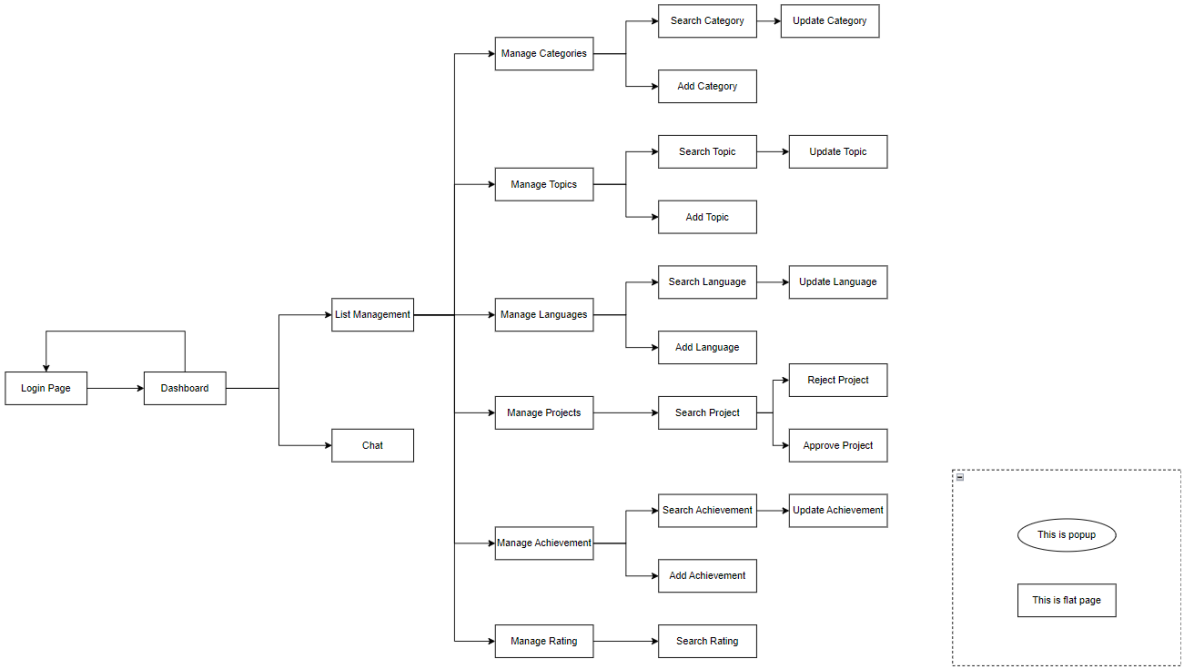


Figure 6 - <Screen flow diagram> Moderator web application

3.1.1.3 <Customer> Web Application Screen Flow

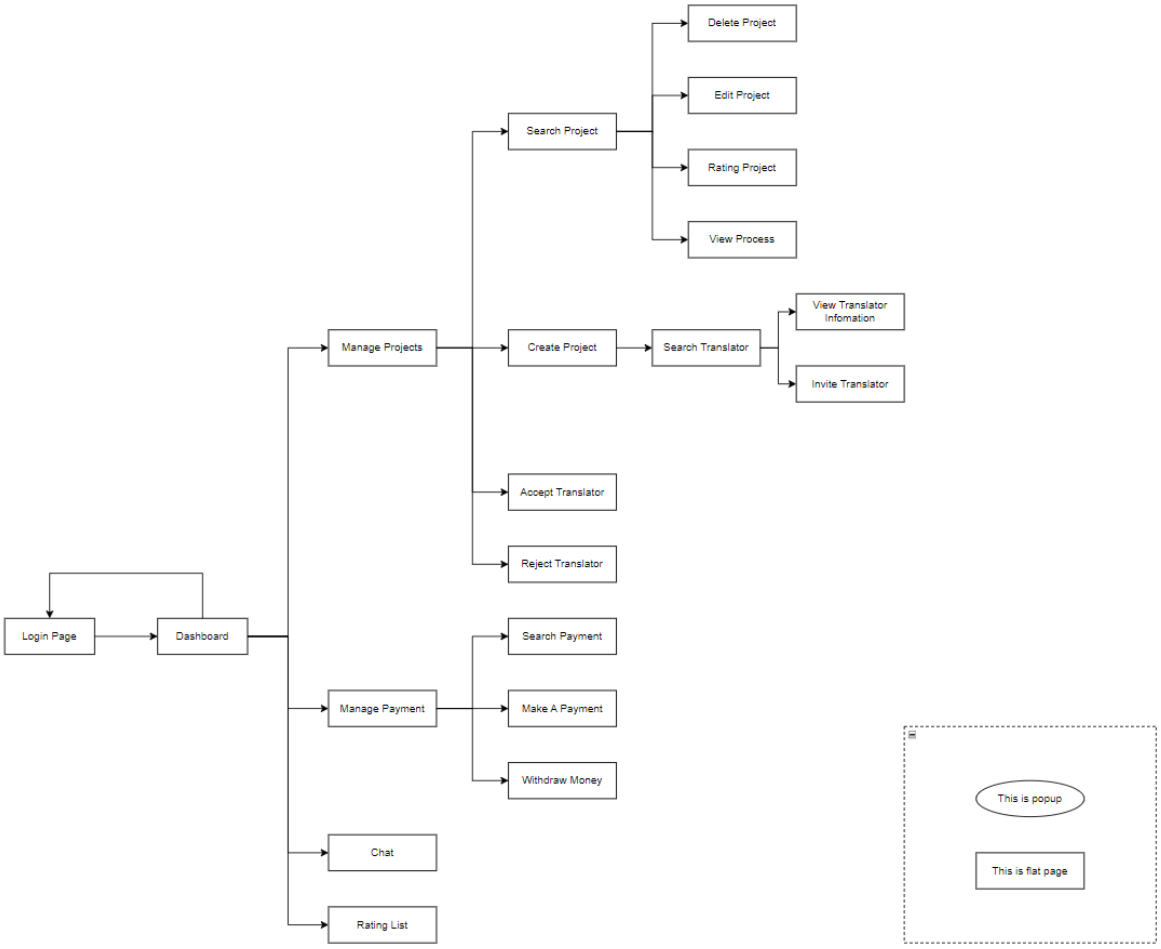


Figure 7 - <Screen flow diagram> Customer web application

3.1.1.4 <Translator> Web Application Screen Flow

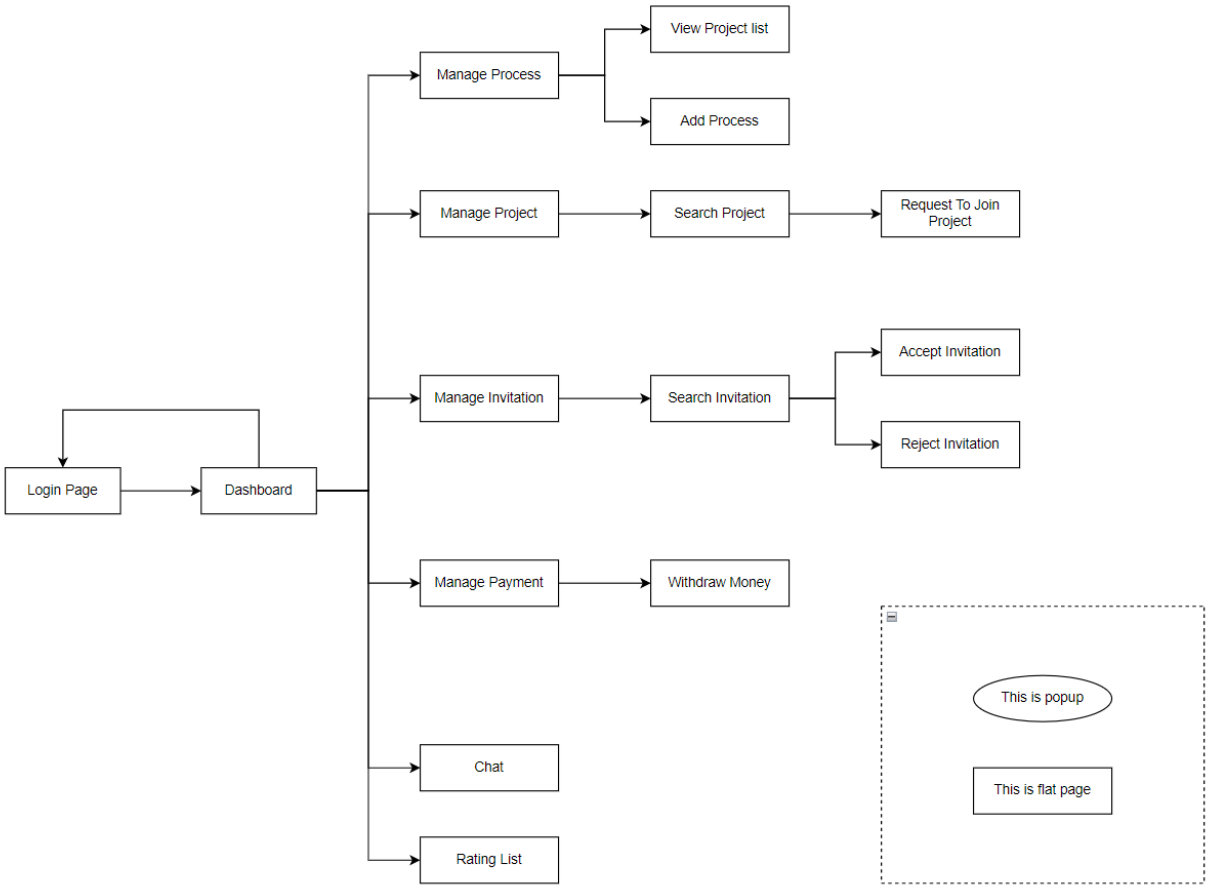


Figure 8 - <Screen flow diagram> Translator web application

3.1.2 Functional requirements

3.1.2.1 Accounts Management

No	*Actor*	*Function Name*	*Purpose*	*Data requirements*	*Data validation*	*Business Rules*
1	Admin	View account	View a account	Account name	Account name: + Type: text field (string) + Max Length= 200	BR-01 BR-35
2	Admin	Ban account	Ban a account	Account Id Account Status		BR-01 BR-36 BR-37 BR-38

Table 15 - Account Management Function

3.1.2.2 Project management

No	*Actor*	*Function Name*	*Purpose*	*Data requirements*	*Data validation*	*Business Rules*
1	Moderator, Customer, Translator	View project	View list project	Project code Project name	Project code: + Type: text field (string) + Max Length= 50 Project name: + Type: text field (string) + Max Length= 200	BR-02 BR-03 BR-04
2	Moderator	Reject project	Reject a project	Project id Project code	Project code: + Type: text field (string) + Max Length= 50	BR-02 BR-46 BR-52 BR-53
3	Moderator	Approve project	Approve a project	Project id		BR-02 BR-47 BR-51 BR-53
4	Translator	Search project	Search project	Project code Project name	Project code: + Type: text field (string) + Max Length= 50	BR-02 BR-59

					Project name: + Type: text field (string) + Max Length= 200	
5	Translator	Request join to project	Create a request	Project Id Project code	Project code: + Type: text field (string) + Max Length= 50	BR-03 BR-69
6	Translator	Update translation	Update the process of project	Project Id Link file	Link file: + Type: text file	BR-03
7	Translator	Send project completion notice	Send project completion notice	Project Id		BR-03
8	Customer	Create project	Create a new project	Project id Project code Project name Category id Topic id Project cost Translator id Source language id Target language id Commission Project start time Project end time Project source Project finished product Project score Customer id Create date Project status	Project id: + Type: text field (string) + Max Length= 50 Project code: + Type: text field (string) + Max Length= 50 Project name: + Type: text field (string) + Max Length= 200 Category id: + Type: text field (string) + Max Length= 50 Topic id: + Type: text field (string) + Max Length= 50 Project cost: + Type: text field (decimal) Translator id: + Type: text field (string) + Max Length= 50 Source language id: + Type: text field (string) + Max Length= 50	BR-04 BR-18 BR-57 BR-64

					Target language id: + Type: text field (string) + Max Length= 50 Commission: + Type: text field (decimal) Project start time: + Type: text field (datetime) Project end time: + Type: text field (datetime) Project source: + Type: text field (string) + Max Length= Max Project finished product: + Type: text field (string) + Max Length= Max Project score: + Type: text field (decimal) Customer id: + Type: text field (string) + Max Length= 50 Create date: + Type: text field (datetime) Project status: + Type: text field (string) + Max Length= 1	
9	Customer	Delete project	Delete a project	Project id		BR-04 BR-55
10	Edit project	Edit project	Edit a project	Project id Project code Project name Category id Topic id Project cost Translator id	Project id: + Type: text field (string) + Max Length= 50 Project code: + Type: text field (string) + Max Length= 50 Project name:	BR-04 BR-54 BR-68

				Source language id Target language id Commission Project start time Project end time Project source Project finished product Project score Customer id Create date	+ Type: text field (string) + Max Length= 200 Category id: + Type: text field (string) + Max Length= 50 Topic id: + Type: text field (string) + Max Length= 50 Project cost: + Type: text field (decimal) Translator id: + Type: text field (string) + Max Length= 50 Source language id: + Type: text field (string) + Max Length= 50 Target language id: + Type: text field (string) + Max Length= 50 Commission: + Type: text field (decimal) Project start time: + Type: text field (datetime) Project end time: + Type: text field (datetime) Project source: + Type: text field (string) + Max Length= Max Project finished product: + Type: text field (string) + Max Length= Max Project score: + Type: text field (decimal) Customer id:	

					+ Type: text field (string) + Max Length= 50 Create date: + Type: text field (datetime) Project status: + Type: text field (string) + Max Length= 1	
11	Customer	Search translator	Search translator for project	Project name Translator name	Project name: + Type: text field (string) + Max Length= 200 Translator name: + Type: text field (string) + Max Length= 200	BR-04
12	Customer	View translator	View translator's information	List translator's information: translator's information table		BR-04
13	Customer	Rating project	Rating for the product of translator	Project id Score	Score: + Type: button	BR-04
14	Customer	View process	View a process of project	Project name	Project name: + Type: text field (string) + Max Length= 200	BR-04

Table 16 - Project Manage Function

3.1.2.3 Authenticate

No	***Actor***	***Function Name***	***Purpose***	***Data requirements***	***Data validation***	***Business Rules***
1	Admin, Moderator, Translator, Customer	Login	Login to website	Username Password	Username: + Type: text field(string) + Max Length= 200 Password: + Type: text field(string) + Max Length= 100	BR-01 BR-02 BR-03 BR-04

Table 17 - Authenticate Function

3.1.2.4 Profile Management

No	*Actor*	*Function Name*	*Purpose*	*Data requirements*	*Data validation*	*Business Rules*
1	Admin, Moderator, Translator, Customer	View profile	View profile	Profile: profile table		BR-01 BR-02 BR-03 BR-04 BR-08
2	Admin, Moderator, Translator, Customer	Update profile	Update new data for profile	Name Email Password Address Phone	Name: + Type: text field(string) + Max Length = 200 Email: + Type: text field(string) + Max Length= 50 Password: + Type: text field(string) + Max Length= 100 Address: + Type: text field(string) + Max Length= 1000 Phone: + Type: text field(string) + Max Length= 12	BR-01 BR-02 BR-03 BR-04 BR-09 BR-23 BR-24 BR-25 BR-26 BR-27 BR-28

Table 18 - Profile Management Function

3.1.2.5 Register

No	*Actor*	*Function Name*	*Purpose*	*Data requirements*	*Data validation*	*Business Rules*
1	Guest	Register	Register account	Name Email Password Address Phone	Name: + Type: text field(string) + Max Length = 200 Email: + Type: text field(string) + Max Length= 50 Password: + Type: text field(string) + Max Length= 100 Address: + Type: text field(string) + Max Length= 1000 Phone: + Type: text field(string) + Max Length= 12	BR-23 BR-24 BR-25 BR-26 BR-27 BR-28 BR-34

Table 19 - Register Function

3.1.2.6 Categories Management

No	*Actor*	*Function Name*	*Purpose*	*Data requirements*	*Data validation*	*Business Rules*
1	Moderator	View category	View a category	Category code Category name	Category code: + Type: text field(string) + Max Length= 50	BR-02

					Category name: + Type: text field(string) + Max Length= 200	
2	Moderator	Create category	Create a new category	Category name Category status Category note	Category name: + Type: text field(string) + Max Length= 200 Category note: + Type: text field(string) + Max Length= 2000 Category status: + Type: text field(string) + Max Length= 1	BR-02 BR-12 BR-13 BR-41 BR-42 BR-43 BR-44
3	Moderator	Update category	Update category	Category name Category note Category status	Category name: + Type: text field(string) + Max Length= 200 Category note: + Type: text field(string) + Max Length= 2000 Category status: + Type: text field(string) + Max Length= 1	BR-02 BR-29 BR-45

Table 20 - Categories Management Function

3.1.2.7 Topics Management

No	*Actor*	*Function Name*	*Purpose*	*Data requirements*	*Data validation*	*Business Rules*
1	Moderator	View topic	View a topic	Topic code Topic name	Topic code: + Type: text field(string) + Max Length= 50 Topic name: + Type: text field(string) + Max Length= 200	BR-02
2	Moderator	Create topic	Create a new topic	Topic name Topic note Topic status	Topic name: + Type: text field(string) + Max Length= 200 Topic note: + Type: text field(string) + Max Length= 2000 Topic status: + Type: text field(string) + Max Length= 1	BR-02 BR-14 BR-15
3	Moderator	Update topic	Update topic	Topic name Topic note Topic status	Topic name: + Type: text field(string) + Max Length= 200 Topic note:	BR-02 BR-30

					+ Type: text field(string) + Max Length= 200 Topic status: + Type: text field(string) + Max Length= 1	

Table 21 - Topic Management Function

3.1.2.8 Languages Management

No	*Actor*	*Function Name*	*Purpose*	*Data requirements*	*Data validation*	*Business Rules*
1	Moderator	View language	View a language	Language name	Language name: + Type: text field(string) + Max Length= 200	BR-02
2	Moderator	Create language	Create a new language	Language name Language commission Language notes Language status	Language name: + Type: text field(string) + Max Length= 200 Language commission: + Type: text field(decimal) Language note: + Type: text field(string) + Max Length= 2000 Language status: + Type: text field(string)	BR-02 BR-16

					+ Max Length= 1	
3	Moderator	Update language	Update language	Language name Language commission Language notes Language status	Language name: + Type: text field(string) + Max Length= 200 Language commission: + Type: text field(decimal) Language note: + Type: text field(string) + Max Length= 2000 Language status: + Type: text field(string) + Max Length= 1	BR-02 BR-31

Table 22 - Languages Management Function

3.1.2.9 Achievement Management

No	***Actor***	***Function Name***	***Purpose***	***Data requirements***	***Data validation***	***Business Rules***
1	Moderator	View achievement	View a achievement	Achievement condition		BR-02
2	Moderator	Create achievement	Create a new achievement	Achievement link Achievement condition Achievement status	Achievement link: + Type: text field(string) Achievement condition:	BR-02 BR-17

					+ Type: text field(string) + Max Length= 200 Achievement status: + Type: text field(string) + Max Length= 1	
3	Moderator	Update achievement	Update achievement	Achievement link Achievement condition Achievement status	Achievement link: + Type: text field(string) Achievement condition: + Type: text field(string) + Max Length= 200 Achievement status: + Type: text field(string) + Max Length= 1	BR-02 BR-32

Table 23 - Achievement Management Function

3.1.2.10 Chat

No	***Actor***	***Function Name***	***Purpose***	***Data requirements***	***Data validation***	***Business Rules***
1	Moderator, Translator, Customer	Chat	Chat with another user in the system	Account id Account name		BR-02 BR-03 BR-04

Table 24 - Chat Function

3.1.2.11 Payment Management

No	*Actor*	*Function Name*	*Purpose*	*Data requirements*	*Data validation*	*Business Rules*
1	Customer	View payment	View the payment	None		BR-04
2	Customer	Make a payment	Make a new payment	Amount	Amount: + Type: text field(decimal)	BR-04
3	Customer, Translator	Withdraw	Withdraw money	Amount Email	Amount: + Type: text field(decimal) Email: + Type: text field(string) + Max Length = 50	BR-03 BR-04

Table 25 - Payment Management Function

3.1.2.12 Invitation Management

No	*Actor*	*Function Name*	*Purpose*	*Data requirements*	*Data validation*	*Business Rules*
1	Translator	View invitation	View the invitation	Project name Translator name	Project name: + Type: text field (string) + Max Length= 200 Translator name: + Type: text field (string) + Max Length= 200	BR-03
2	Translator	Accept invitation	Accept the invitation of customer	Translator id Project id		BR-03
3	Translator	Reject invitation	Reject the invitation of customer	Translator id Project id		BR-03

Table 26 - Invitation Management Function

3.1.3 Screen Authorization

3.1.3.1 Web Application Screen Authorization

Screen	Admin	Moderator	Customer	Translator	Guest
Register					x
Login	x	x	x	x	
Dashboard	x	x	x	x	x
Category		x			
Language		x			
Topic		x			
Achievement		x			
Project		x	x	x	
Process			x	x	
Account	x				
Invitation			x	x	
Request			x	x	
Banking	x		x	x	

Table 27 - <Screen authorization> Web application

3.1.4 Non-Screen Functions

#	System function	Description
1	Push notification	System push notification to customers and translator

Table 28 - Non-screen functions

3.2 Entity Relationship Diagram

- Conceptual ERD

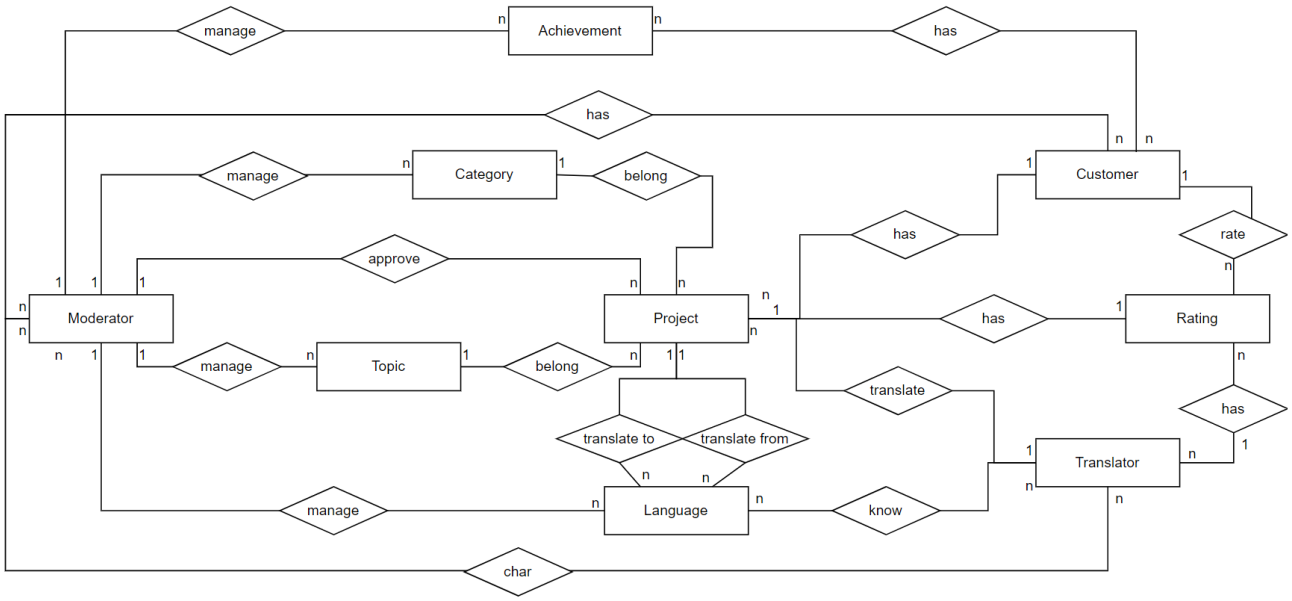


Figure 9 - Conceptual entity relationship diagram

- Logical ERD

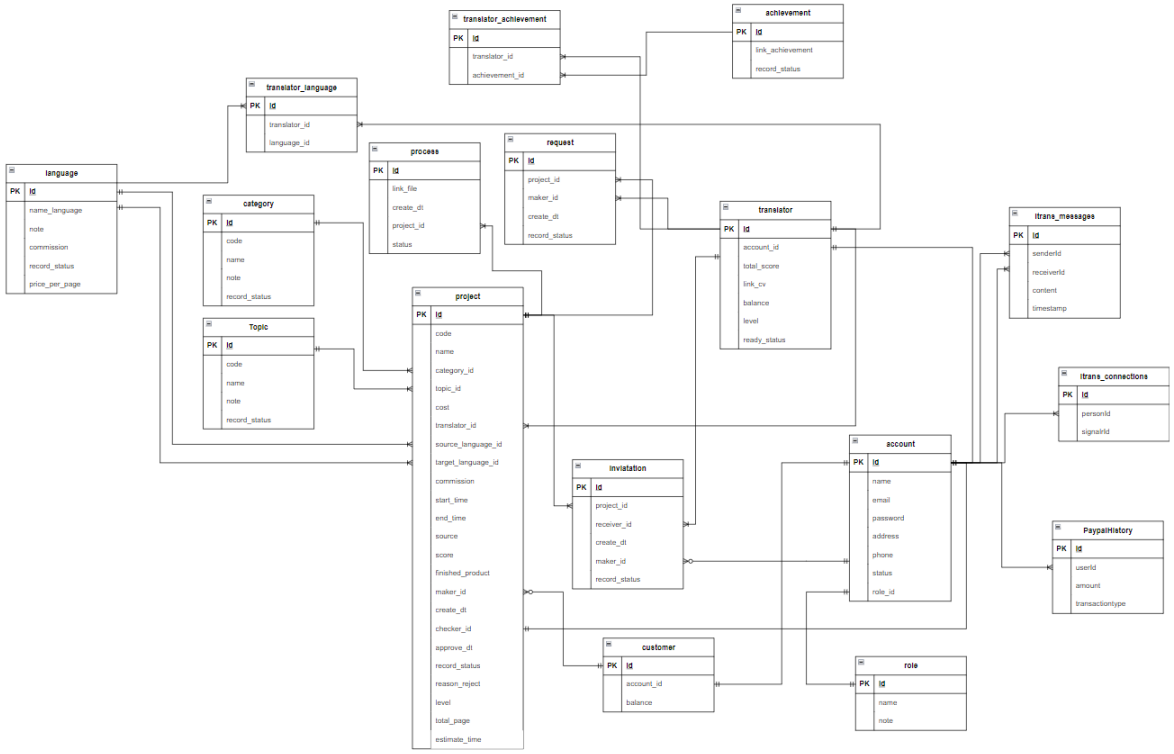


Figure 10 - Logical entity relationship diagram

Entity Name	Attributes	Description
account	tlname_id {PK}	Unique identifier of system account
	name	Name's account
	email	Registered email
	password	Account's password
	address	User's role
	phone	Account's status
	role_id	Unique identifier of system role
	status	Account's status
project	id{PK}	Unique identifier of project

	code	Code’s project
	name	Name’s project
	category_id	Unique identifier of system category
	topic_id	Unique identifier of system topic
	cost	Cost of project
	translator_id	Unique identifier of system translator
	source_language_id	Unique identifier of system language
	target_language_id	Unique identifier of system language
	commission	Commission of project
	start_time	Start time of project
	end_time	End time of project
	source	Source of project
	finished_product	Product of translator for project
	score	Score of project
	maker_id	Unique identifier of system customer
	create_dt	project creation date
	checker_id	Unique identifier of system account
	approve_dt	project approval date
	record_status	Status of project
	total_page	Total page of project

	estimate_time	Estimate time
category	id{PK}	Unique identifier of category
	code	Code of category
	name	Name of category
	note	Note of category
	record_status	Status of category
topic	id{PK}	Unique identifier of topic
	code	Code of topic
	name	Name of topic
	note	Note of topic
	record_status	Status of topic
language	id{PK}	Unique identifier of language
	name	Name of language
	note	Note of language
	commission	Commission of language
	price_per_page	Price per page
	record_status	Status of language
rating	id{PK}	Unique identifier of rating
	note	Note of rating
	project_id	Unique identifier of system project

	maker_id	Unique identifier of system customer
	create_dt	Rating creation date
	record_status	Status of rating
customer	id{PK}	Unique identifier of customer
	account_id	Unique identifier of system account
	amount	amount of customer
translator	id{PK}	Unique identifier of translator
	account_id	Unique identifier of system account
	amount	Amount of translator
	score	Score of the translator
	link_cv	Link cv of translator
	ready_status	Ready status
process	id{PK}	Unique identifier of process
	link_file	Link file of process
	create_dt	Process update date
	project_id	Unique identifier of system project
	maker_id	Unique identifier of system translator
	notes	Note of process
	status	Status of process
achievement	id{PK}	Unique identifier of achievement

	name	name of achievement
	record_status	Status of achievement
request	id{PK}	Unique identifier of request
	project_id	Unique identifier of system project
	maker_id	Unique identifier of system translator
	create_dt	Request creation date
	record_status	Status of request
invitation	id{PK}	Unique identifier of invitation
	project_id	Unique identifier of system project
	receiver_id	Unique identifier of system translator
	create_dt	Invitation creation date
	record_status	Status of invitation
role	id{PK}	Unique identifier of role
	name	Name of role
	note	Note of role
translator_achievement	id{PK}	Unique identifier of translator’s achievement
	translator_id	Unique identifier of system translator
	achievement _id	Unique identifier of system achievement
translator_language	id{PK}	Unique identifier of translator’s language
	translator_id	Unique identifier of system translator

	language_id	Unique identifier of system language
ITRANS_MESSAGES	id{PK}	Unique identifier of itrans_messages
	senderId	Foreign key, account identifier
	receiverId	Foreign key, account identifier
	content	Content of chat
	timeStamp	Time of chat
ITRANS_CONNECTIONS	id{PK}	Unique identifier of itrans_connections
	personId	Foreign key, account identifier
	signalrId	Id of signalR
	timestamp	Time of create
PaypalHistory	id{PK}	Unique identifier of paypal history
	userId	Unique identifier of system account
	amount	amount of transaction
	transactionType	Type of transaction
	createDate	Transaction creation date

Table 29 - Logic entities and attributes description

4. Non-Functional Requirements

4.1 External Interfaces

4.1.1 User Interfaces

- All the texts, labels and should be written in Vietnamese.

4.2 Quality Attributes

4.2.1 Usability

- Admin, moderator should need less than one week of training to be productive in using the system.

4.2.2 Reliability

- Admin, moderator should need training to be productive in using the system.

4.2.3 Performance

- All common response time is under 5 seconds (except define pickup points feature).

4.2.4 Design constraints

- Application use Vietnamese

4.2.5 Security

- Each role of user has a specific permission to interact with system

5. Requirement Appendix

5.1 Business Rules

ID	Rule Definition
BR-01	Admin have to login by admin account.
BR-02	Moderator have to login by moderator account.
BR-03	Translators have to login by translator account.
BR-04	Customers have to login by customer account.
BR-05	There is one admin account in system.
BR-06	There is one moderator account in system.
BR-07	When logged out successfully, the system will redirect to the login page.
BR-08	User can view profile after login to system
BR-09	User can update profile after login to system
BR-10	Username must be unique.
BR-11	Id category will be generated by the system.
BR-12	Code category will be generated by the system.
BR-13	Id topic will be generated by the system.
BR-14	Code topic will be generated by the system.
BR-15	Id language will be generated by the system.
BR-16	Id achievement will be generated by the system.
BR-17	Id project will be generated by the system.

BR-18	Code project will be generated by the system.
BR-19	Id process will be generated by the system.
BR-20	Status default of new account translator is unactive
BR-21	Username cannot be updated
BR-22	Id category cannot change when updated
BR-23	Id topic cannot change when updated
BR-24	Id language cannot change when updated
BR-25	Id achievement cannot change when updated
BR-26	Id account cannot change when updated
BR-27	Customer can search only translator account's information
BR-28	Customer can view detail only translator account's information
BR-29	Only the admin can ban account
BR-30	When a banned account login in system. The system has show message
BR-31	Moderator reject the project must write the reason
BR-32	Only the moderator can create category
BR-33	Only the moderator can update category
BR-34	Only the moderator can reject project
BR-35	The account has two status: active and unactive
BR-36	The account has translator role has four status: ready, free, busy, unactive
BR-37	The start time of project must be earlier than the end time
BR-38	Moderator is expert in translation
BR-39	Customer can only edit project when no translator joined the project
BR-40	Customer can only delete project when no translator joined the project
BR-41	When a customer deletes the project, the salary deducted will be refunded to Customer's balance.

BR-42	Customer can only create project when their balance more than project's cost
BR-43	Modeator add translator appropriate to project
BR-44	A login session with the web application lasts until the browser is closed and the users log out of the system.
BR-45	Type of date/datetime must be valid with format : dd/mm/yyyy
BR-46	User can't login with account is banned
BR-47	One project just has one translator
BR-48	If a project is started. It can't be deleted
BR-49	One translator can join multiple project at the same time.
BR-50	After the translator submit the product, the system notify the moderator to approve
BR-51	The commission of the system depend on language
BR-52	Translator can withdraw money through the paypal
BR-53	Translator can view their translate experience
BR-54	Translator can view their translated project
BR-55	Customer can deposit their balance through paypal
BR-56	1$ = 1 cent in systems
BR-57	When project is completed, if moderator does not accept the final product of translator, the money will be refunded to the customer
BR-58	Project have 5 type of level: 1, 2 , 3, 4 ,5
BR-59	Level 1 depend on the rate of total page of project divide estimate time <= 5
BR-60	Level 2 depend on the rate of total page of project divide estimate time <= 15
BR-61	Level 3 depend on the rate of total page of project divide estimate time <= 25
BR-62	Level 4 depend on the rate of total page of project divide estimate time <= 35
BR-63	Level 5 depend on the rate of total page of project divide estimate time <= 50
BR-64	Project has the rate of total page of project divide estimate time > 50 will be reject

Table 30 - Business rules

5.2 Application Messages List

#	Message code	Message Type	Context	Content
1	MSG01	Toast message	Added new record successfully	Tạo mới * thành công
2	MSG02	Toast message	Updated record successfully	Cập nhật * thành công
3	MSG03	Toast message	Deleted record successfully	Cập nhật trạng thái thành công
4	MSG04	Toast message	Authentication failed	Tài khoản hoặc mật khẩu không đúng
5	MSG05	Toast message	Request is not valid	* không thành công
6	MSG06	Delete confirm message	Click delete record	Bạn có chắc muốn xóa * này?
7	MSG07	Error message, text color red, beside label of text box	Input-required fields are not valid	* không đúng
8	MSG08	Error message, color red, under text box	Input-required fields are empty	* không được bỏ trống
9	MSG09	Error message, response	Interval Server Error, Request is not valid and still send to server	* lỗi hệ thống

Table 31 - Application messages

IV. Software Design Description

1. System Design

1.1 System Architecture Overview

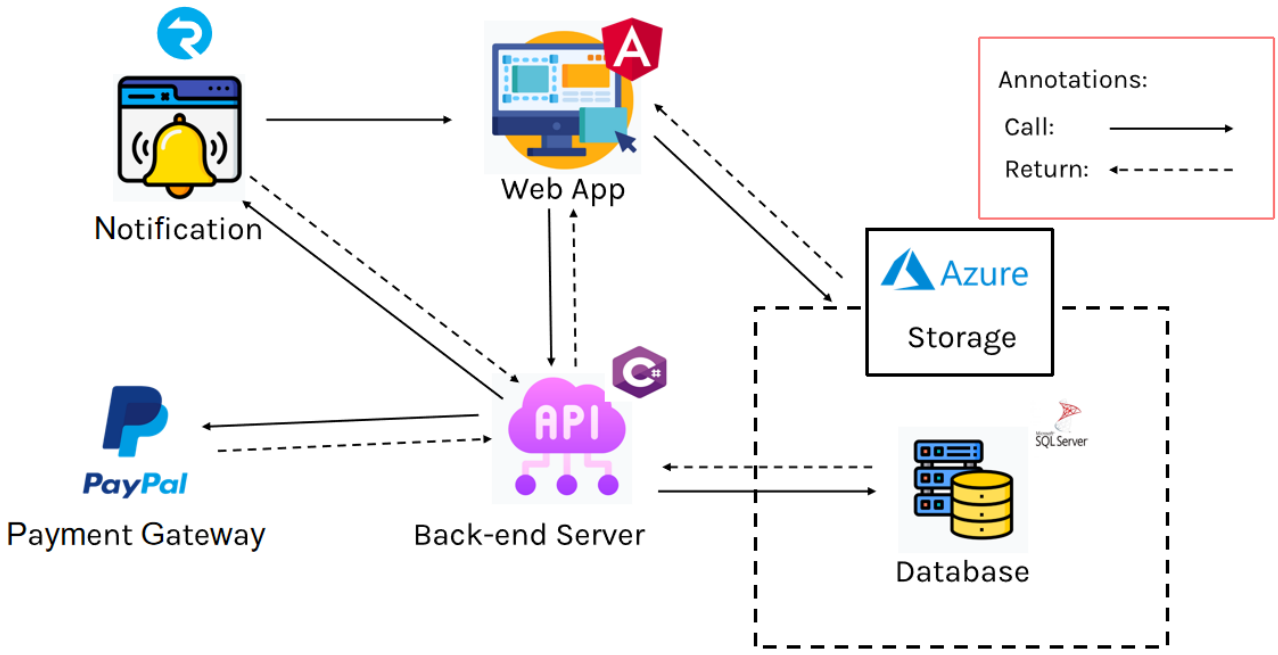


Figure 11 - System architecture

- The iTranslator system consists of two components, the .NET Core web server provides APIs, Angular is used for the web application. Angular is used for web applications. Web application is provide for admin, moderator, customer and translator.
- iTranslator System uses several third-party services such as Azure Storage to store user's database, SignalR for push notification and Paypal for payment

1.2 Package Diagram

1.2.1 Web API Package Diagram

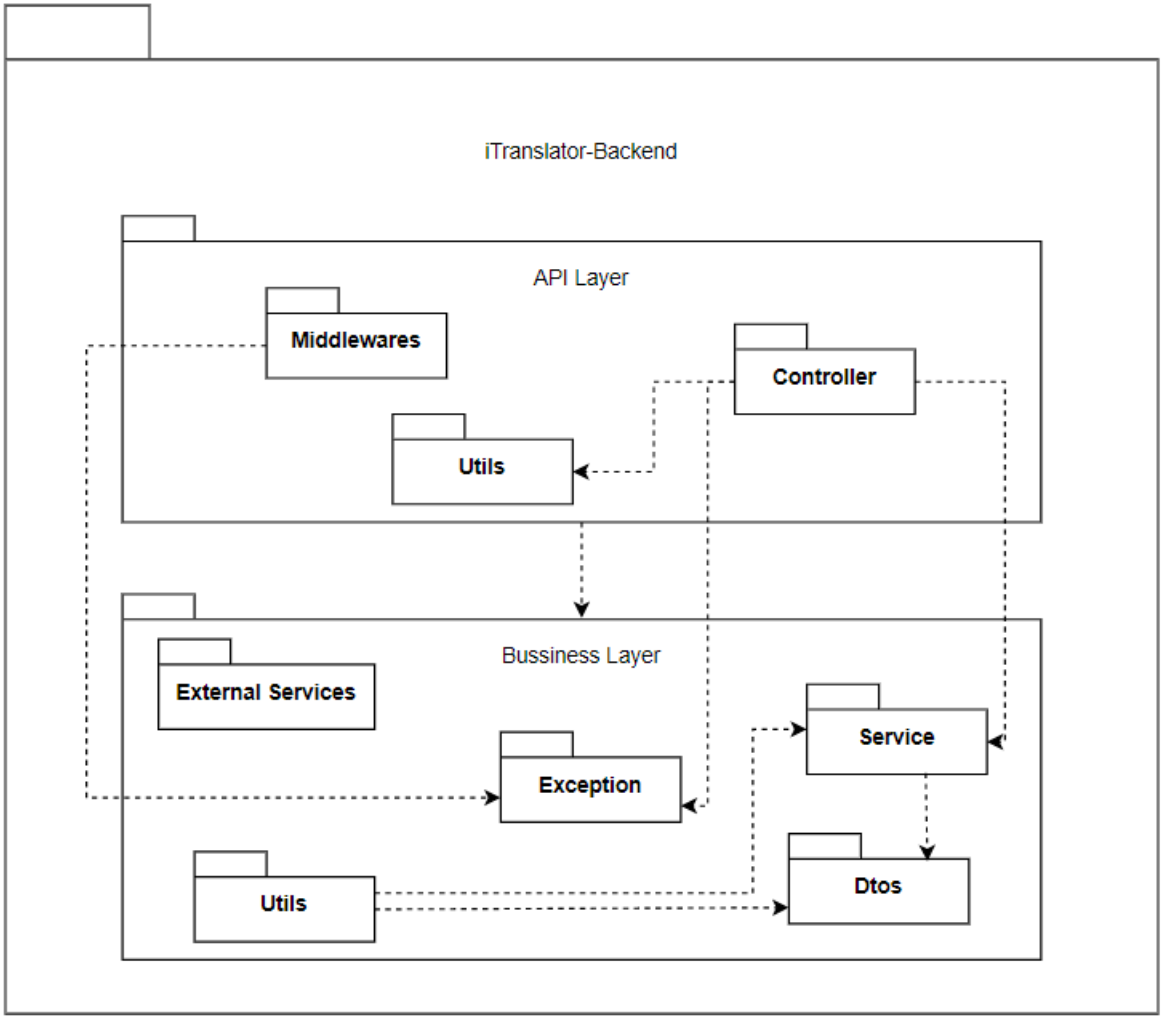


Figure 12 - Web API Package Diagram

1.2.2 Web Application Package Diagram

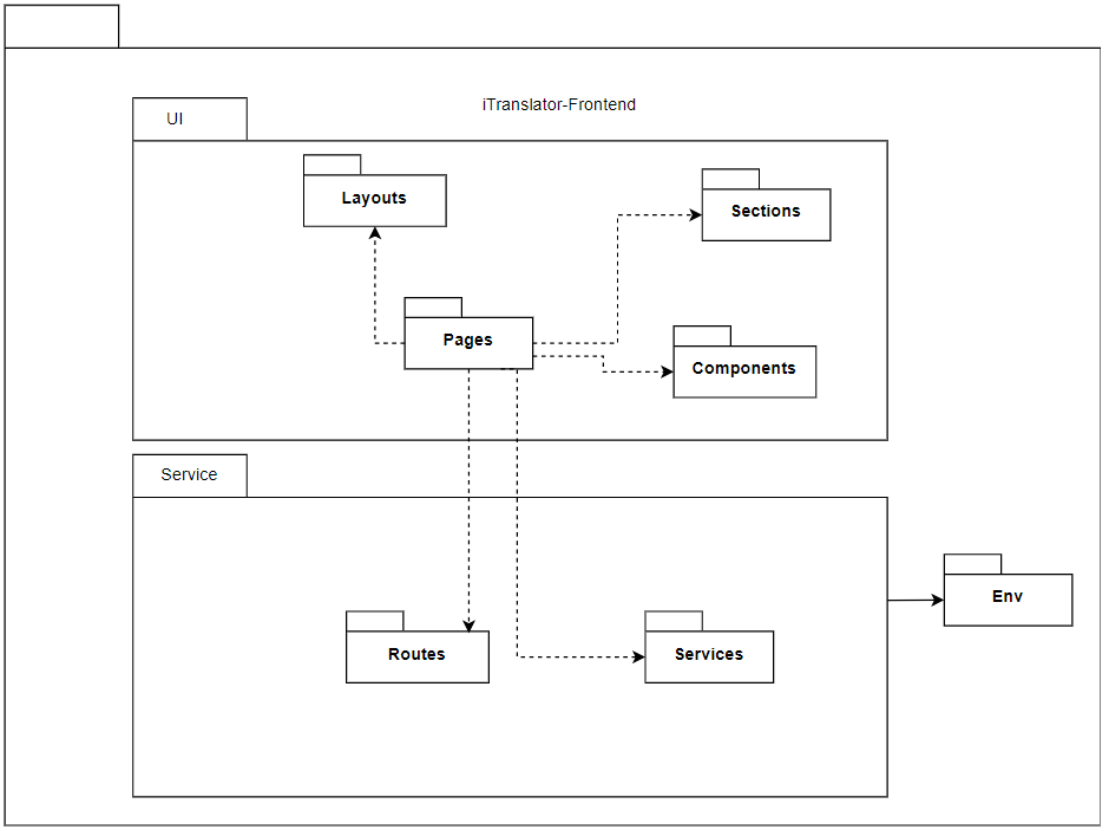


Figure 13 - Web Application Package Diagram

2. Database Design

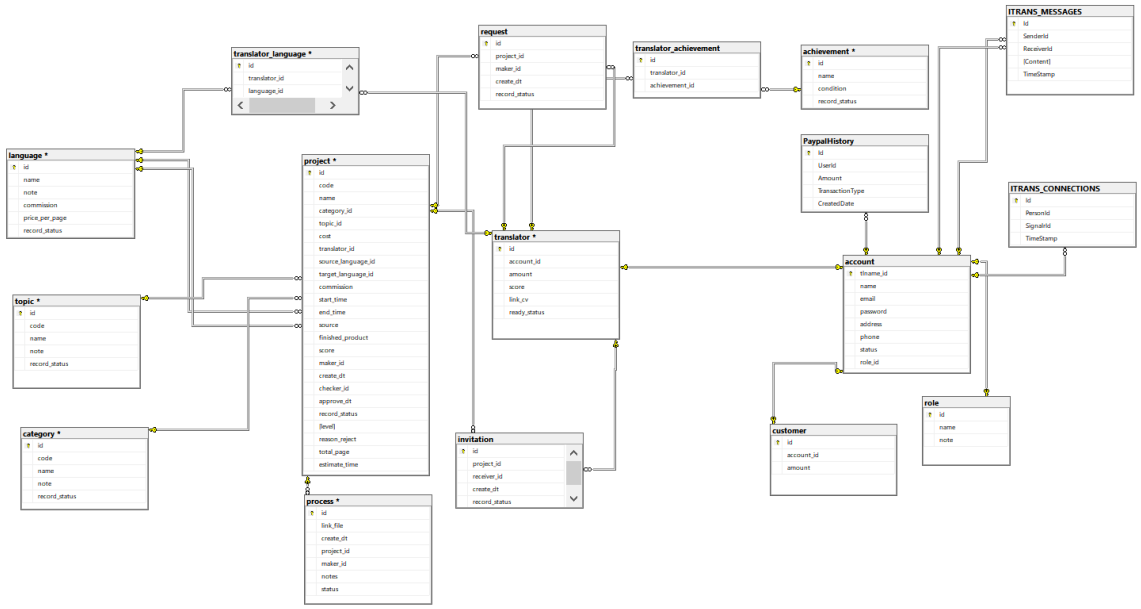


Figure 14 - Database Design

Table Descriptions

No	Table	Description
01	account	Describe account in the system
02	project	Describe project in the system
03	category	Describe category in the system
04	topic	Describe topic in the system
05	Language	Describe language in the system
06	rating	Describe rating in the system
07	customer	Describe customer in the system
08	translator	Describe translator in the system
09	process	Describe process in the system
10	achievement	Describe achievement in the system
11	request	Describe request in the system
12	invitation	Describe invitation in the system
13	role	Describe role in the system
14	translator_achievement	Describe achievement for translator in the system

15	translator_language	Describe language of translator in the system
16	ITRANS_MESSAGES	Describe message between mod and translator or customer
17	ITRANS_CONNECTIONS	Describe who is online
18	PaypalHistory	Describe money transaction of customer and translator in the system

Table 32 - Table Data Descriptions

Attribute Data Dictionary

Entity Name	Attributes	Description	Domain	Null
account	tlname_id {PK}	Unique identifier of system account	varchar(50)	No
	name	Name's account	nvarchar(200)	No
	email	Registered email	nvarchar(50)	No
	password	Account's password	varchar(100)	No
	address	User's role	nvarchar(1000)	No
	phone	Account's status	varchar(12)	No
	role_id	Foreign key, role identifier	nvarchar(10)	No
	status	Account's status	int	No
project	id{PK}	Unique identifier of project	varchar(50)	No
	code	Code's project	varchar(50)	No
	name	Name's project	nvarchar(200)	No
	category_id	Foreign key, category identifier	varchar(50)	No
	topic_id	Foreign key, topic identifier	varchar(50)	No
	cost	Cost of project	decimal(32,2)	No

	translator_id	Foreign key, translator identifier	varchar(50)	No
	source_language_id	Foreign key, language identifier	varchar(50)	No
	target_language_id	Foreign key, language identifier	varchar(50)	No
	commission	Commission of project	decimal(32,2)	No
	start_time	Start time of project	datetime	No
	end_time	End time of project	datetime	No
	source	Source of project	nvarchar(max)	No
	finished_product	Product of translator for project	nvarchar(max)	No
	score	Score of project	decimal(32,2)	No
	maker_id	Foreign key, customer identifier	varchar(50)	No
	create_dt	Project creation date	datetime	No
	checker_id	Foreign key, account identifier	nvarchar(50)	No
	approve_dt	Project approval date	datetime	No
	record_status	Status of project	nvarchar(1)	No
	level	Level of project	nvarchar(10)	No
	reason_reject	Reason of rejection	nvarchar(max)	No
	total_page	Total page	decimal(32,2)	No
	estimate_time	Estimate time	decimal(32,2)	No
category	id{PK}	Unique identifier of category	varchar(50)	No

	code	Code of category	varchar(50)	No
	name	Name of category	nvarchar(200)	No
	note	Note of category	nvarchar(200)	Yes
	record_status	Status of category	nvarchar(1)	No
topic	id{PK}	Unique identifier of topic	varchar(50)	No
	code	Code of topic	varchar(50)	No
	name	Name of topic	nvarchar(200)	No
	note	Note of topic	nvarchar(200)	No
	record_status	Status of topic	nvarchar(1)	No
language	id{PK}	Unique identifier of language	varchar(50)	No
	name	Name of language	nvarchar(200)	No
	note	Note of language	nvarchar(200)	Yes
	commission	Commission of language	decimal(32,2)	No
	price_per_page	Price per page	decimal(32,2)	No
	record_status	Status of language	nvarchar(1)	No
rating	id{PK}	Unique identifier of rating	varchar(50)	No
	note	Note of rating	nvarchar(200)	Yes
	project_id	Foreign key, project identifier	varchar(50)	No
	maker_id	Foreign key, customer identifier	nvarchar(50)	No

	create_dt	Rating creation date	datetime	No
	record_status	Status of rating	nvarchar(1)	No
customer	id{PK}	Unique identifier of customer	varchar(50)	No
	account_id	Foreign key, account identifier	varchar(50)	No
	amount	amount of customer	decimal(32,2)	No
translator	id{PK}	Unique identifier of translator	varchar(50)	No
	account_id	Foreign key, account identifier	varchar(50)	No
	amount	Amount of translator	decimal(32,2)	No
	score	Score of the translator	float	No
	link_cv	Link cv of translator	varchar(500)	No
	ready_status	Status of translator	nvarchar(50)	No
process	id{PK}	Unique identifier of process	varchar(50)	No
	link_file	Link file of process	varchar(500)	No
	create_dt	Process update date	datetime	No
	project_id	Foreign key, project identifier	varchar(50)	No
	maker_id	Foreign key, translator identifier	varchar(50)	No
	status	Status of process	Nvarchar(1)	No
	notes	Note of process	nvarchar(2000)	Yes
achievement	id{PK}	Unique identifier of achievement	varchar(50)	No

	link_achievement	Link of achievement	varchar(500)	No
	record_status	Status of achievement	nvarchar(1)	No
request	id{PK}	Unique identifier of request	varchar(50)	No
	project_id	Foreign key, project identifier	varchar(50)	No
	maker_id	Foreign key, translator identifier	varchar(50)	No
	create_dt	Request creation date	datetime	No
	record_status	Status of request	nvarchar(1)	No
invitation	id{PK}	Unique identifier of invitation	varchar(50)	No
	project_id	Foreign key, project identifier	varchar(50)	No
	receiver_id	Foreign key, translator identifier	varchar(50)	No
	create_dt	Invitation creation date	datetime	No
	record_status	Status of invitation	nvarchar(1)	No
role	id{PK}	Unique identifier of role	nvarchar(20)	No
	name	Name of role	varchar(50)	No
	note	Note of role	varchar(200)	Yes
translator_achievement	id{PK}	Unique identifier of translator's achievement	varchar(50)	No
	translator_id	Foreign key, translator identifier	varchar(50)	No
	achievement _id	Foreign key, achievement identifier	varchar(50)	No

translator_language	id{PK}	Unique identifier of translator's language	varchar(50)	No
	translator_id	Foreign key, translator identifier	varchar(50)	No
	language_id	Foreign key, language identifier	varchar(50)	No
ITRANS_MESSAGES	id{PK}	Unique identifier of itrans_messages	varchar(50)	No
	senderId	Foreign key, account identifier	varchar(50)	No
	receiverId	Foreign key, account identifier	varchar(50)	No
	content	Content of chat	nvarchar(max)	No
	timeStamp	Time of chat	datetime	No
ITRANS_CONNECTIONS	id{PK}	Unique identifier of itrans_connections	varchar(50)	No
	personId	Foreign key, account identifier	varchar(50)	No
	signalrId	Id of signalR	varchar(255)	No
	timestamp	Time of create	datetime	No
PaypalHistory	id{PK}	Unique identifier of paypal history	varchar(50)	No
	userId	Foreign key, account identifier	varchar(50)	No
	amount	amount of transaction	decimal(18,2)	No
	transactionType	Type of transaction	nvarchar(50)	No
	createDate	Transaction creation date	datetime	No

Table 33 - Attribute Data Description

3. Detailed Design

3.1 Project Management Feature

3.1.1 Class Diagram

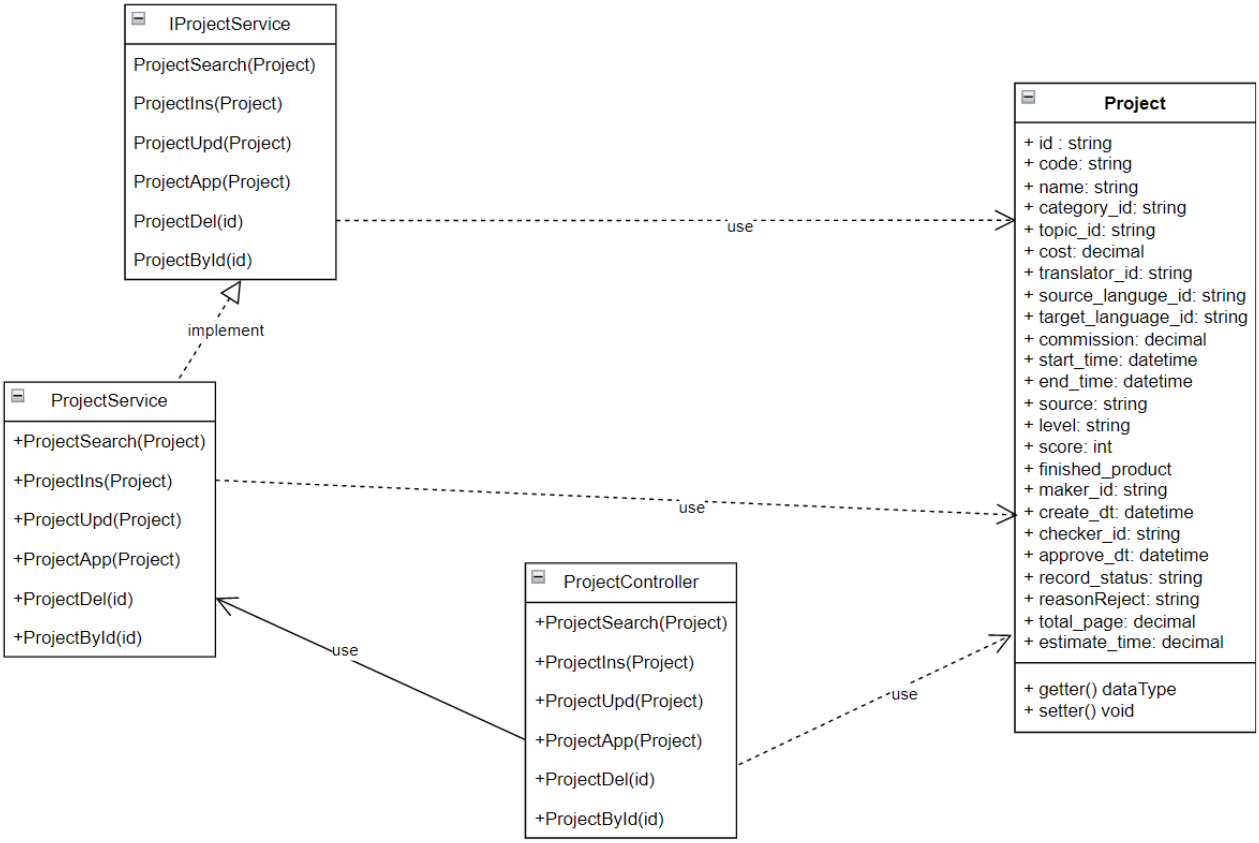


Figure 15 - Project Management Feature Class Diagram

3.1.2 Class Specification

a. Project Service

No	Method	Description
1	searchProject(parameterSearchProject)	handle get project by conditions
2	createProject(project)	handles create new project
3	viewProject(project)	handles view project information
4	deletProject(projectId: string)	handles delete project
5	editProject(project)	handles update project information
6	searchTranslator(parameterSearchTranslator)	handle get translator by conditions
7	viewTranslator(translator)	handles view translator information
8	inviteTranslator(invitationId: string)	handles invite translator

9	acceptTranslator(requestId: string)	handles accept translator
10	rejectTranslator(requestId: string)	handles reject translator
11	viewProcess(process)	handles view process information

Table 34 - Project Service Class Specification

3.1.3 Activity Diagram

3.1.3.1 Create Project Activity Diagram

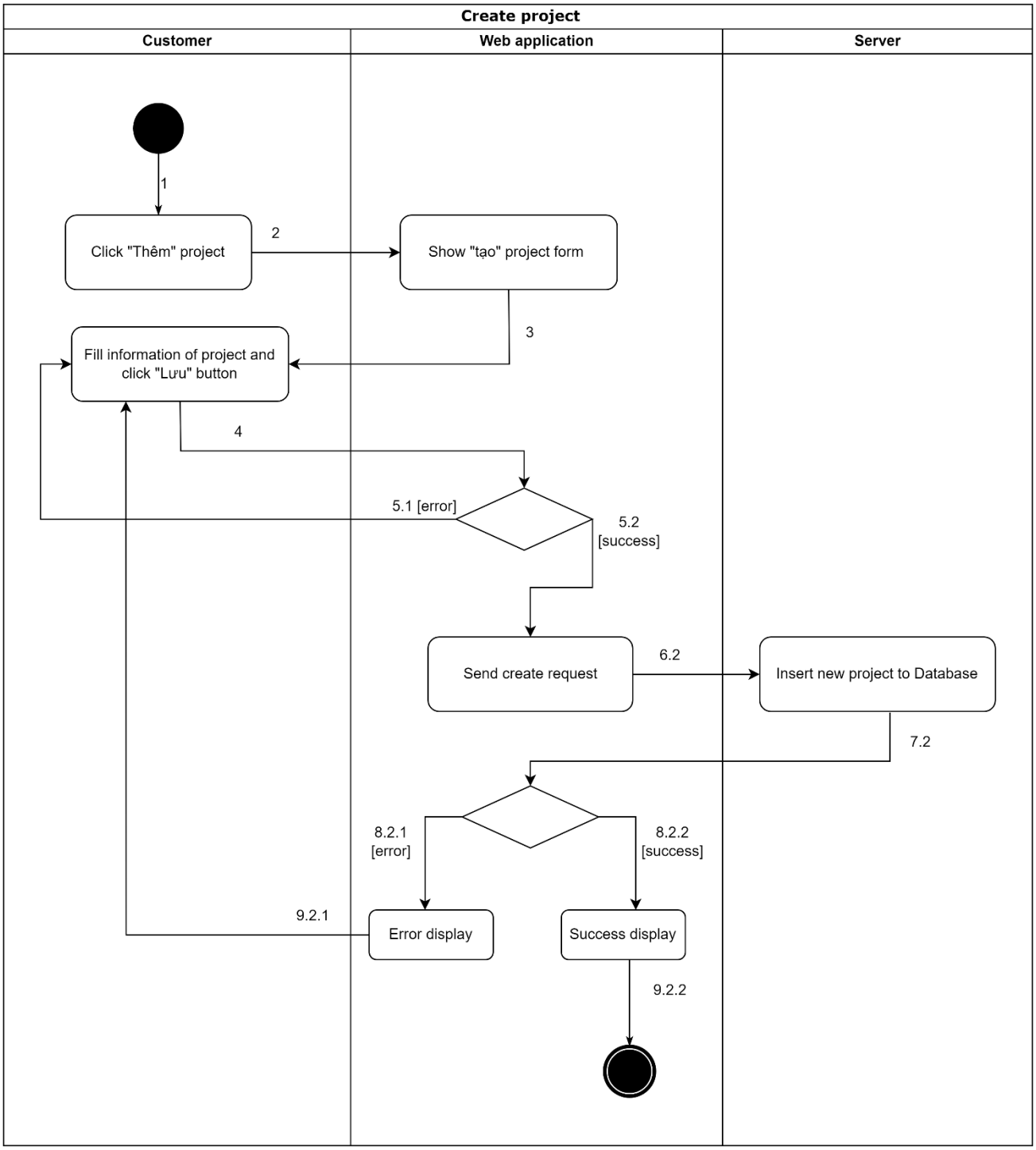


Figure 16 - Create Project Activity Diagram

3.1.3.2 View Project Activity Diagram

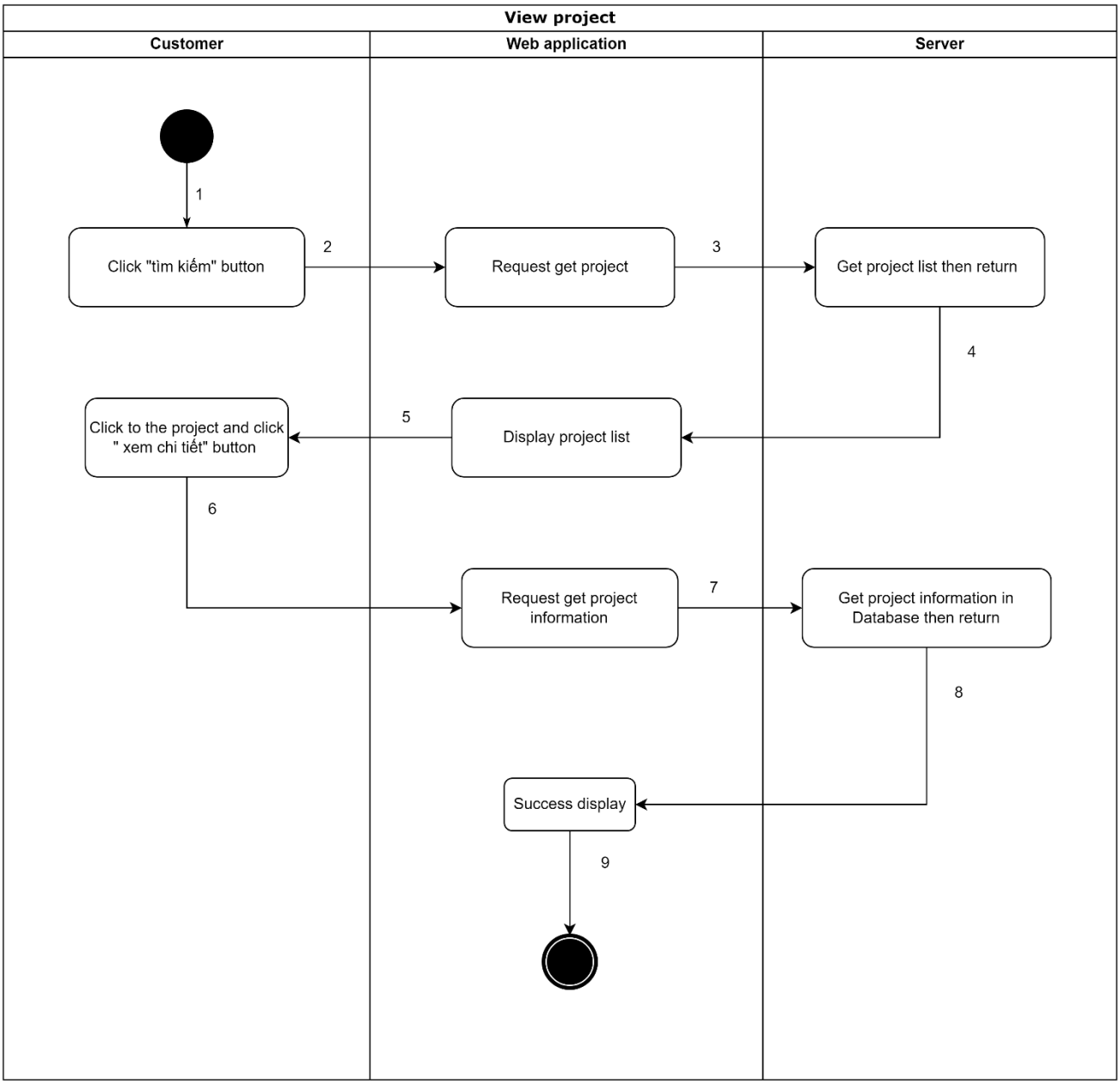


Figure 17 - View Project Activity Diagram

3.1.3.3 Edit Project Activity Diagram

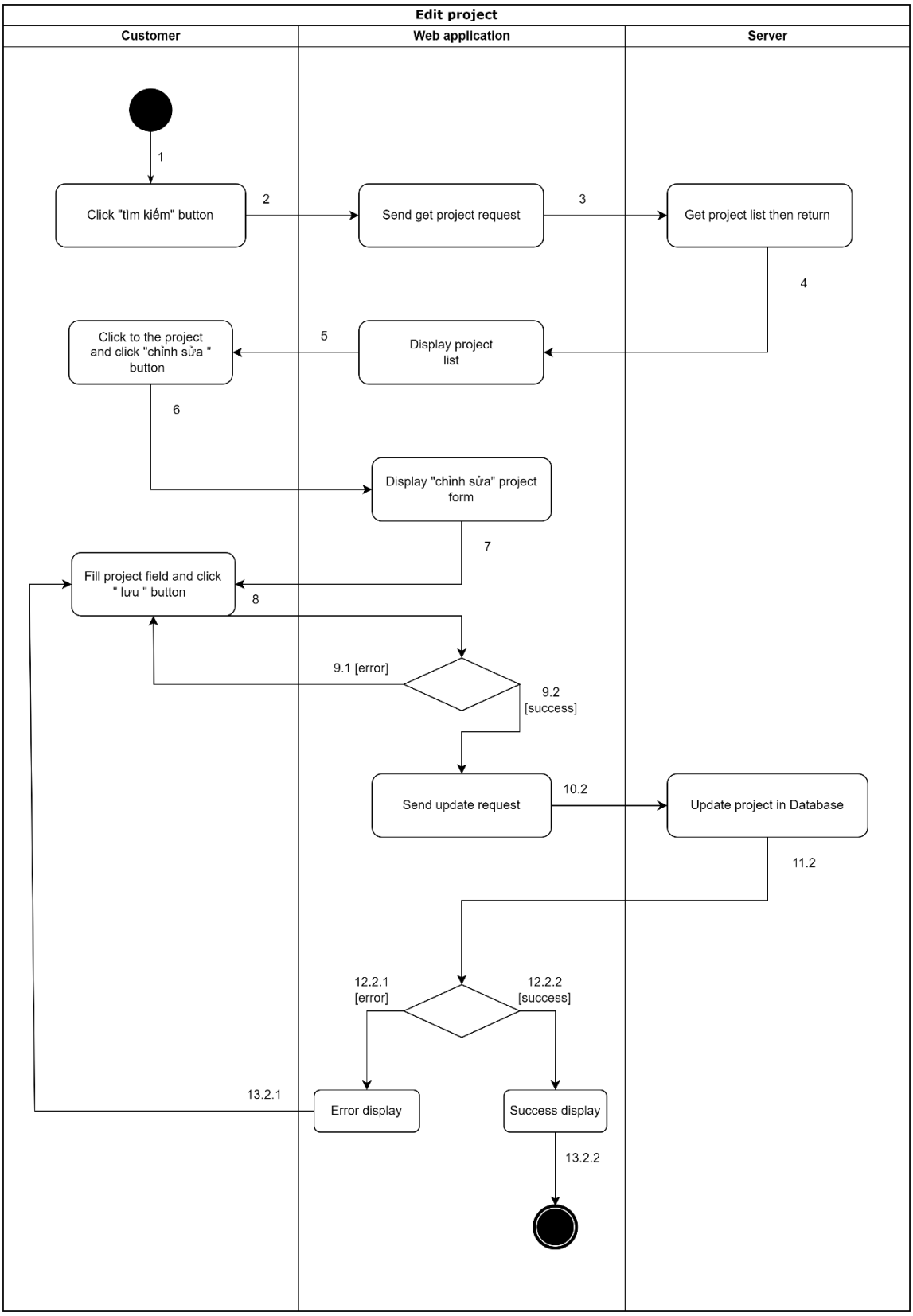


Figure 18 - Edit Project Activity Diagram

3.1.3.4 Delete Project Activity Diagram

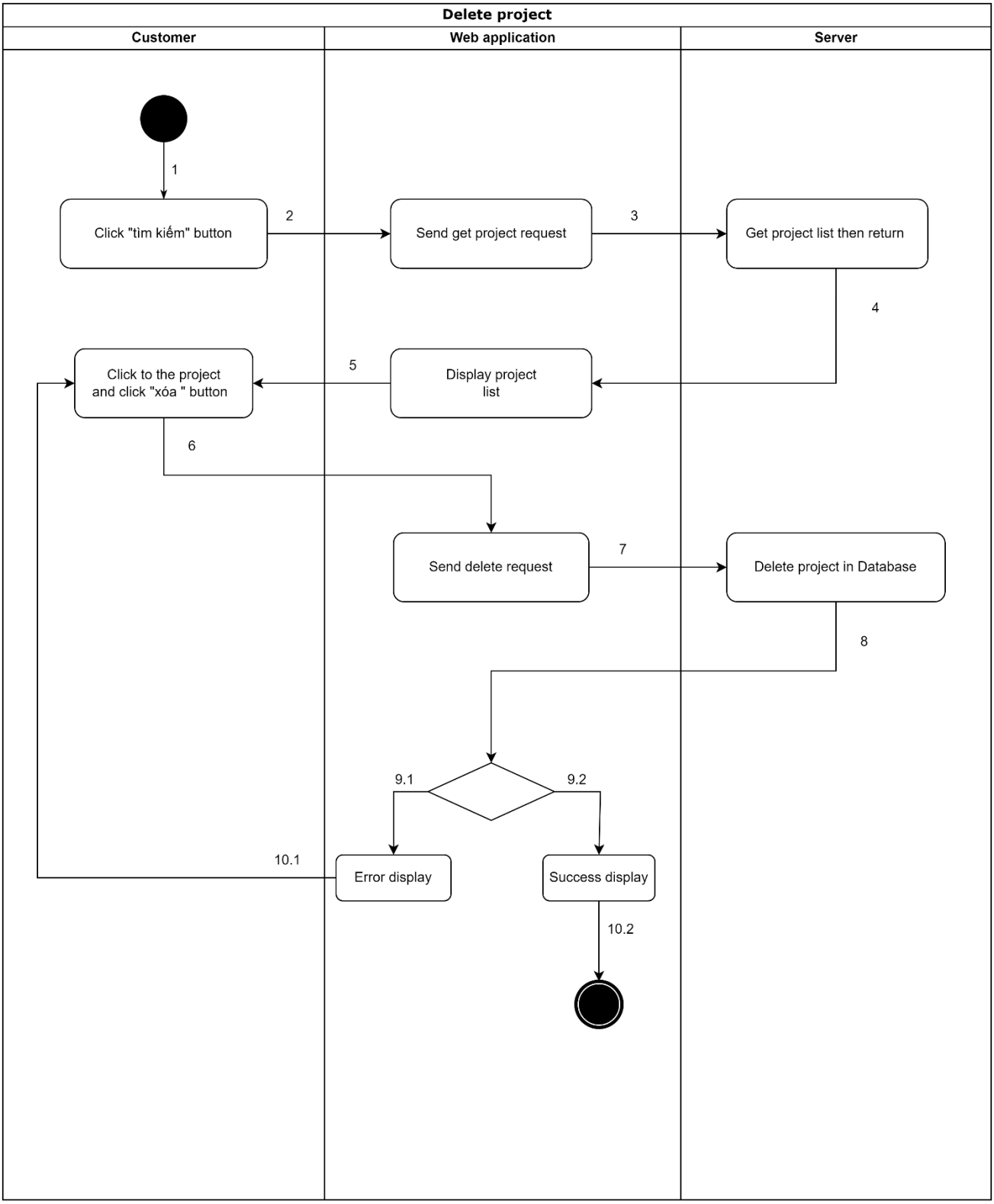


Figure 19 - Delete Project Activity Diagram

3.1.4 Create Project Sequence Diagram

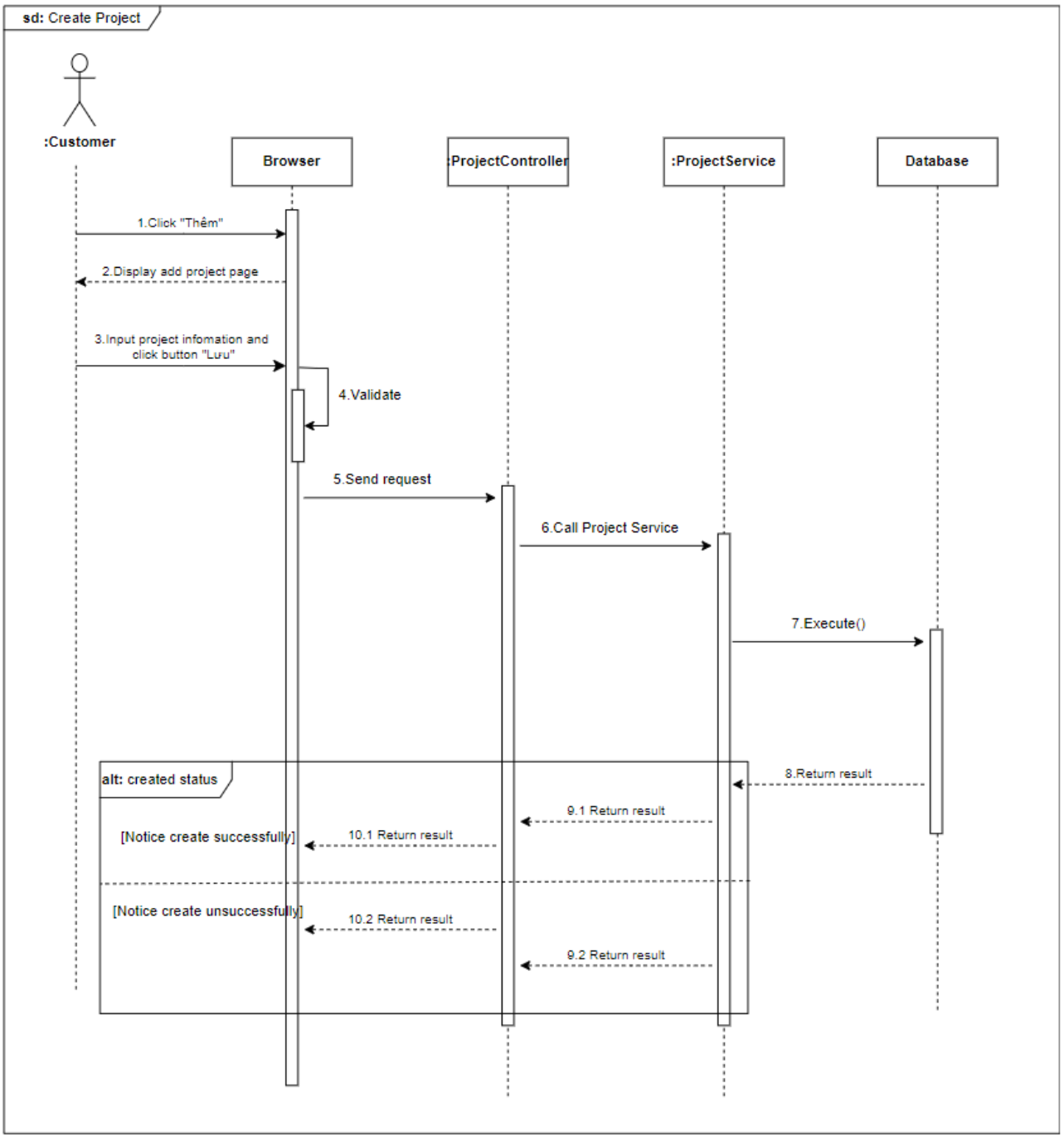


Figure 20 - Create Project Sequence Diagram

3.1.5 Create Project State Machine Diagram

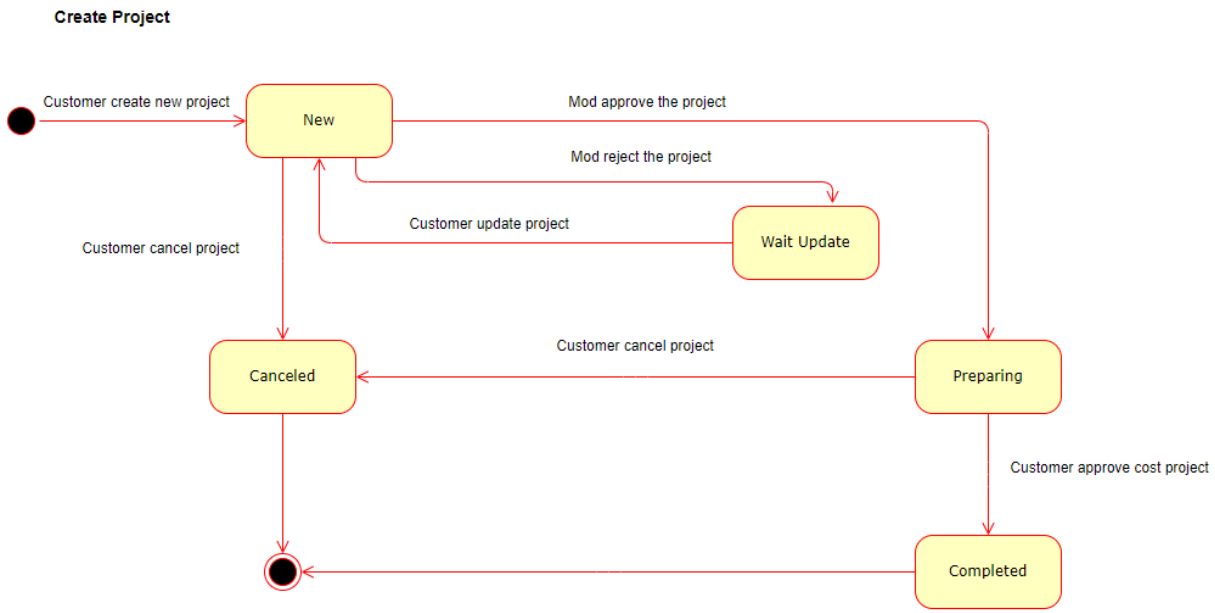


Figure 21 - Create Project State Machine Diagram

3.2 Process Management Feature

3.2.1 Class Diagram

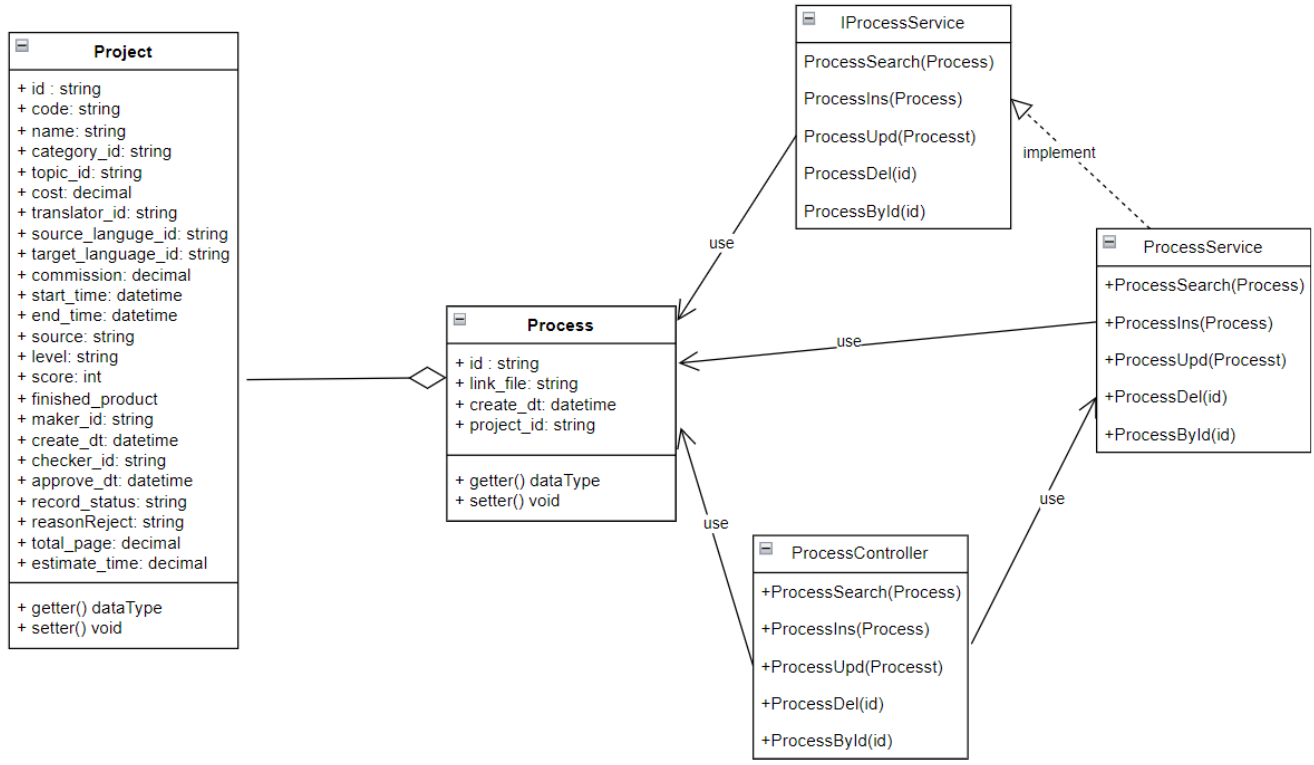


Figure 22 - Process Management Feature Class Diagram

3.2.2 Class Specification

a. Process Service

No	Method	Description
1	viewProcess(process)	handles view process information
2	updateTranslation(process)	handles update process information

Table 35 - Process Service Class Specification

3.2.3 Activity Diagram

3.2.3.1 View Process Activity Diagram

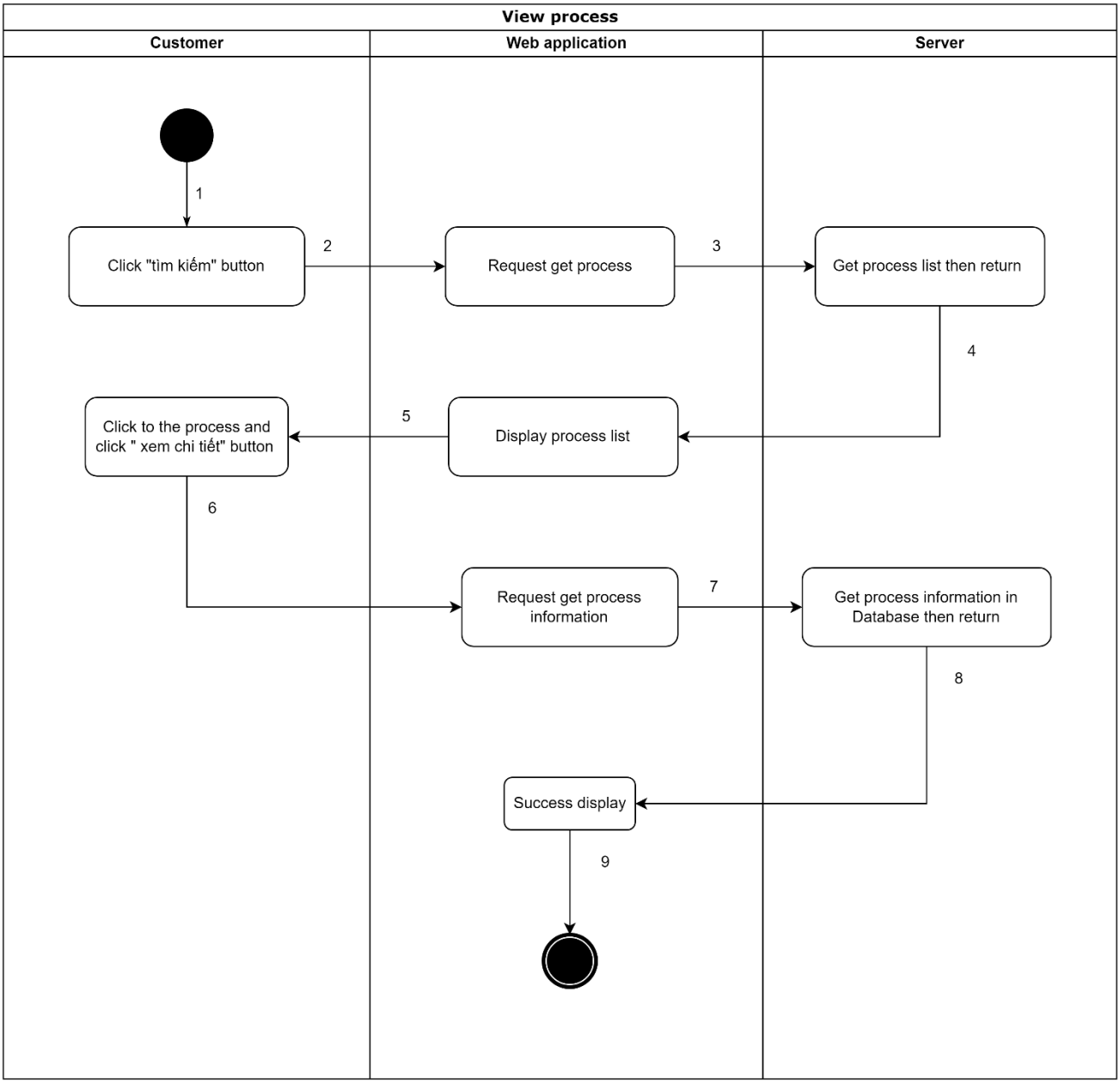


Figure 23 - View Process Activity Diagram

3.2.4 Create Process Sequence Diagram

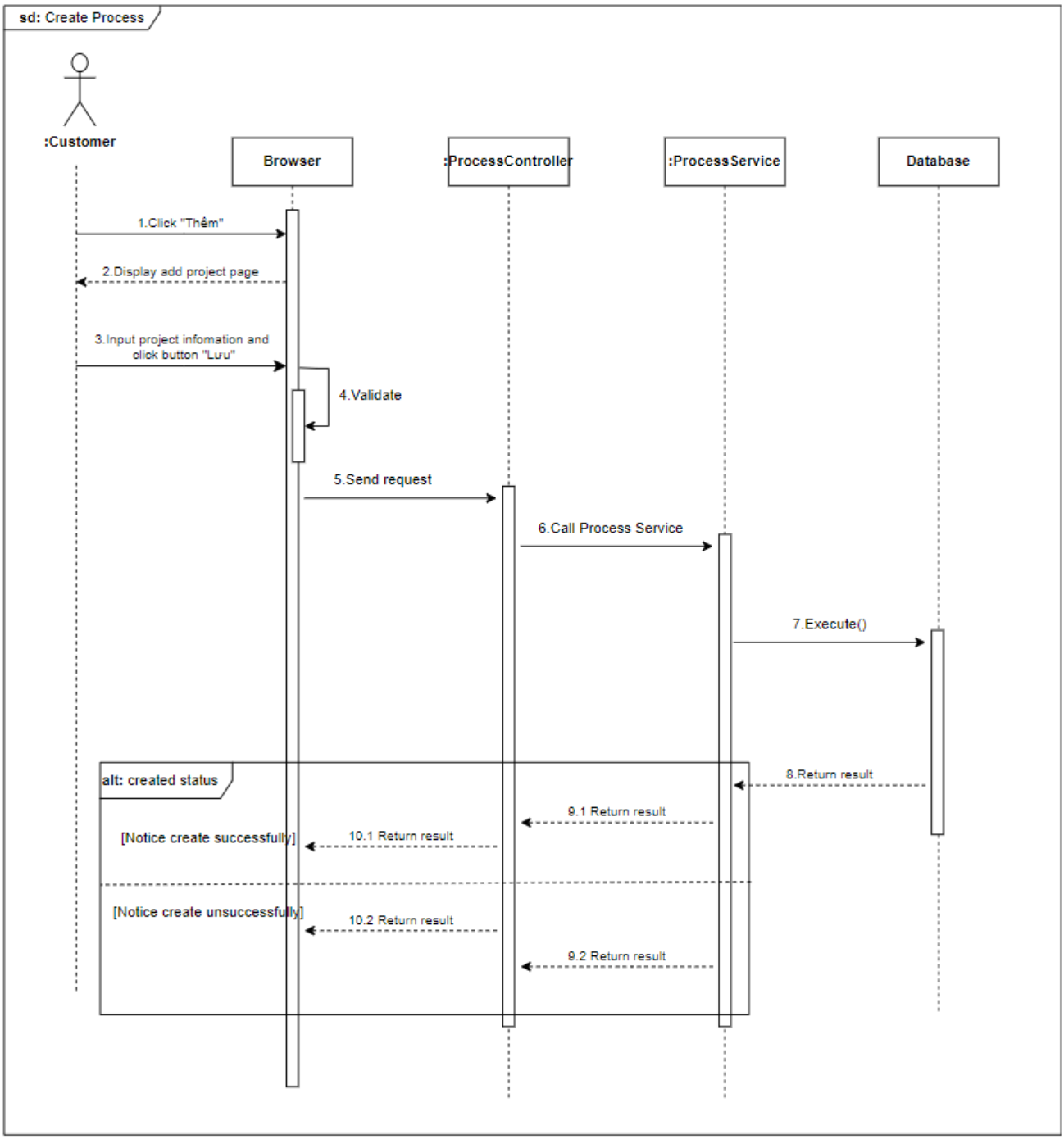


Figure 24 - Create Process Sequence Diagram

3.3 Payment Management Feature

3.3.1 Class Diagram

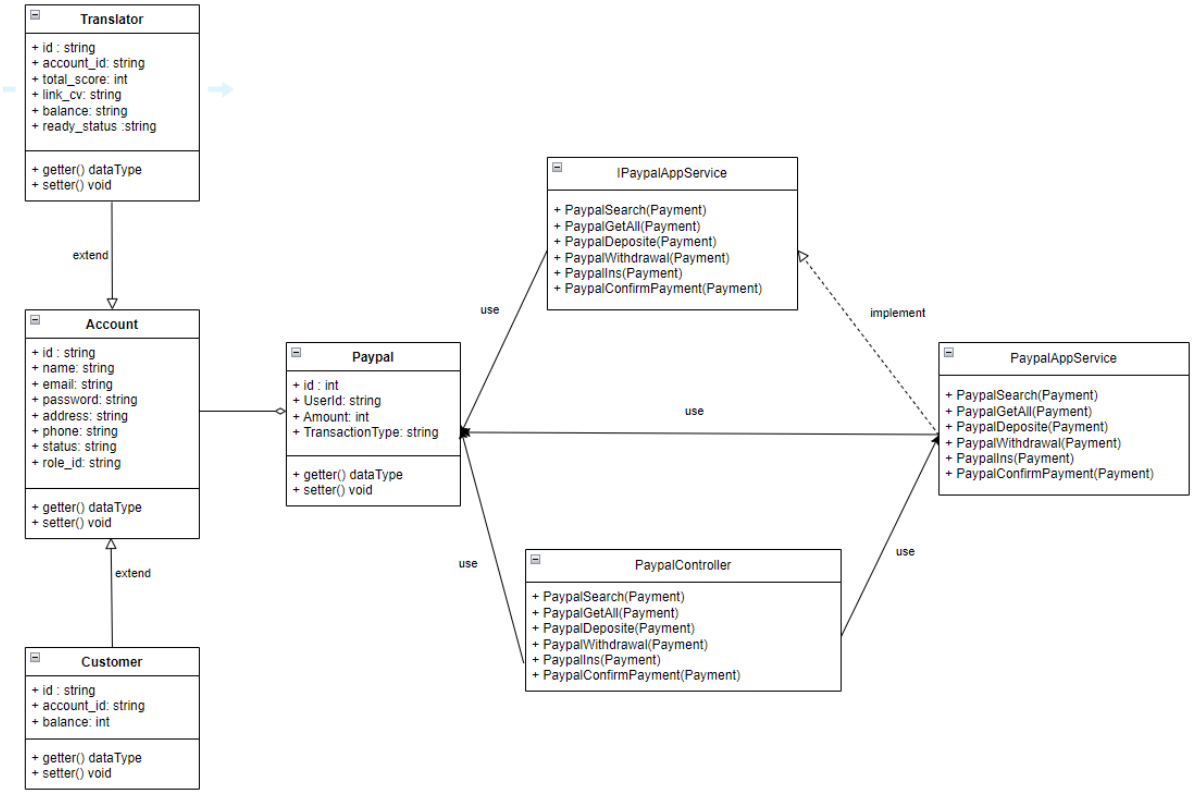


Figure 25 - Payment Management Feature Class Diagram

3.3.2 Class Specification

a. Payment Service

No	Method	Description
1	viewPayment(paypalHistory)	handles view payment information
2	deposit(paypal)	handles make a payment
3	withdraw(paypal)	handles withdraw money

Table 36 - Process Service Class Specification

3.3.3 Activity Diagram

3.3.3.1 View Payment History Activity Diagram

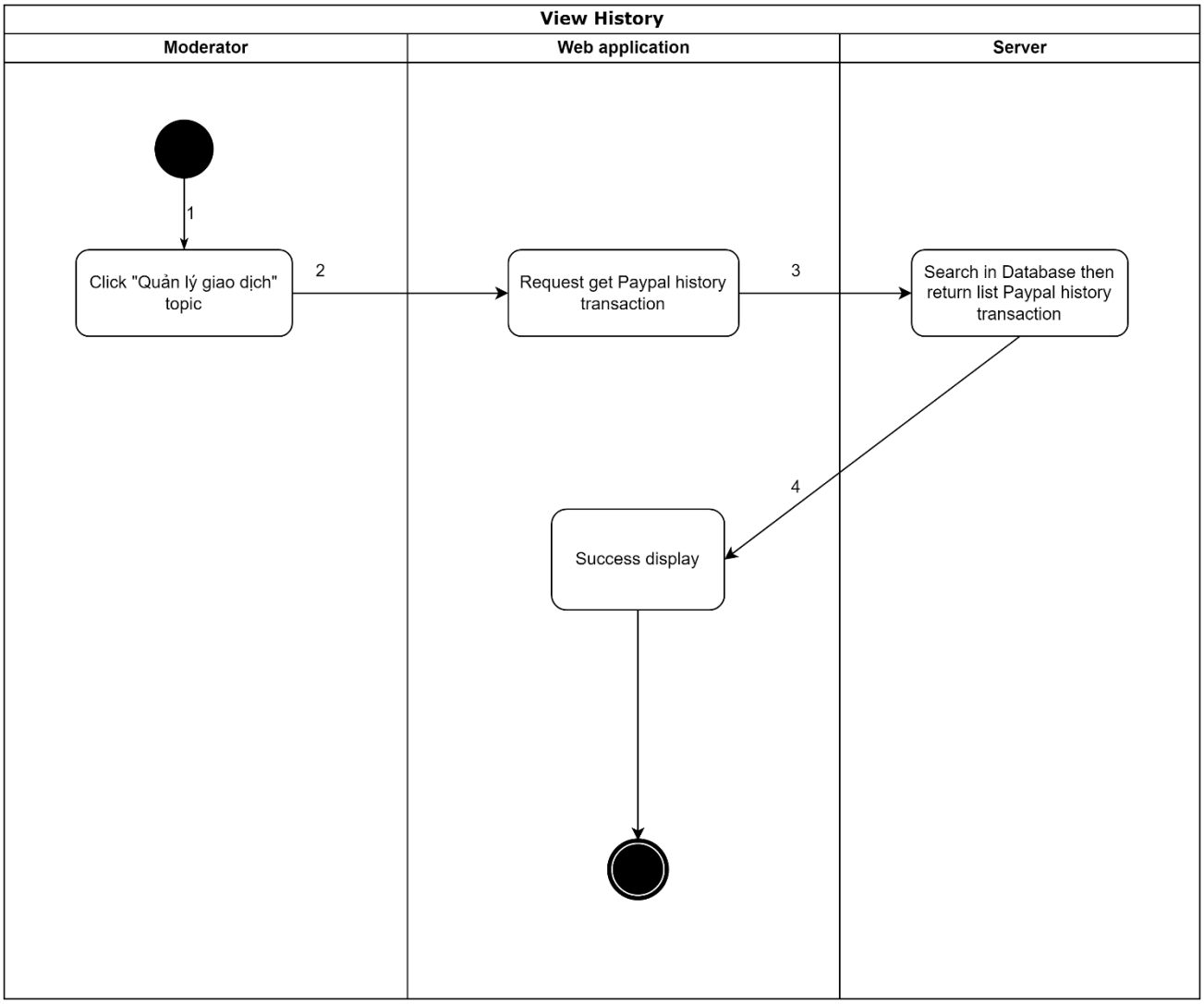


Figure 26 - View Payment History Activity Diagram

3.3.3.2 Deposit Activity Diagram

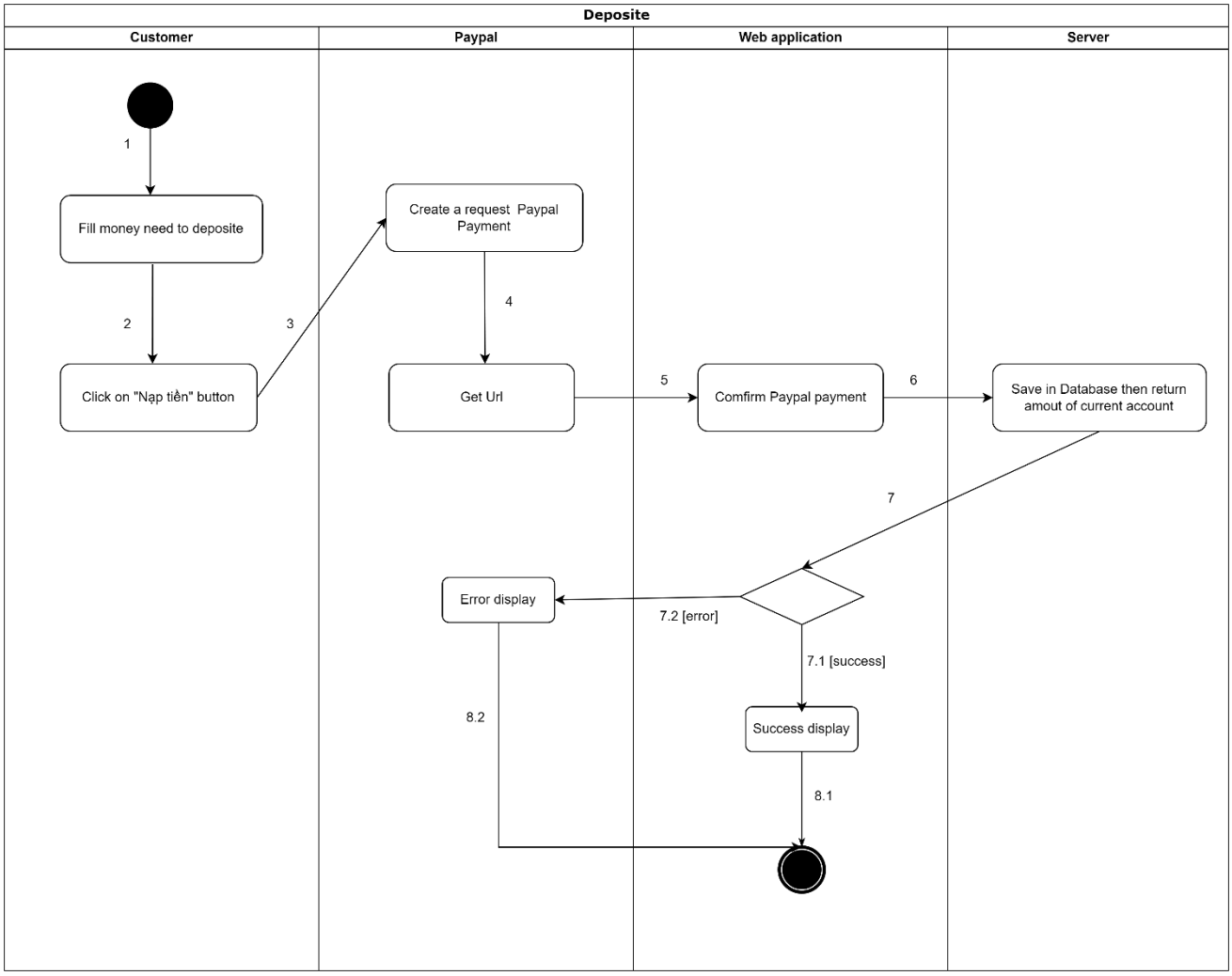


Figure 27 - Deposit Activity Diagram

3.3.3.3 Withdraw Activity Diagram

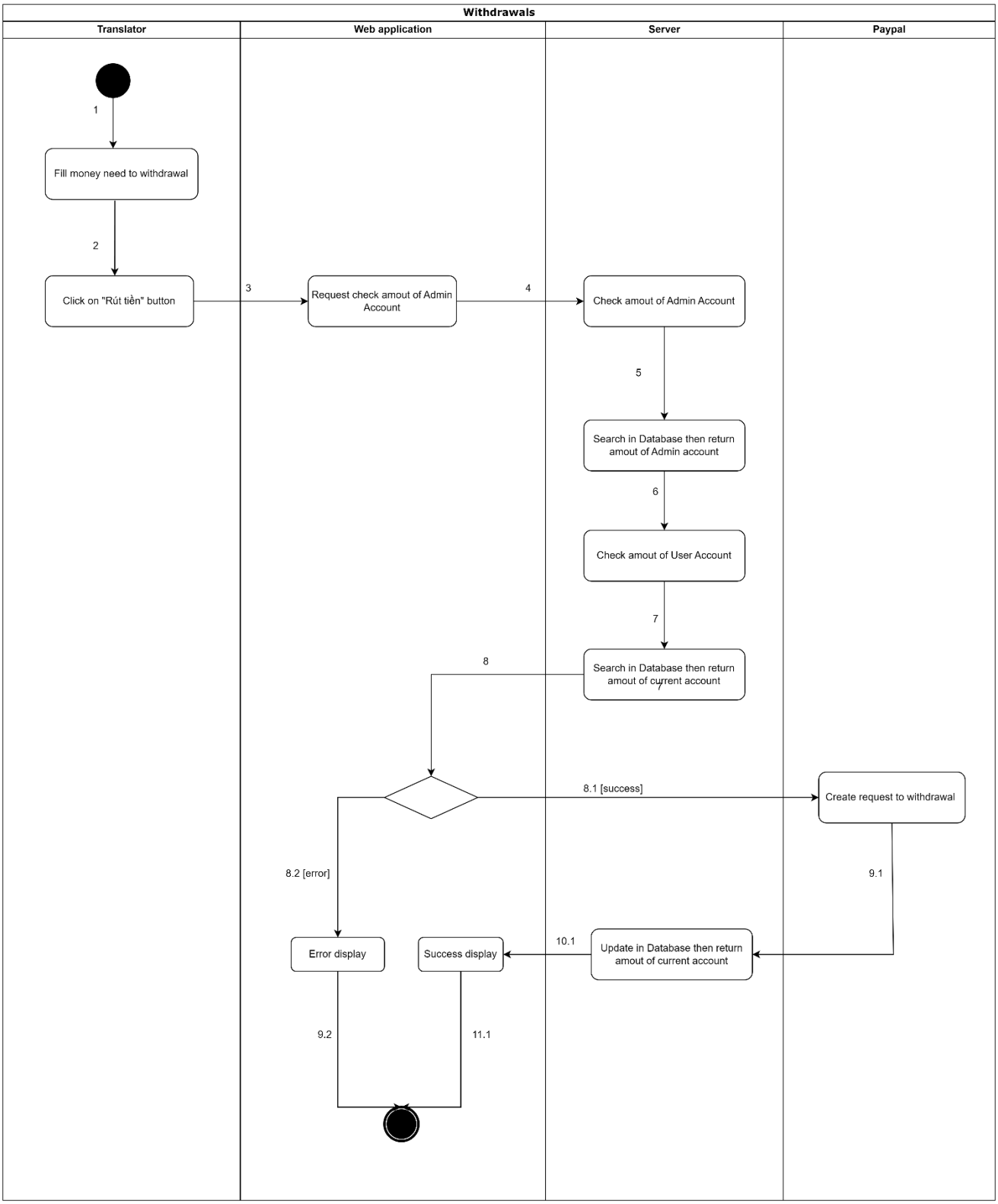


Figure 28 - Withdraw Activity Diagram

3.3.4 Deposit Sequence Diagram

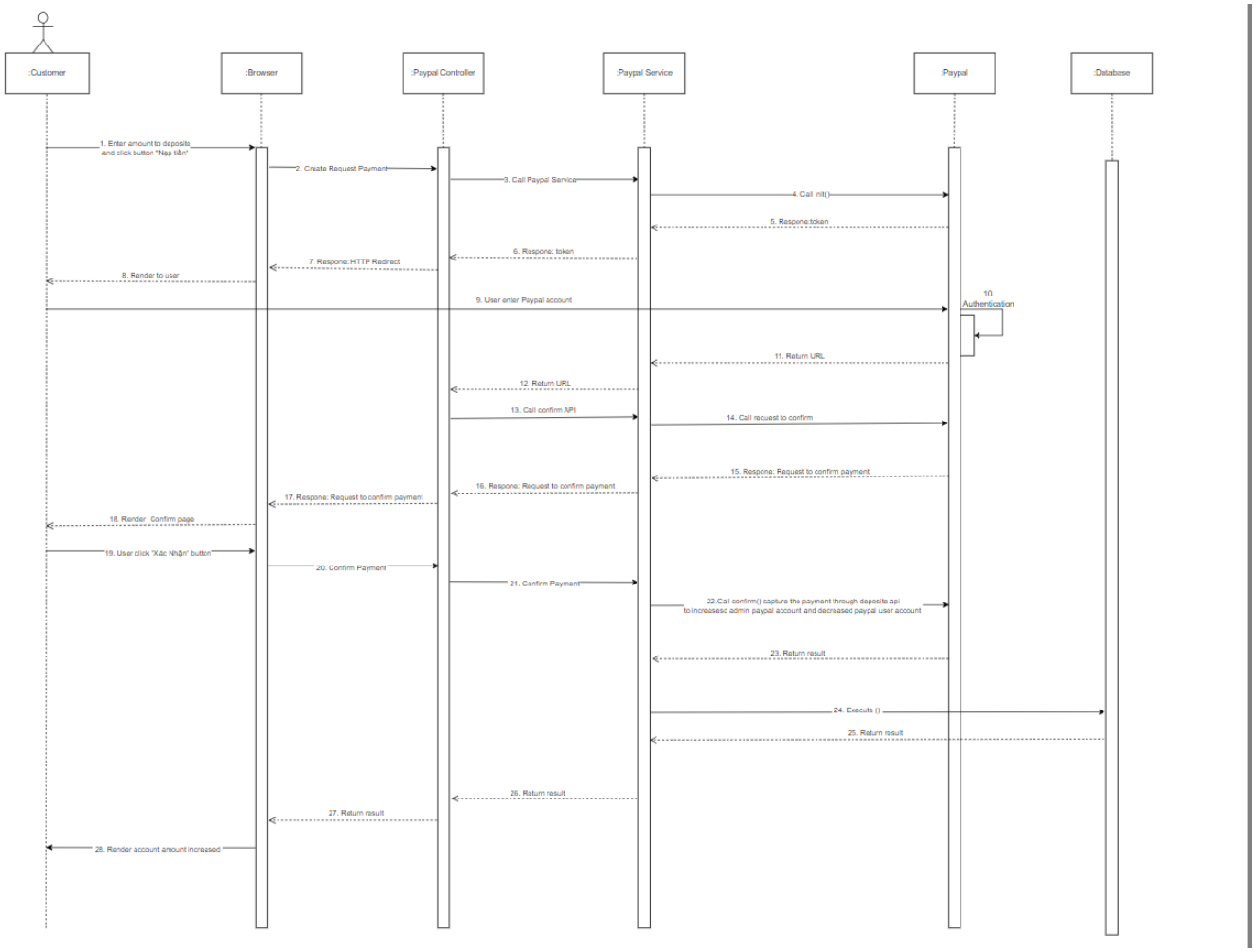


Figure 29 - Deposit Sequence Diagram

3.3.5 Withdraw Sequence Diagram

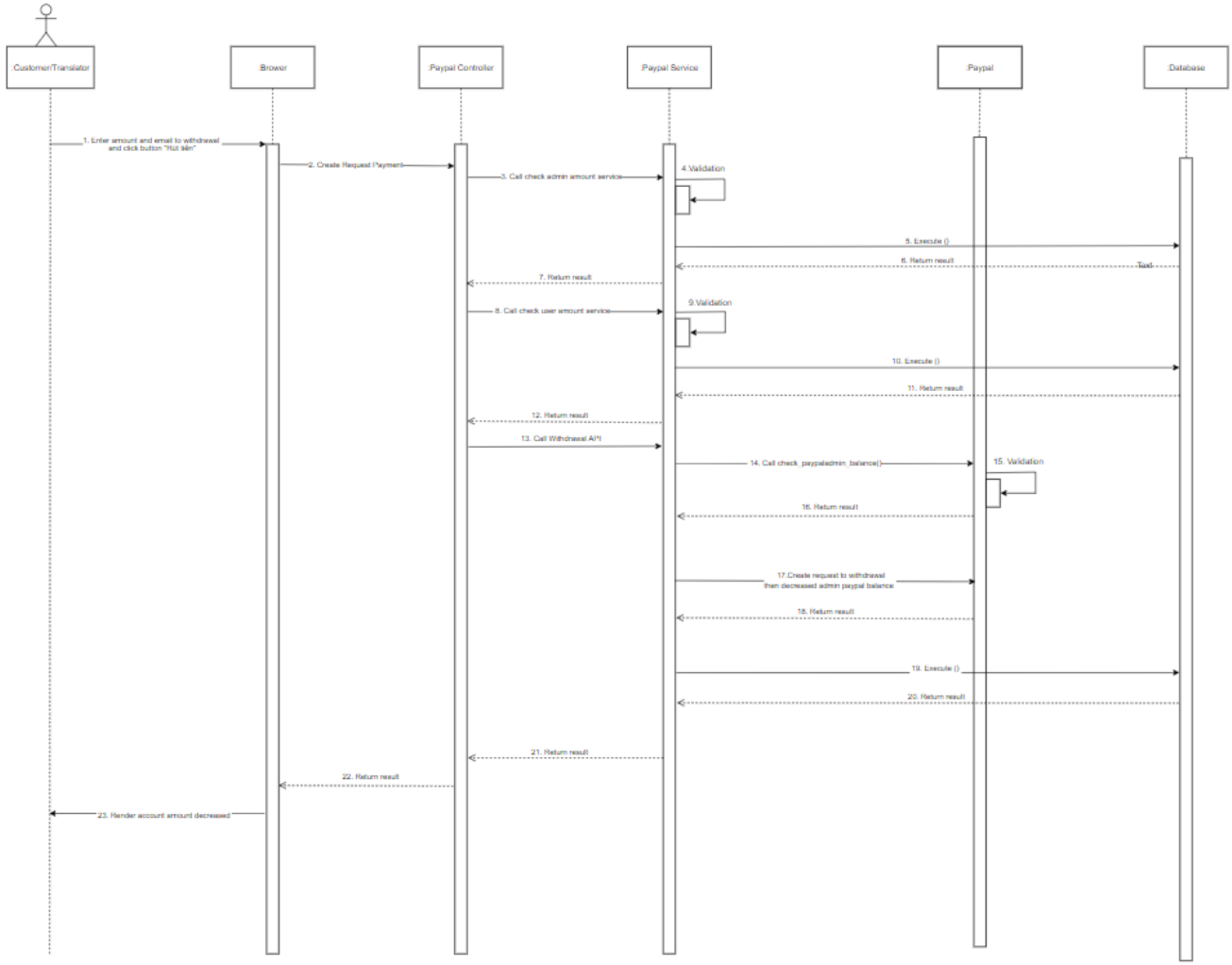


Figure 30 - Withdraw Sequence Diagram

3.4 Activity Diagram Order

3.4.1 <Admin> View account

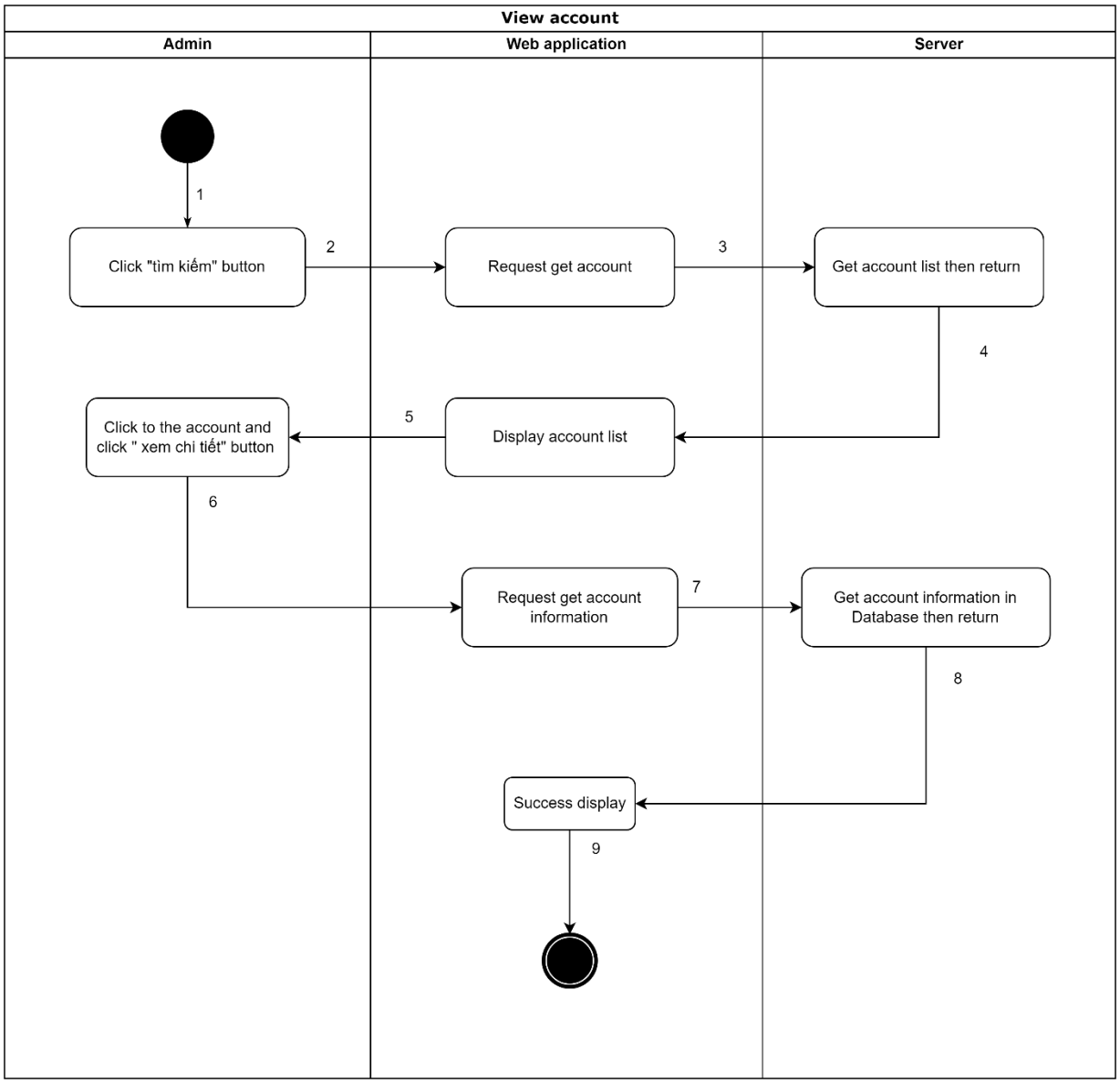


Figure 31 - <Activity Diagram> Admin view account

3.4.2 <Admin> Ban account

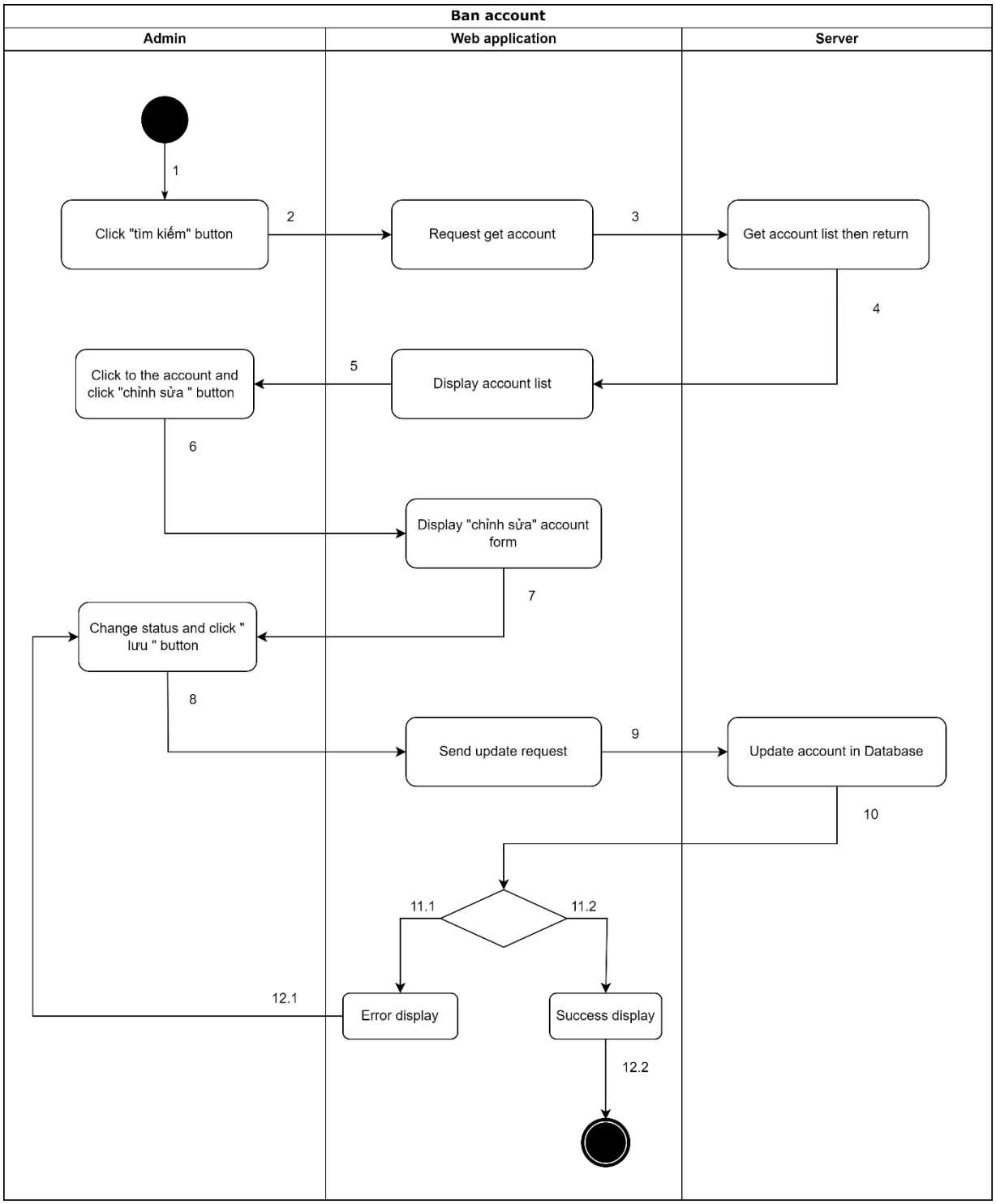


Figure 32 - <Activity Diagram> Admin Ban account

3.4.3 <Mod> Create category

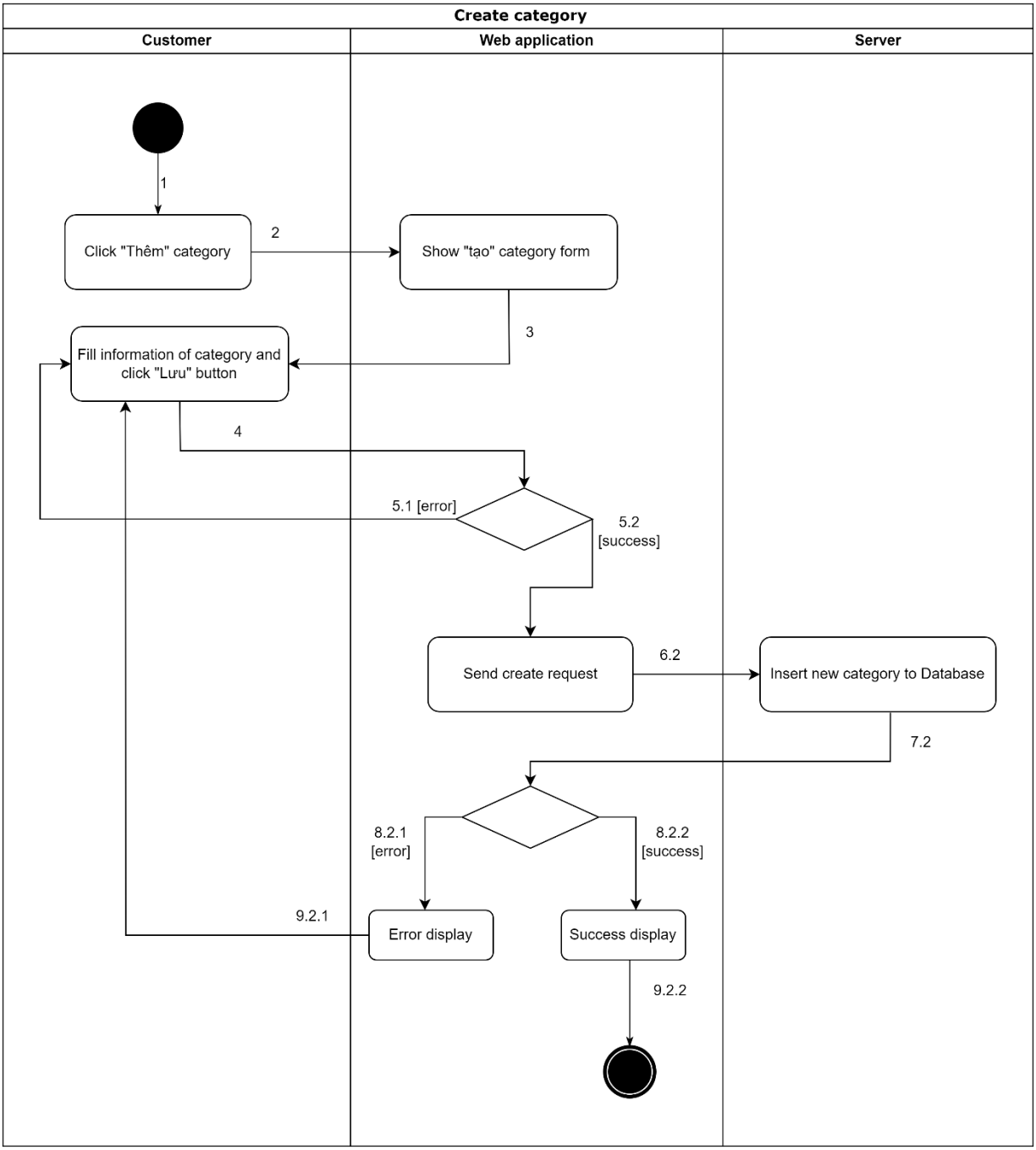


Figure 33 - <Activity Diagram> Mod create category

3.4.4 <Mod> Create topic

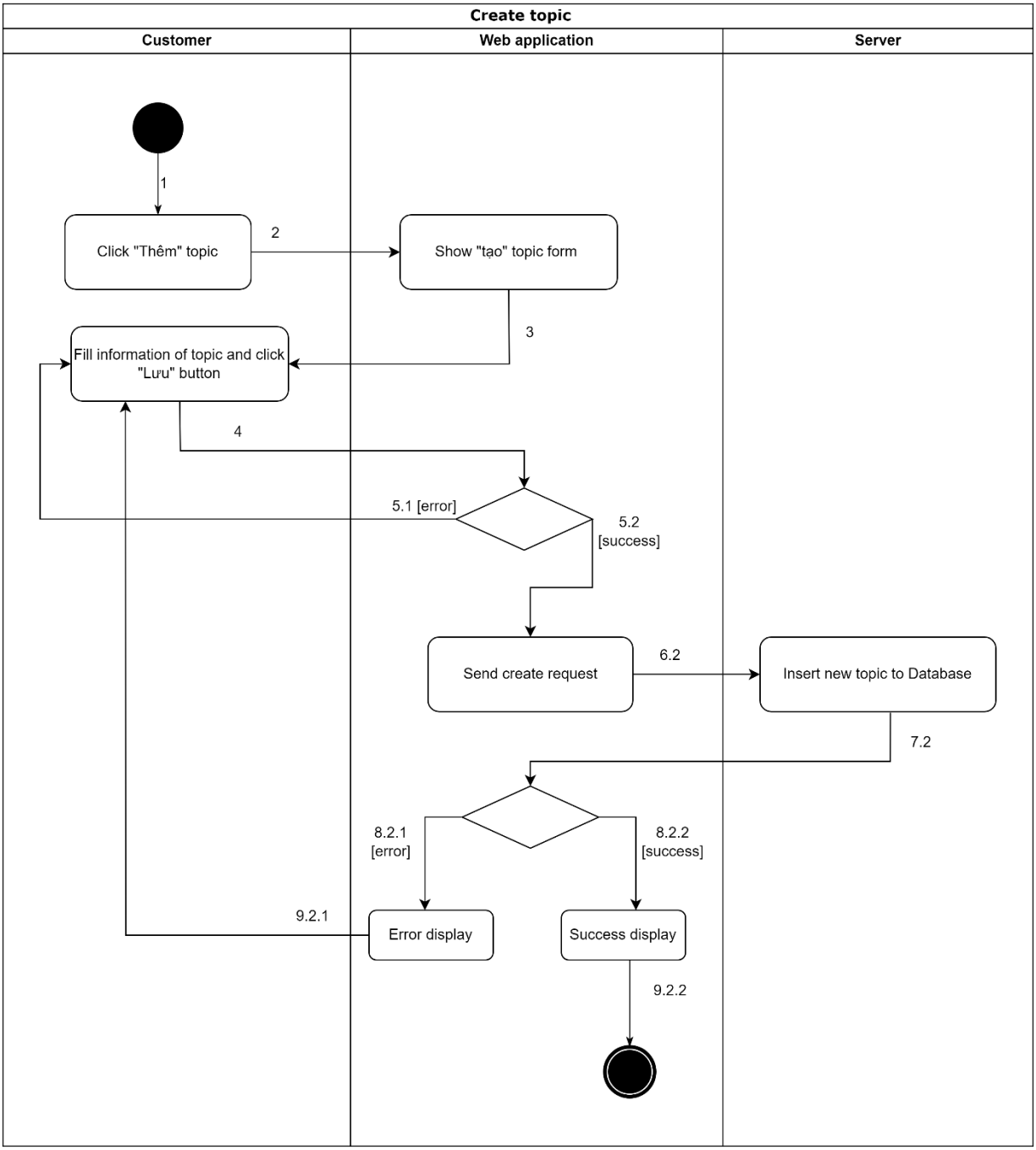


Figure 34 - <Activity Diagram> Mod create topic

3.4.5 <Mod> Create language

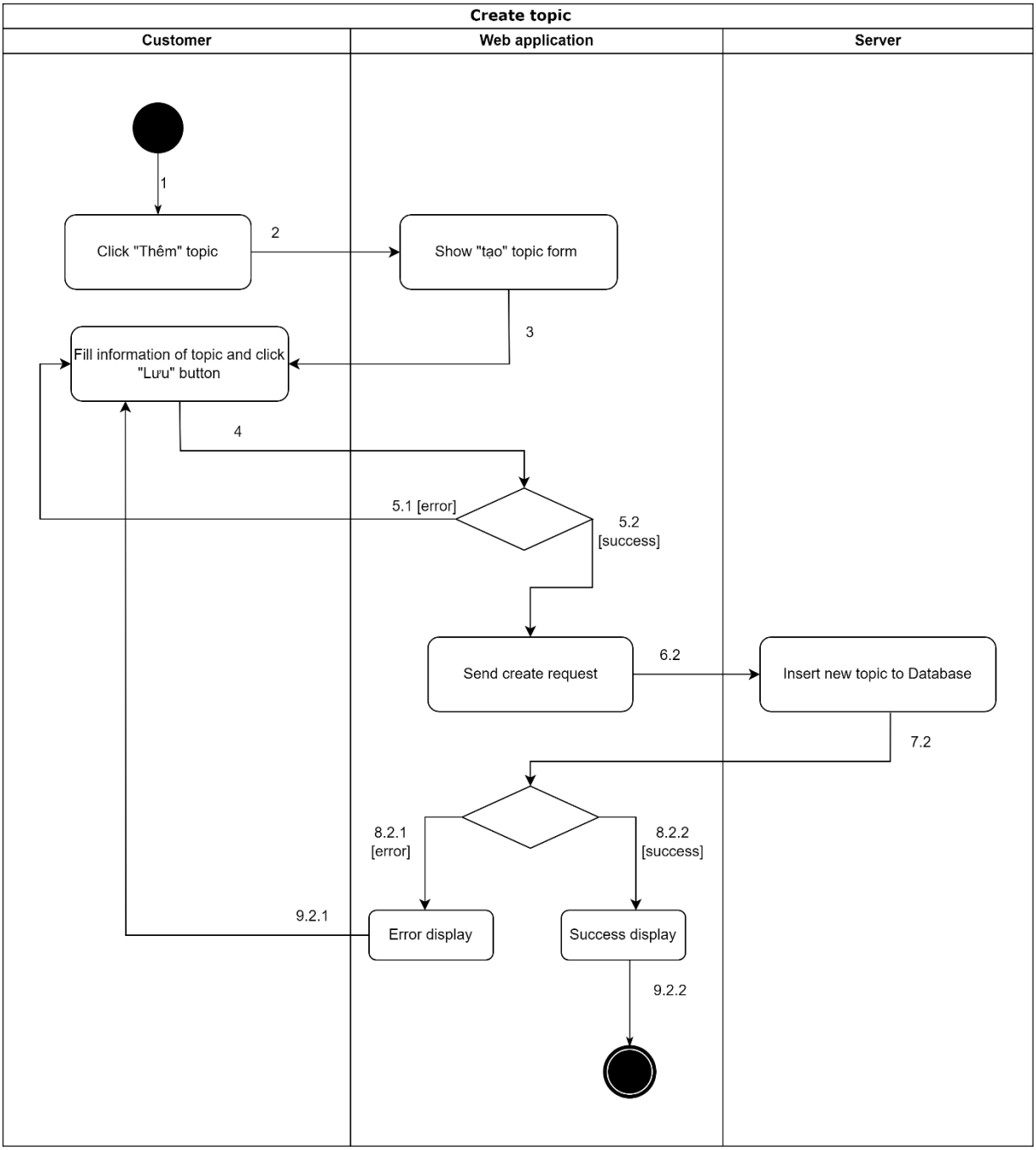


Figure 35 - <Activity Diagram> Mod create language

3.4.6 <Mod> Create achievement

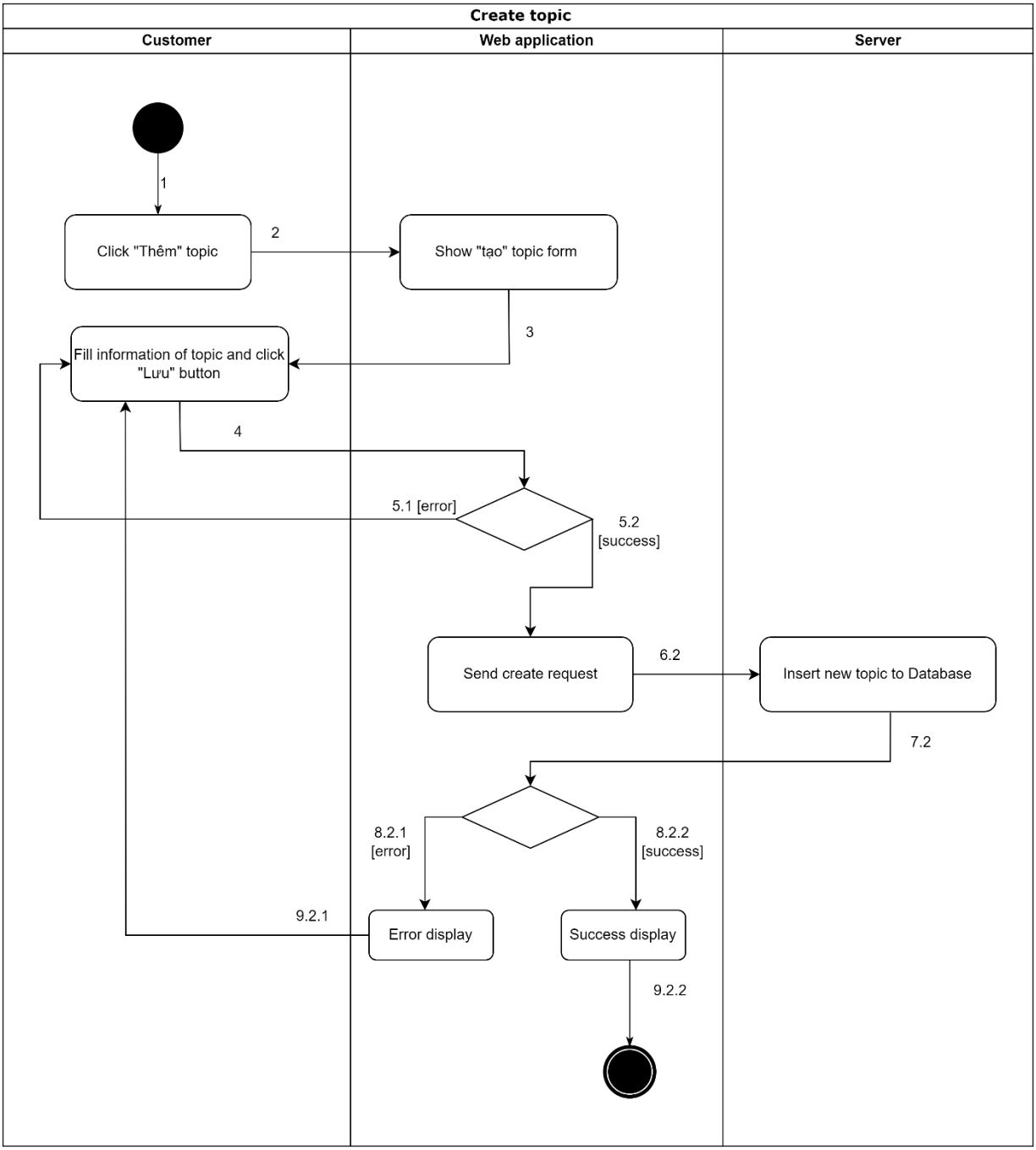


Figure 36 - <Activity Diagram> Mod create achievement

3.4.7 <Mod> View project

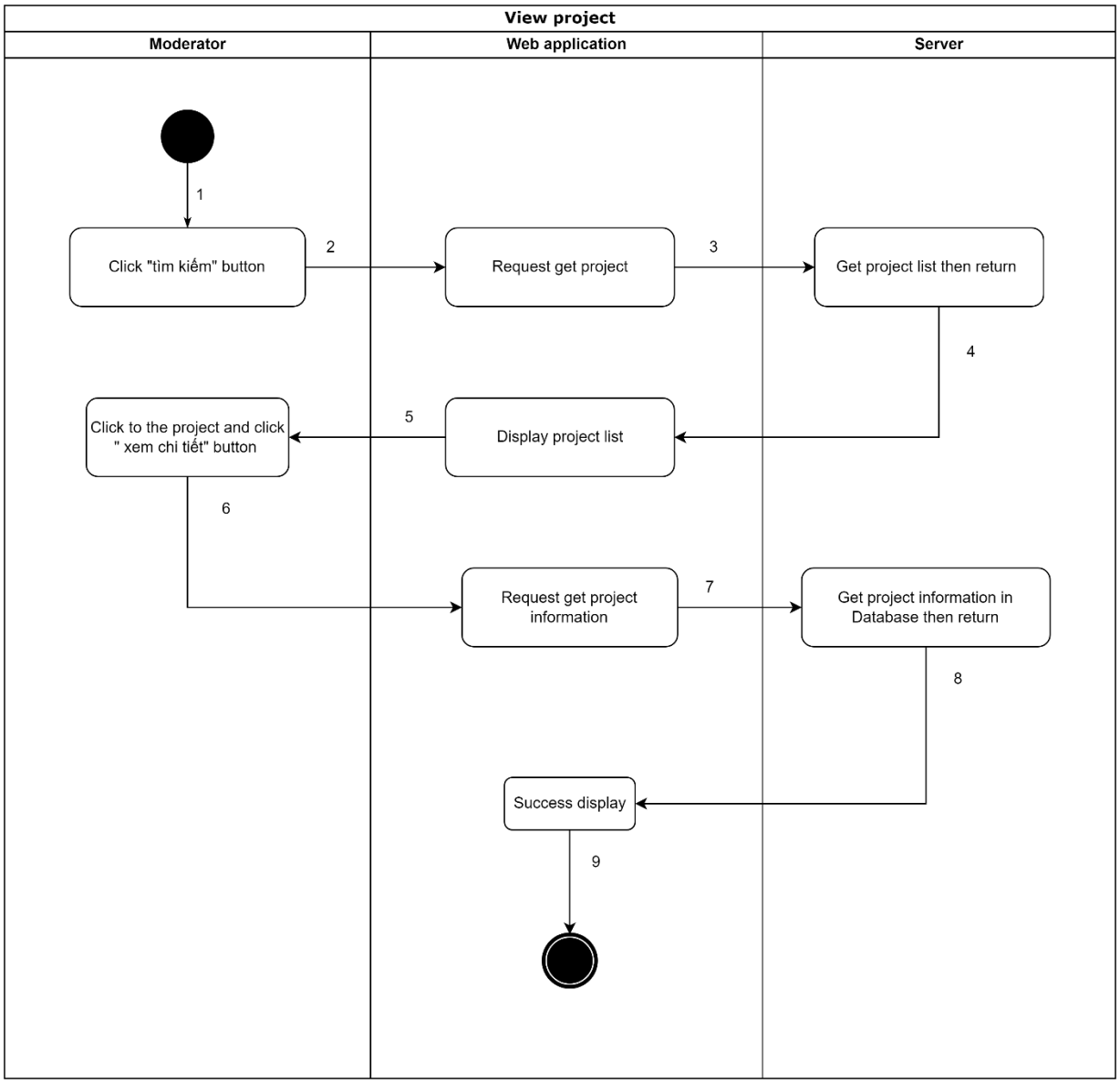


Figure 37 - <Activity Diagram> Mod view project

3.4.8 <Mod> View topic

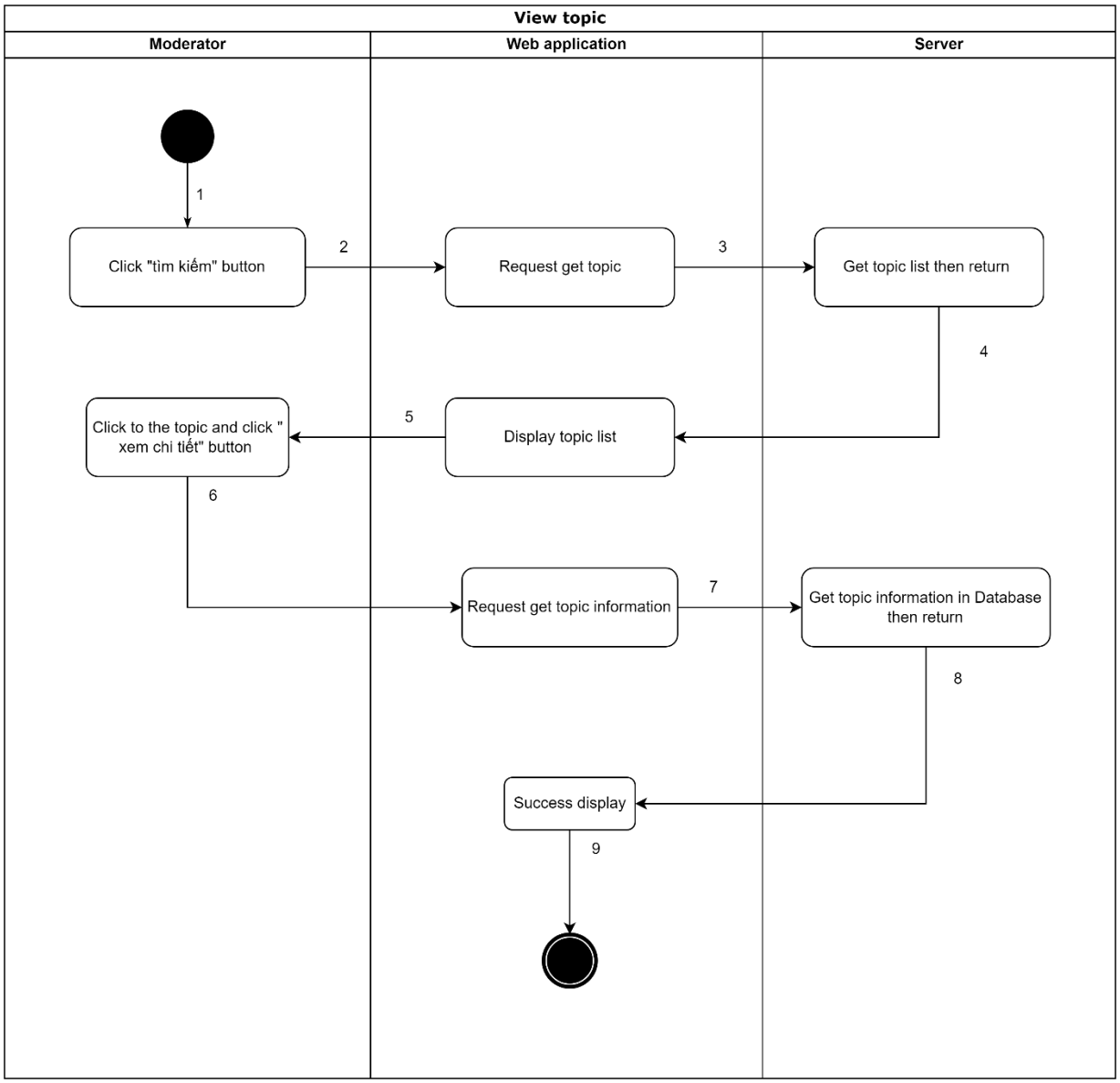


Figure 38 - <Activity Diagram> Mod view topic

3.4.9 <Mod> View category

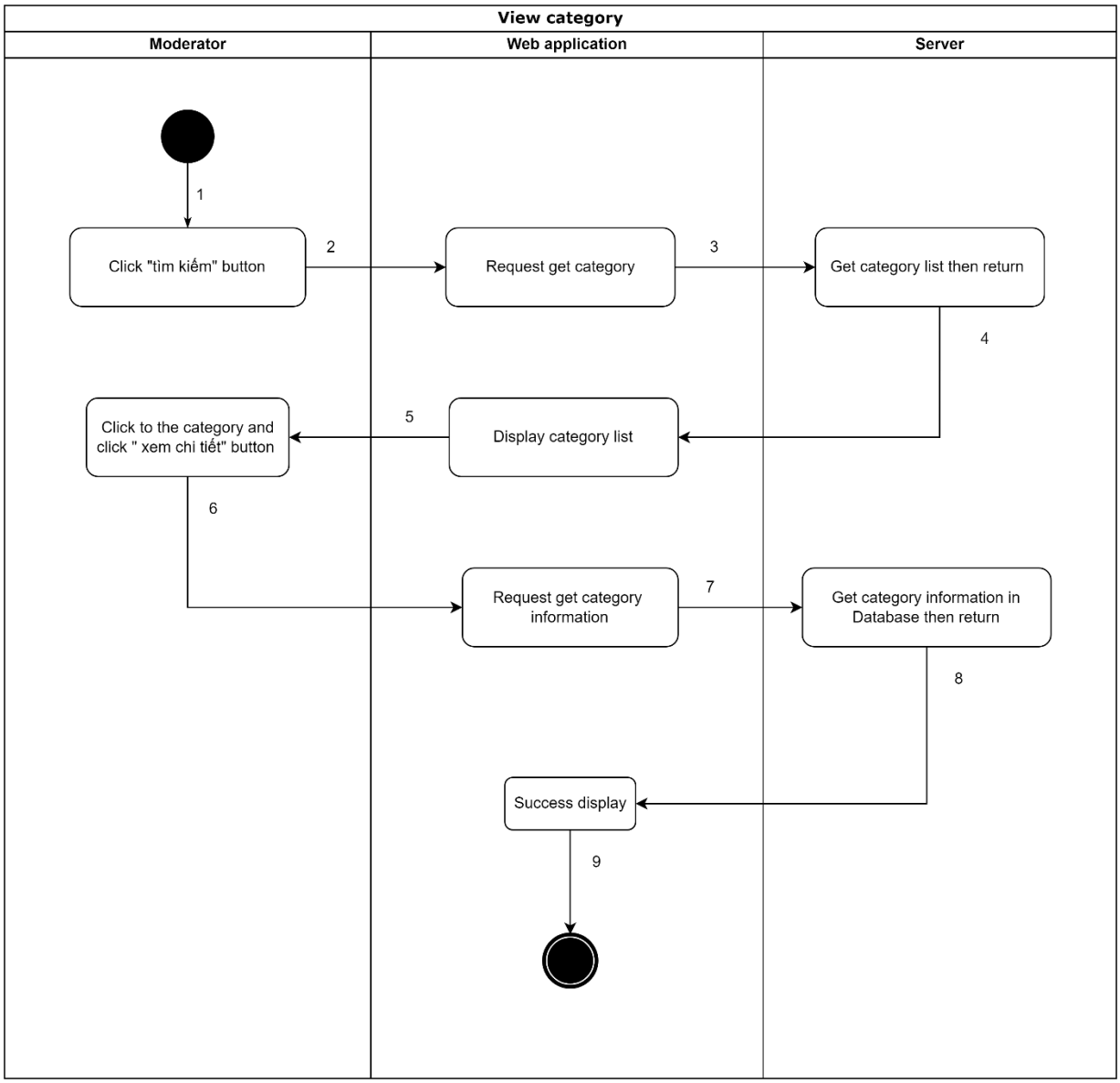


Figure 39 - <Activity Diagram> Mod category

3.4.10 <Mod> View language

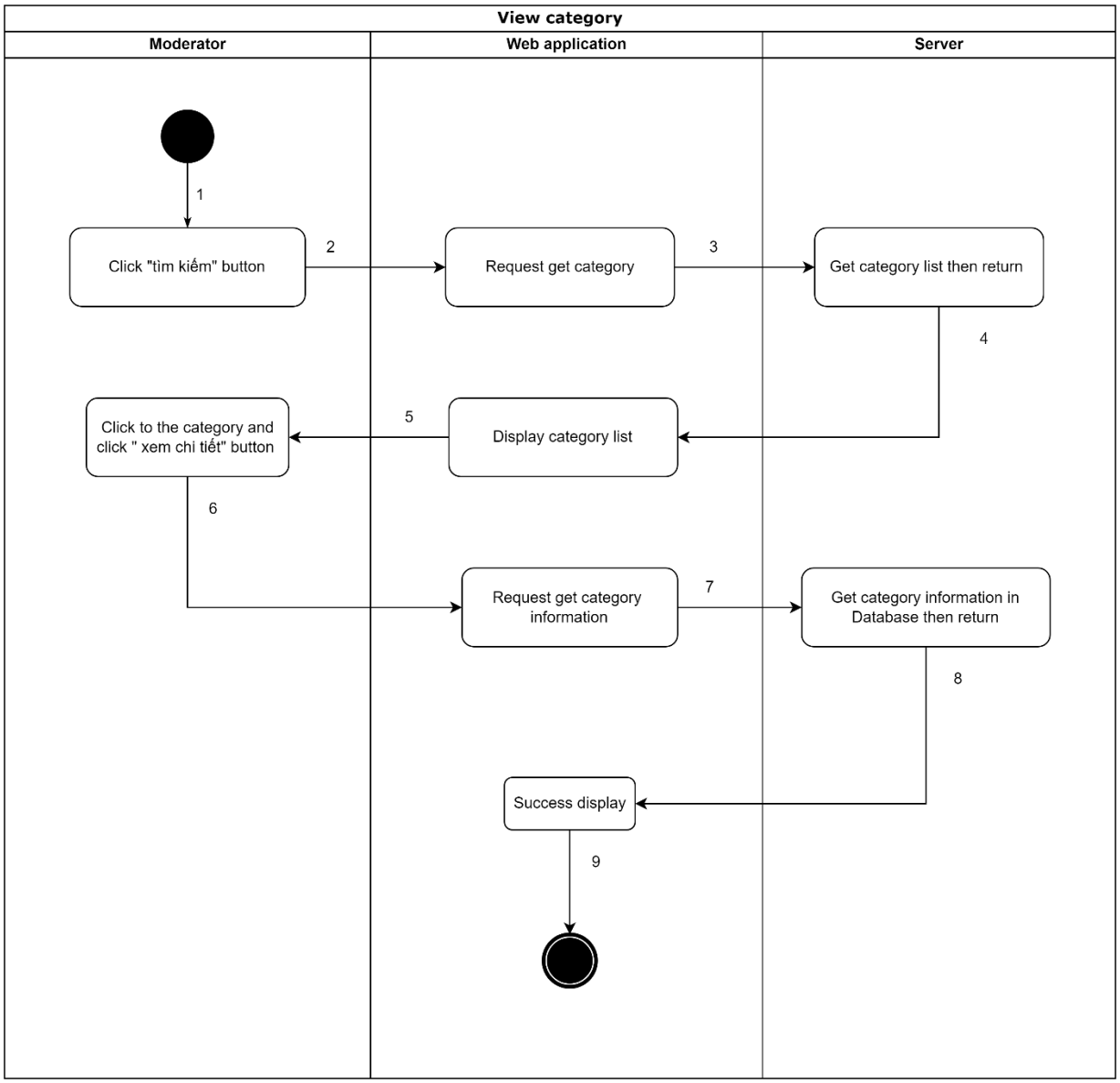


Figure 40 - <Activity Diagram> Mod view language

3.4.11 <Mod> View achievement

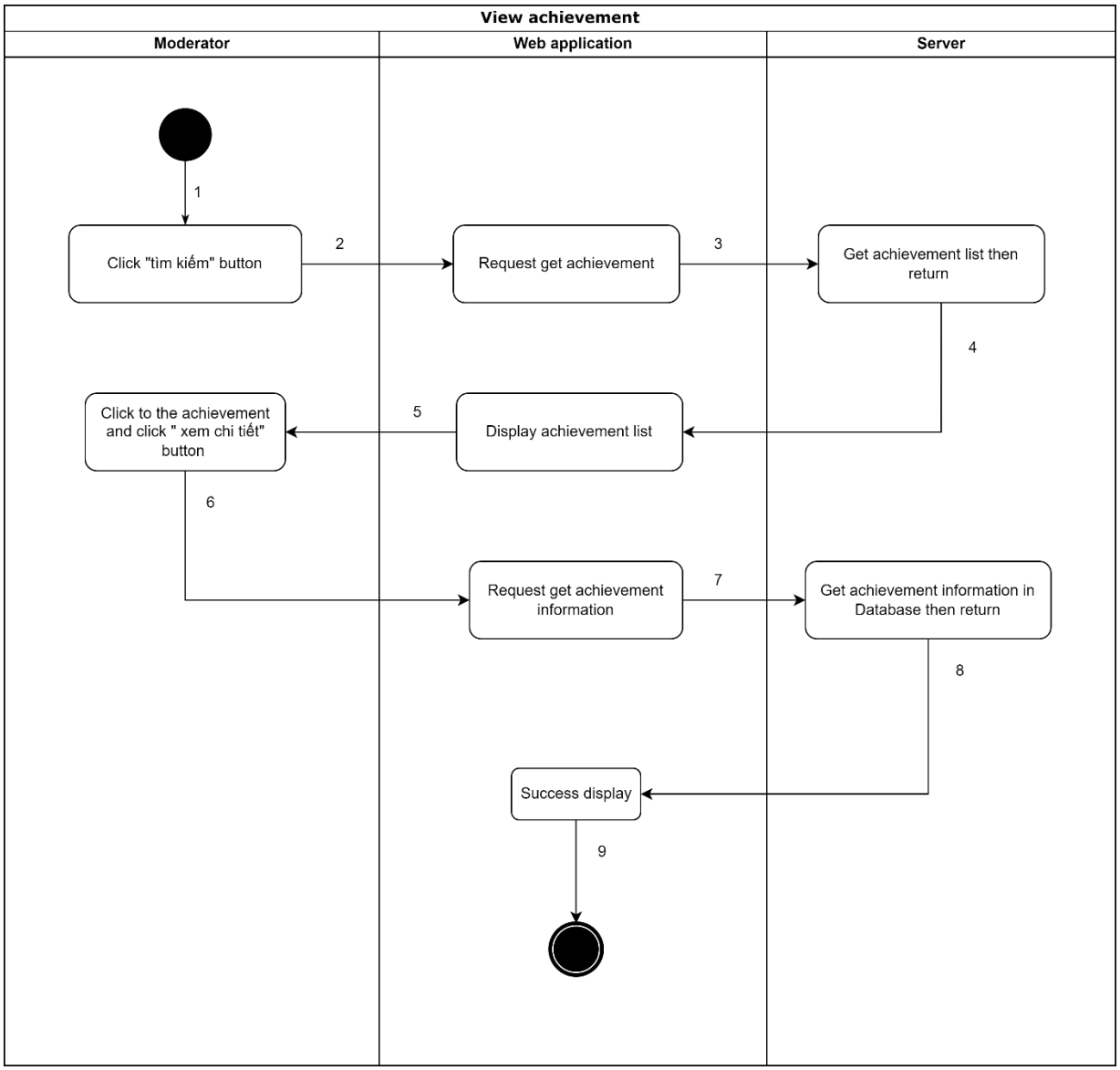


Figure 41 - <Activity Diagram> Mod view achievement

3.4.12 <Mod> Approve project

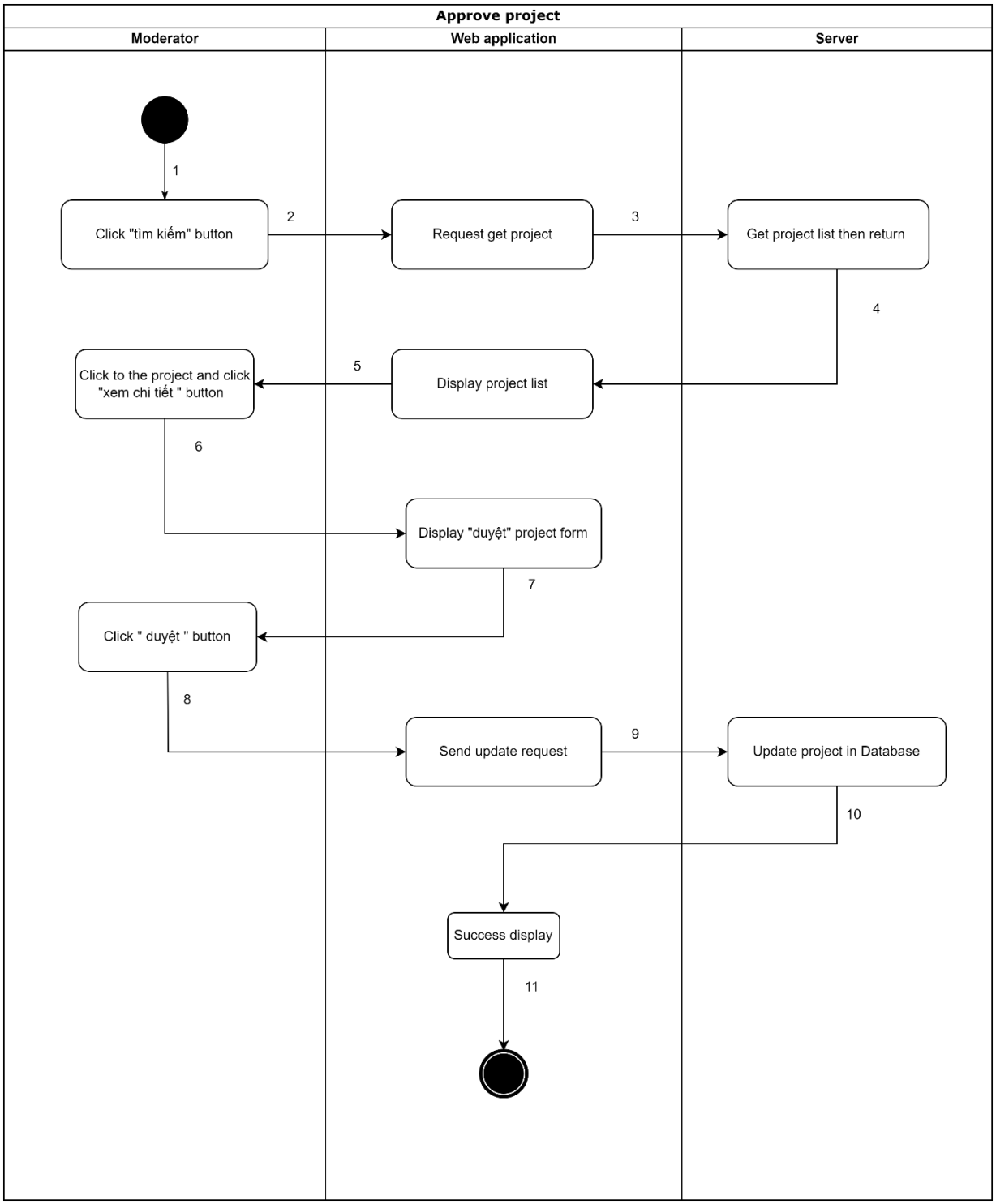


Figure 42 - <Activity Diagram> Mod approve project

3.4.13 <Mod> Reject project

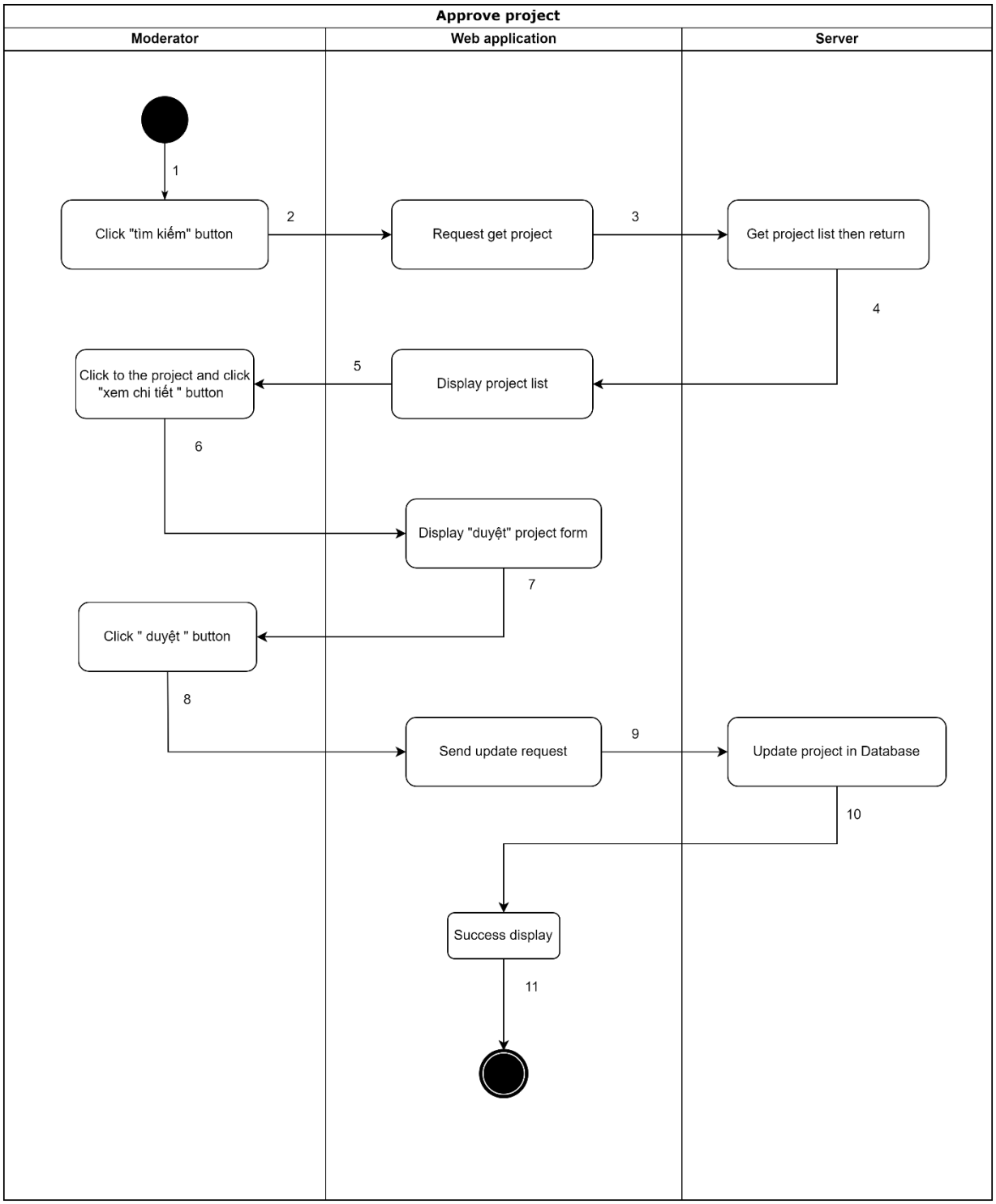


Figure 43 - <Activity Diagram> Mod reject project

3.4.14 <Mod> Update topic

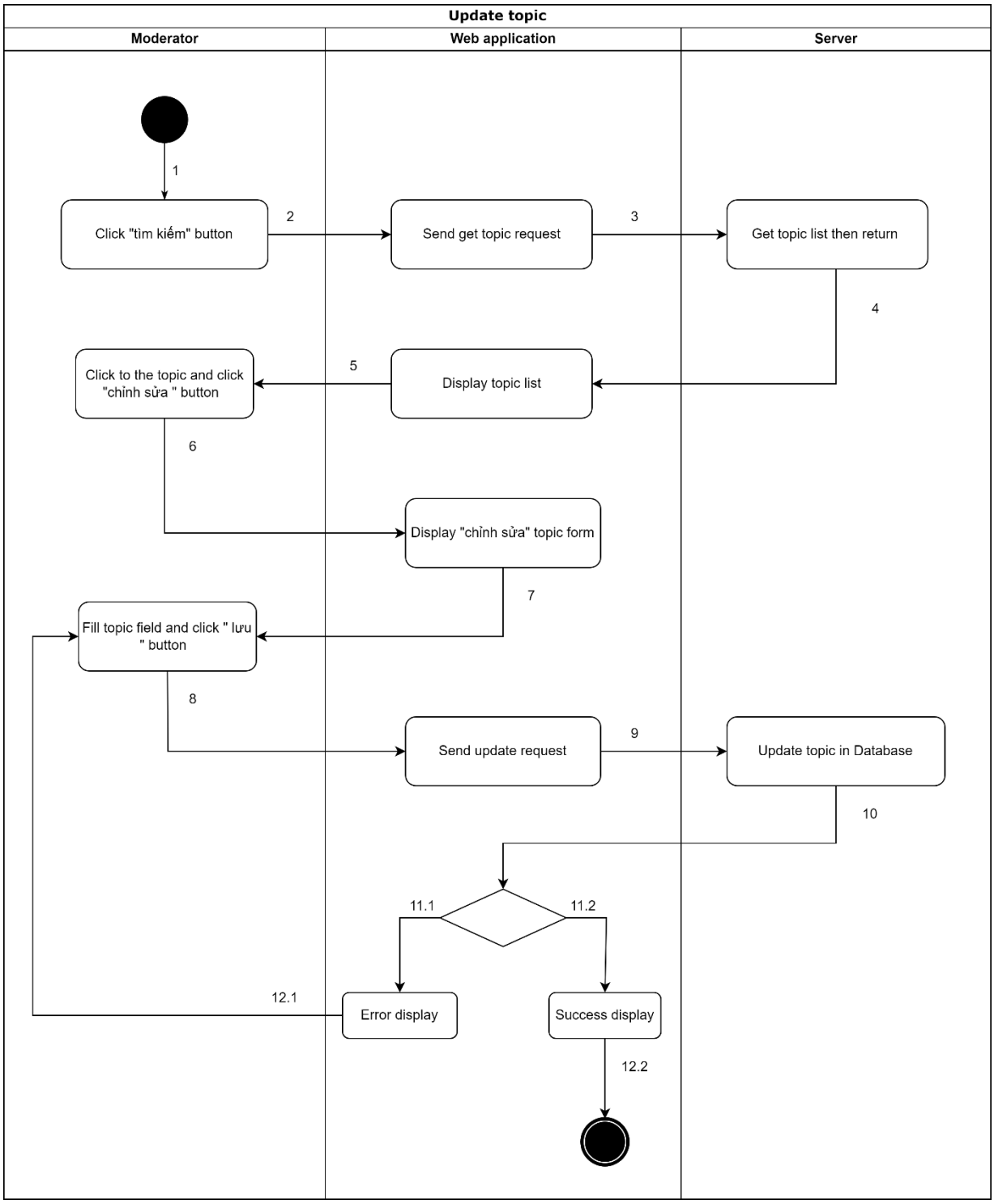


Figure 44 - <Activity Diagram> Mod update topic

3.4.14 <Customer> Create project

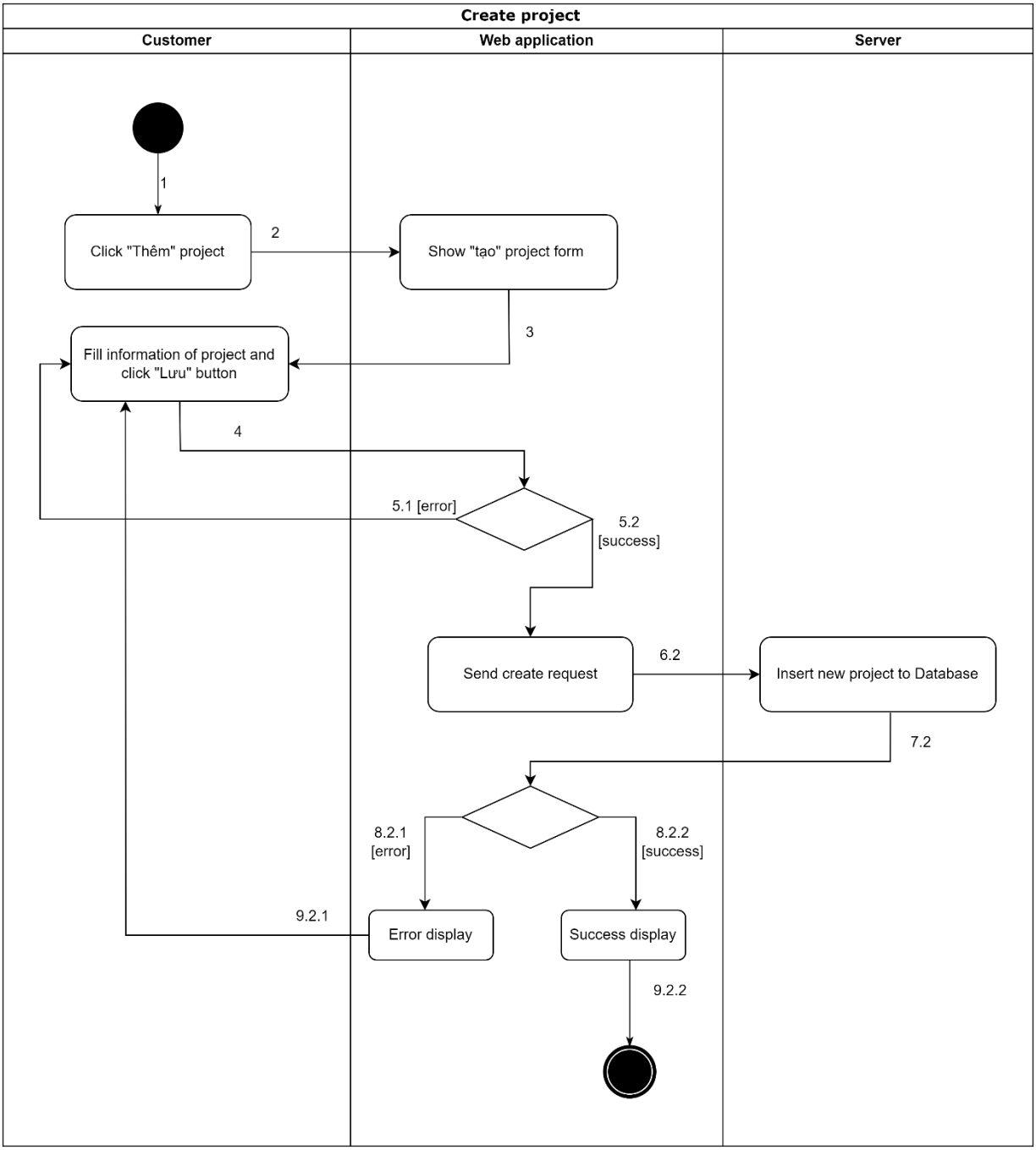


Figure 45 - <Activity Diagram> Customer create project

3.4.15 <Customer> View project

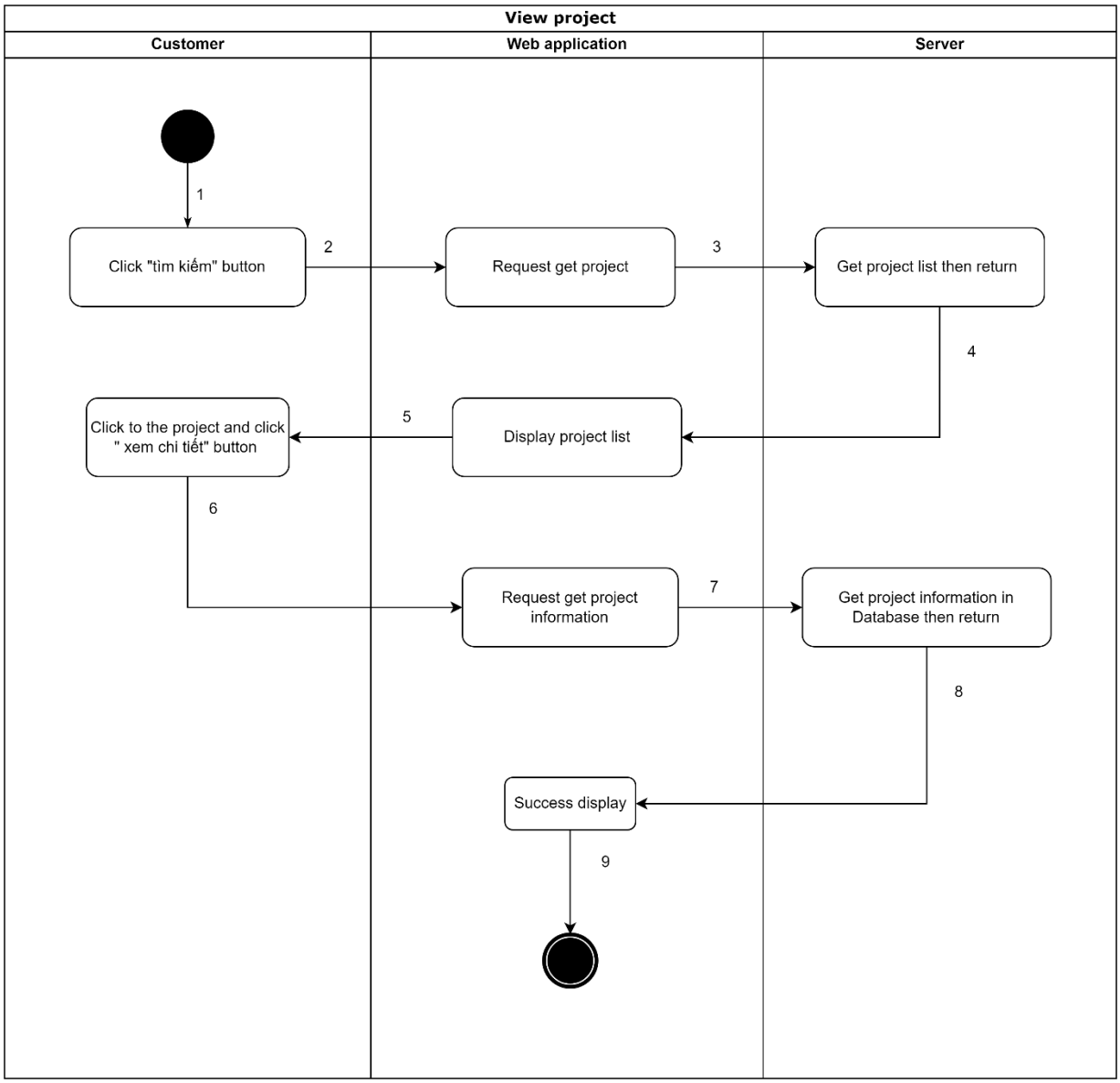


Figure 46 - <Activity Diagram> Customer view project

3.4.16 <Customer> View translator

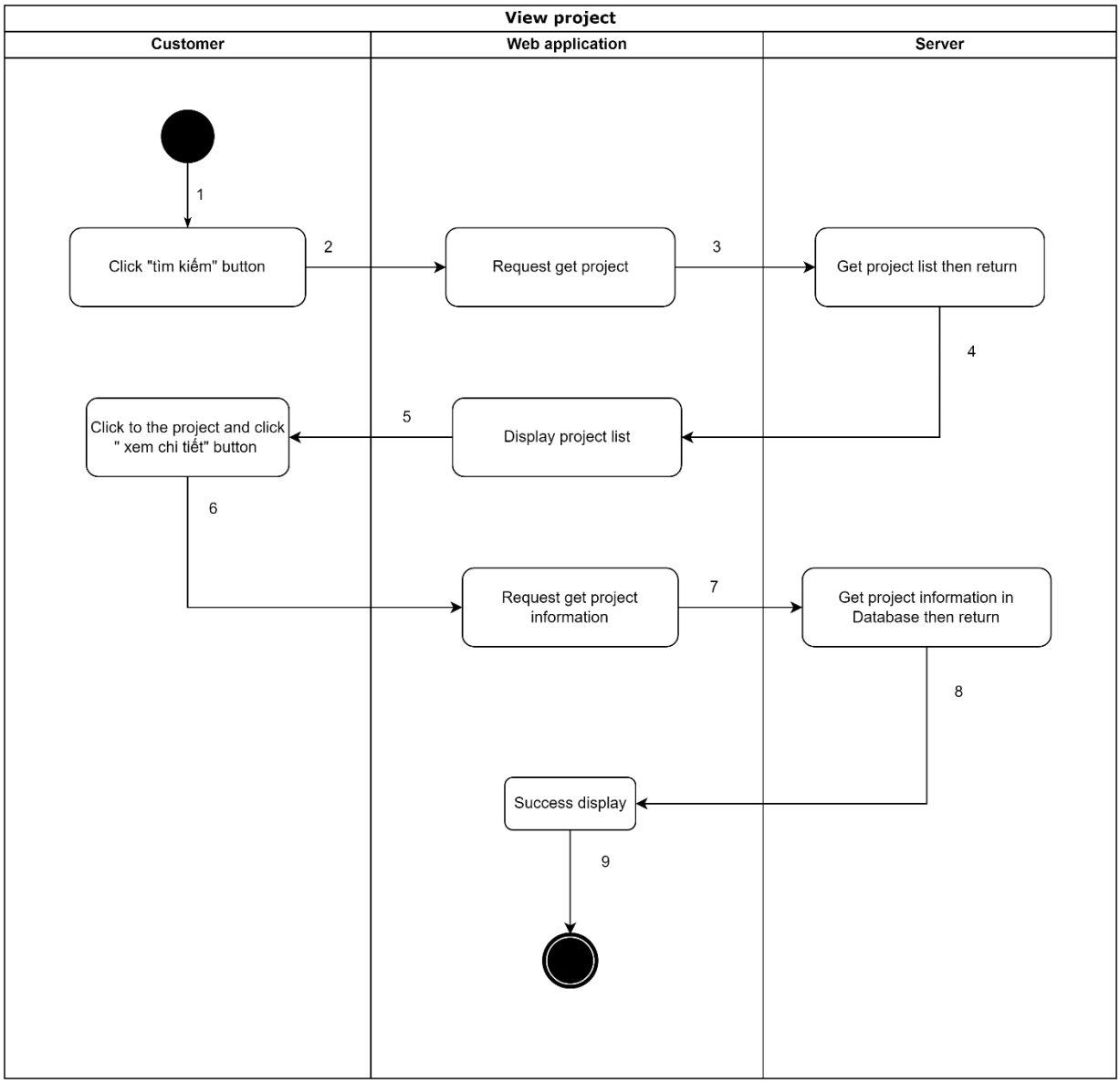


Figure 47 - <Activity Diagram> Customer view translator

3.4.17 <Customer> View process

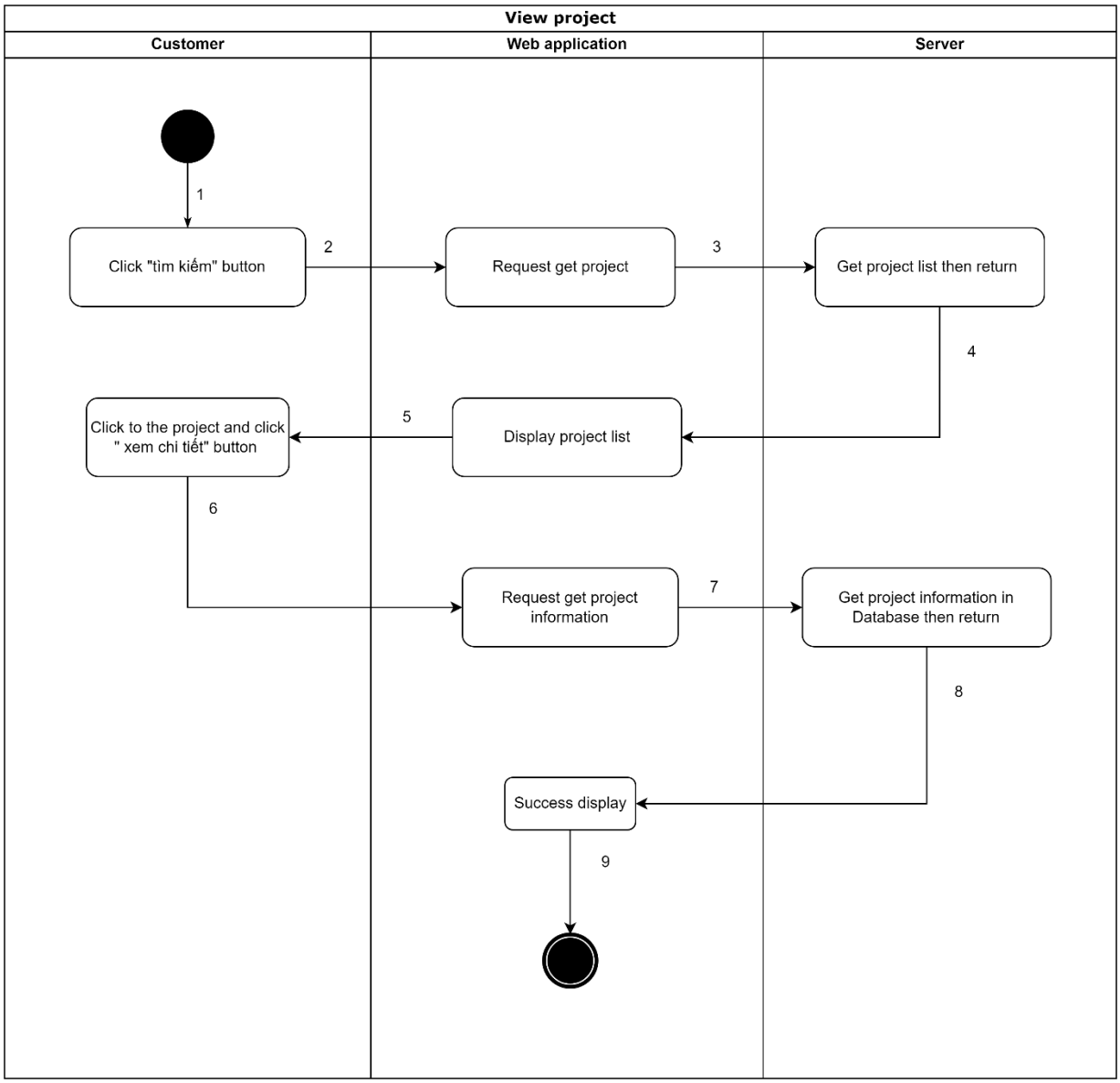


Figure 48 - <Activity Diagram> Customer view process

3.4.18 <Customer> View history rating

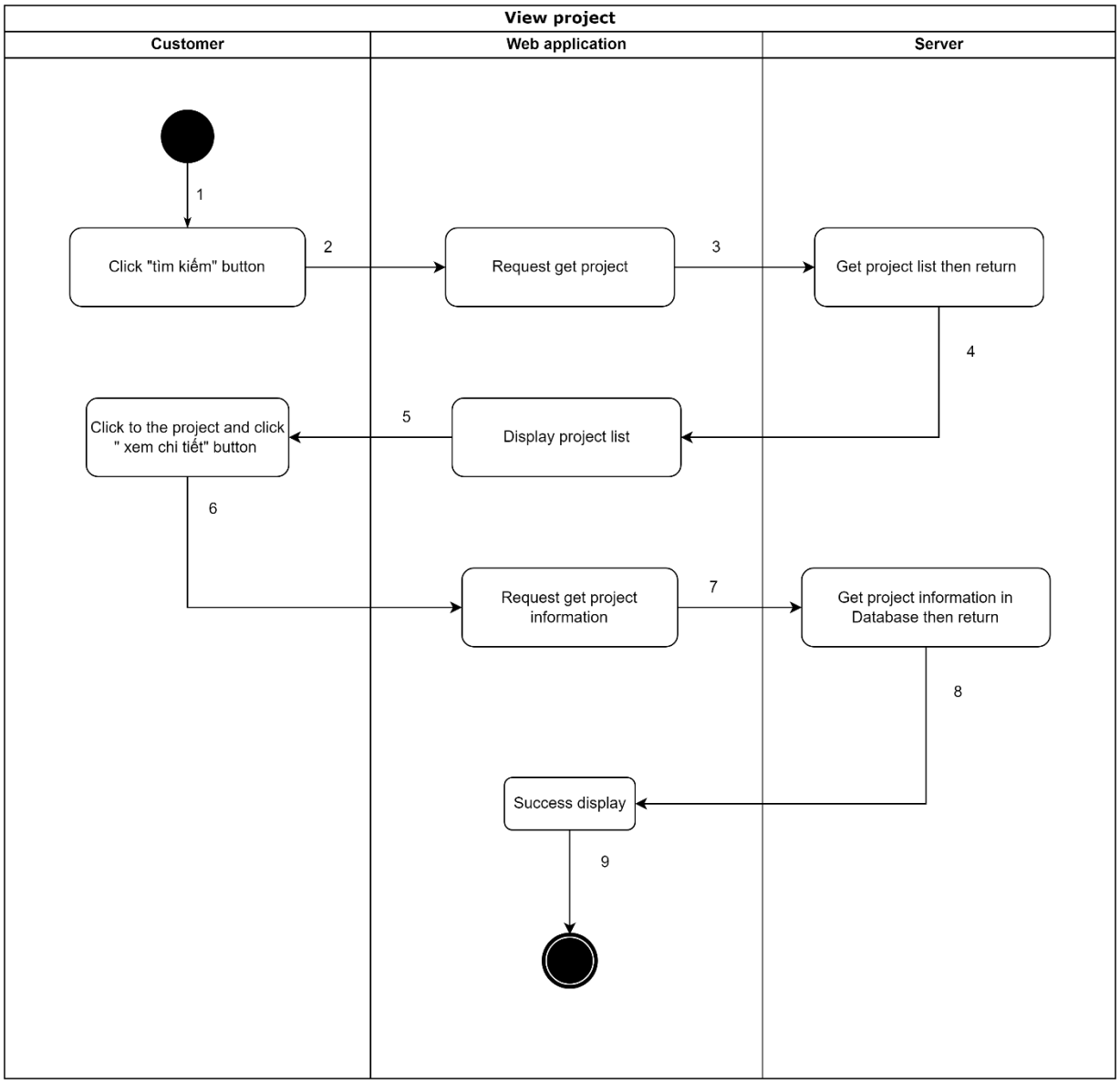


Figure 49 - <Activity Diagram> Customer view history rating

3.4.19 <Customer> Accept translator into project

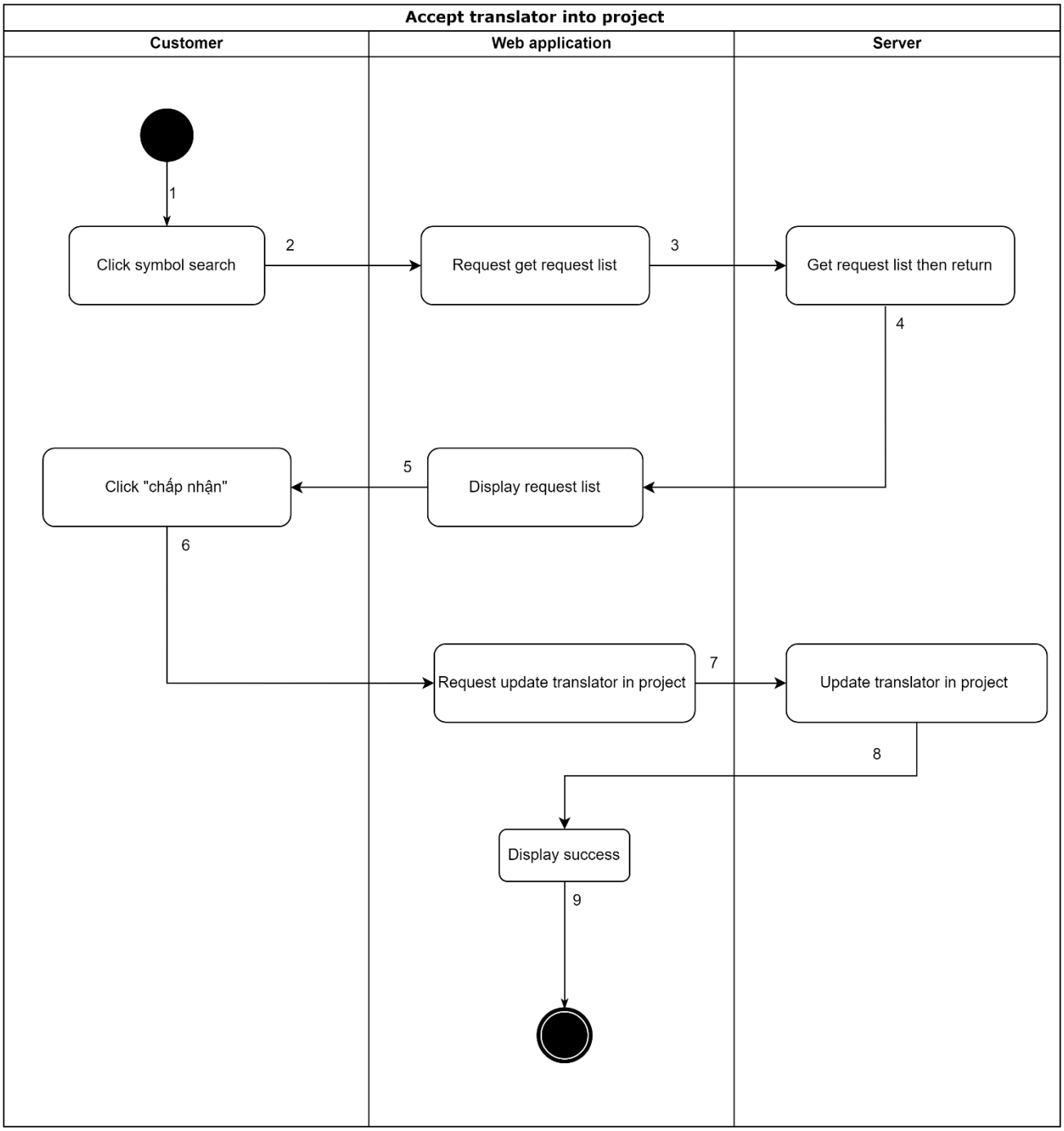


Figure 50 - <Activity Diagram> Customer accept translator into project

3.4.19 <Customer> Reject translator into project

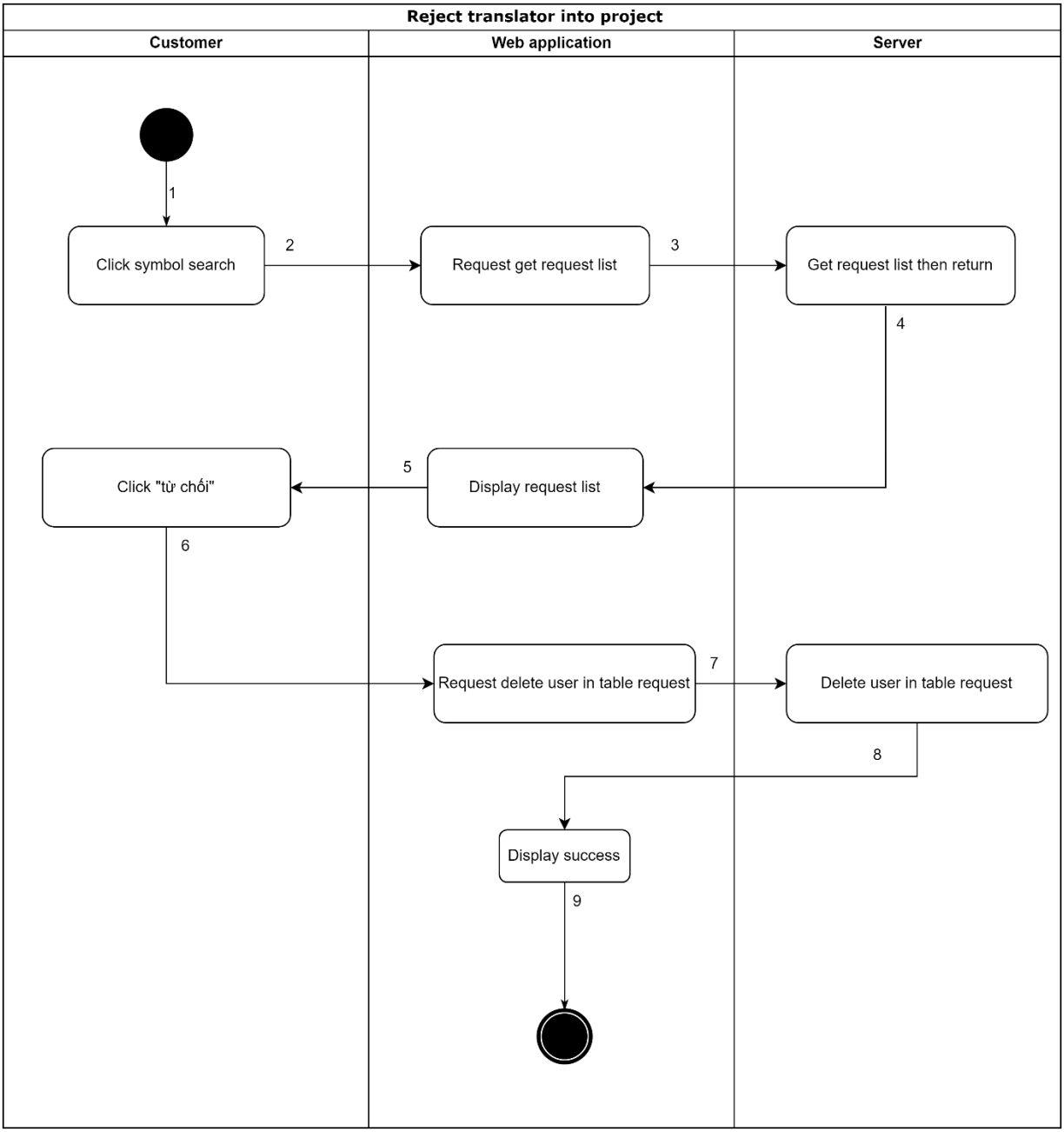


Figure 51 - <Activity Diagram> Customer reject customer into project

3.4.20 <Customer> Delete project

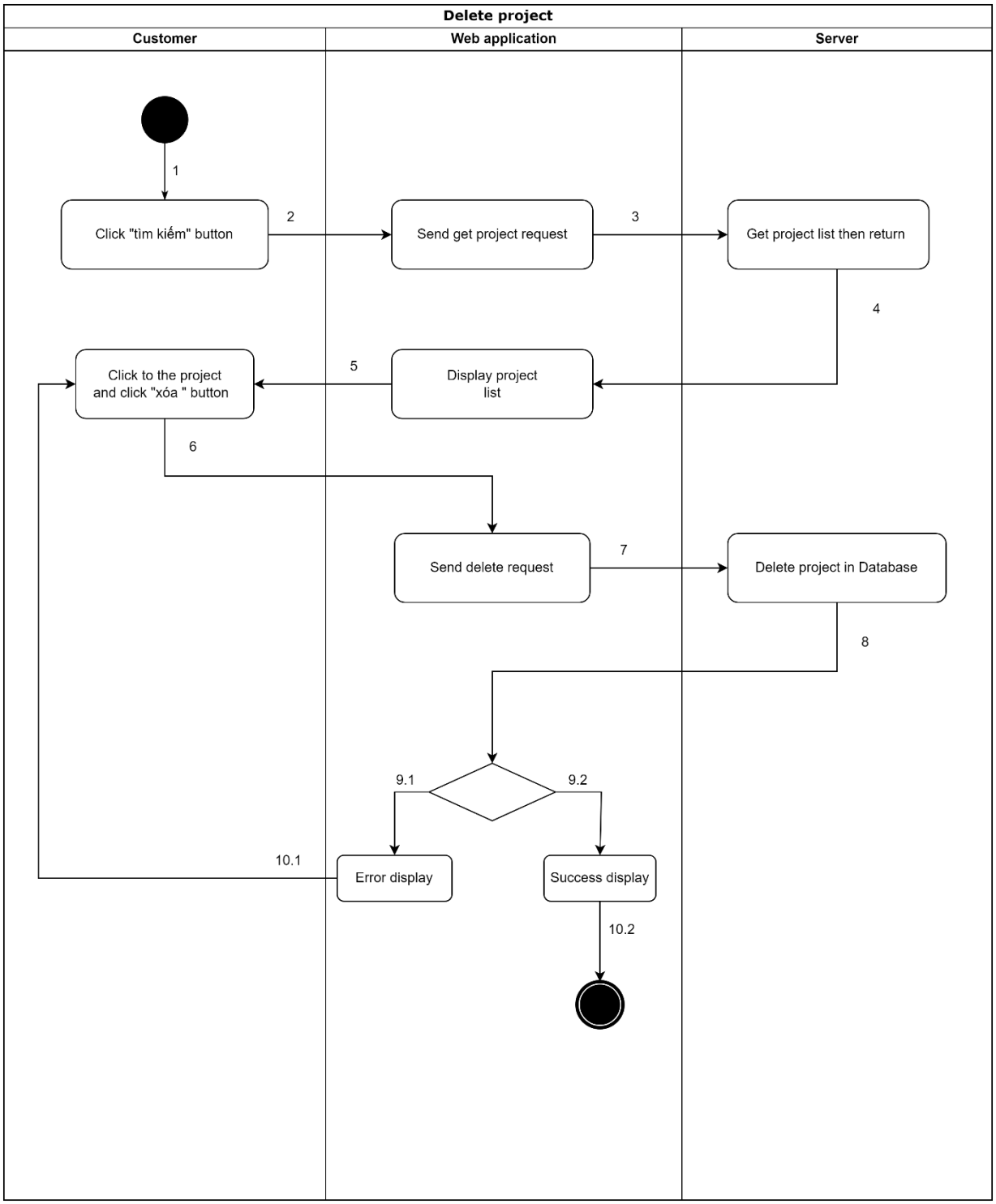


Figure 52 - <Activity Diagram> Customer delete project

3.4.21 **<Customer>Login**

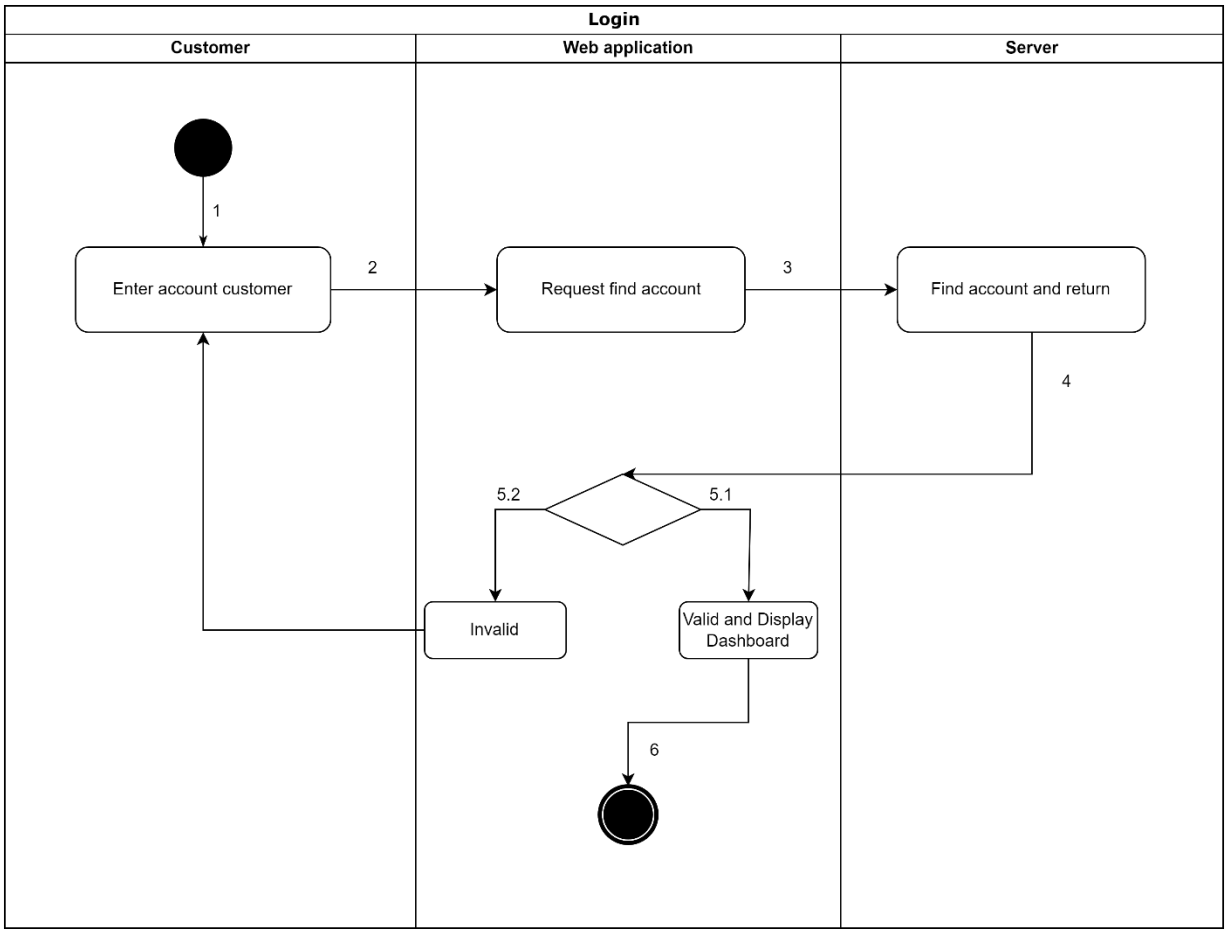


Figure 53 - <Activity Diagram> Customer login

3.4.21 <Customer> Logout

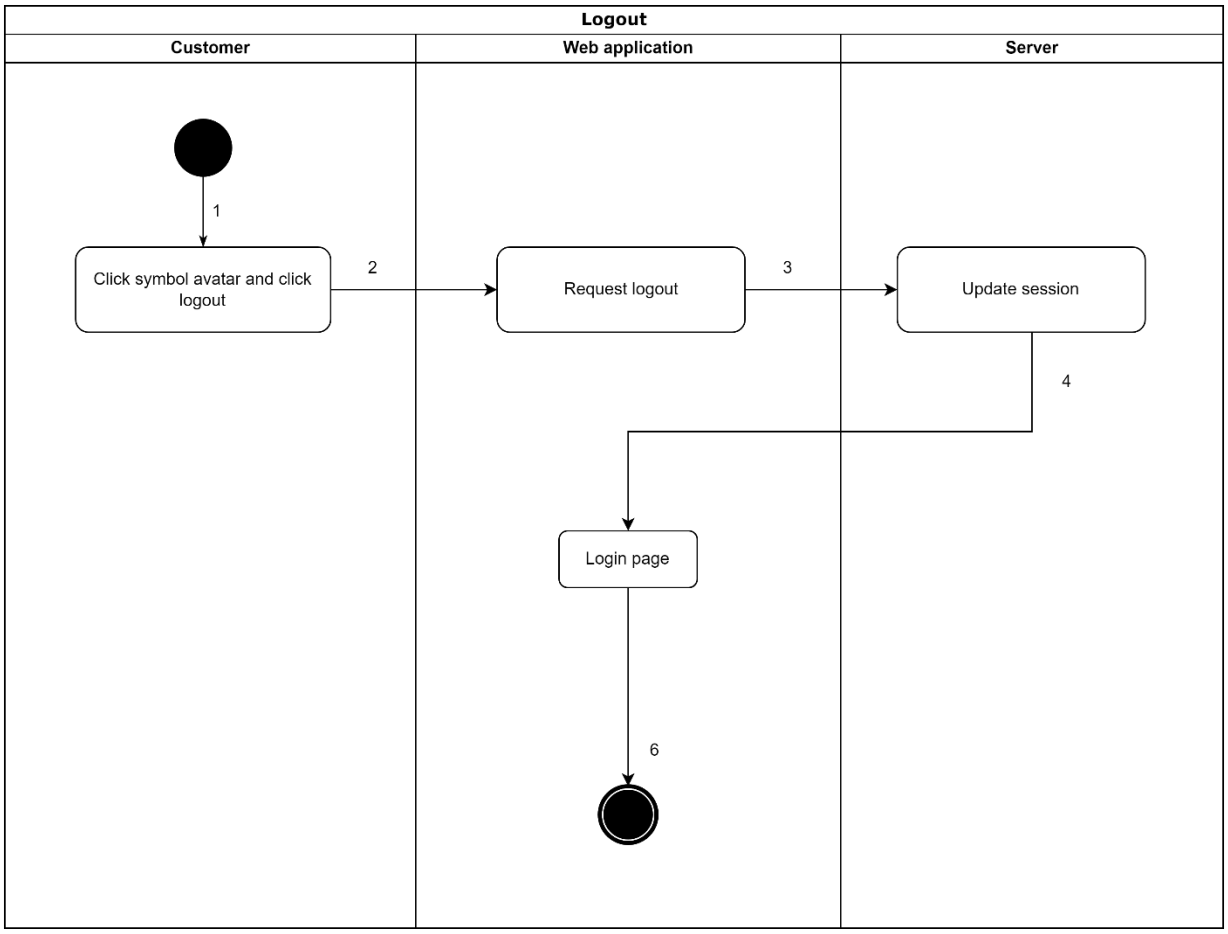


Figure 54 - <Activity Diagram> Customer logout

3.4.22 <Translator> View history rating

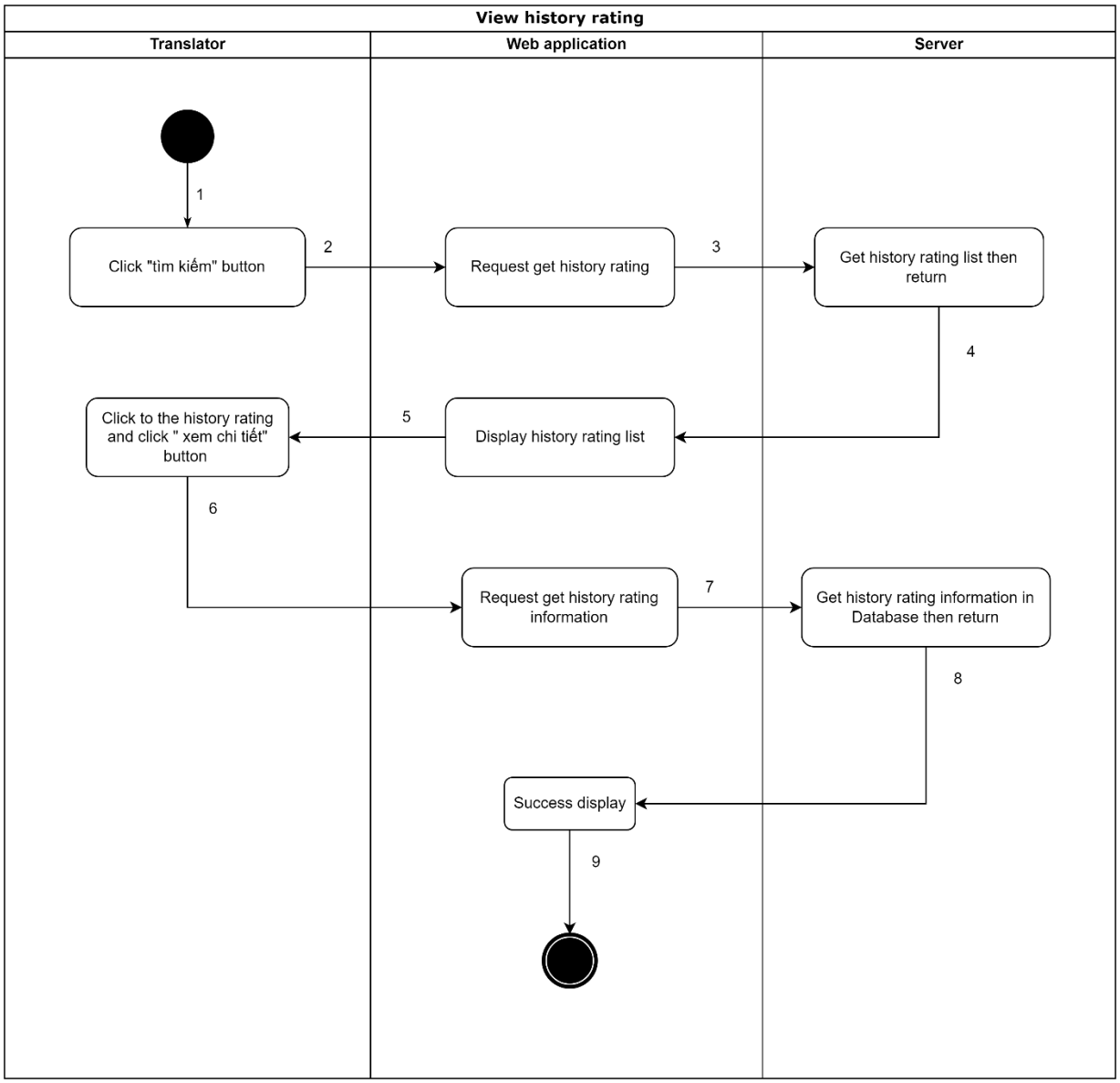


Figure 55 - <Activity Diagram> Translator view history rating

3.4.23 <Translator> View invitation

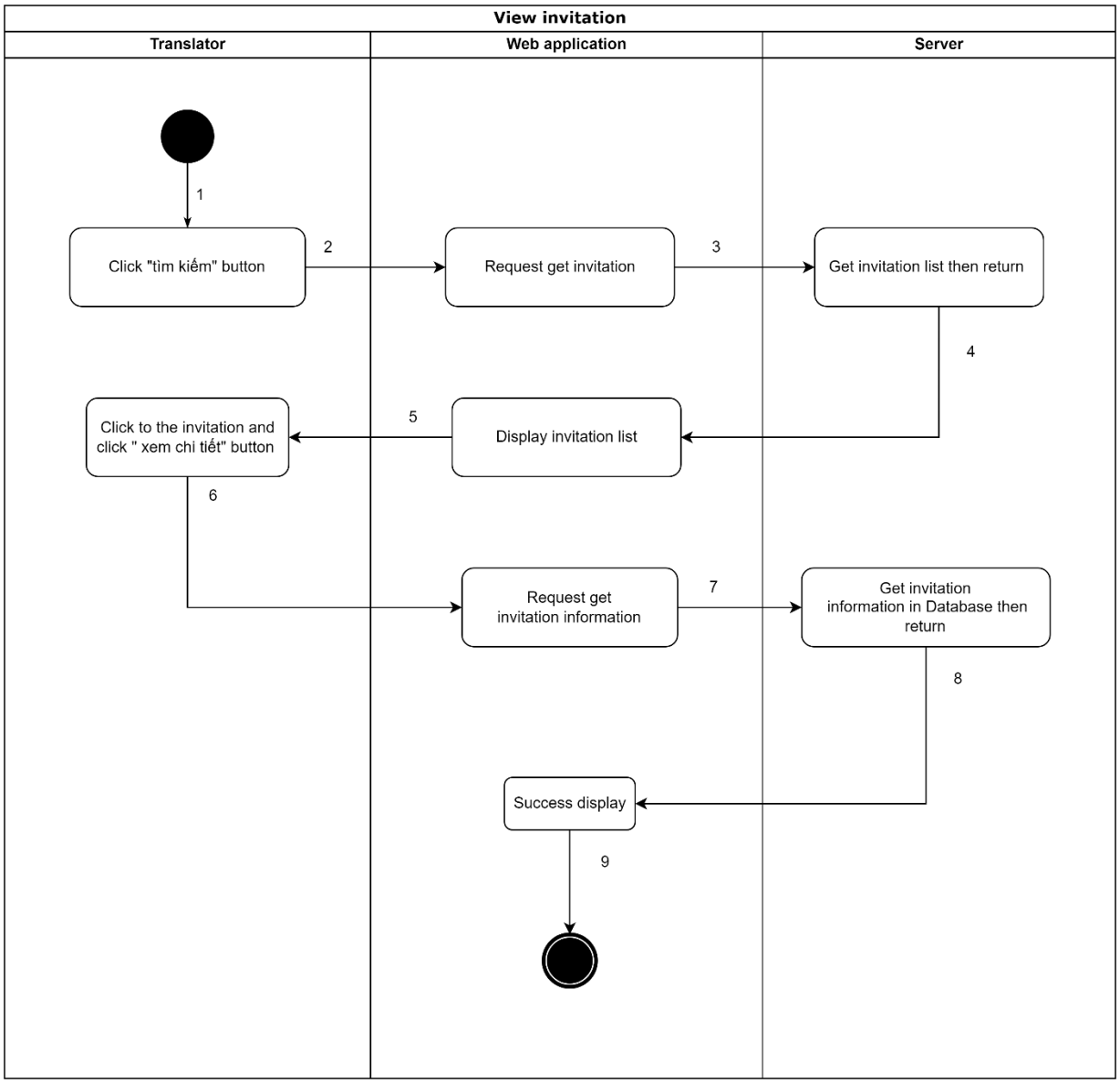


Figure 56 - <Activity Diagram> Translator view invitation

3.4.24 <Translator> View project

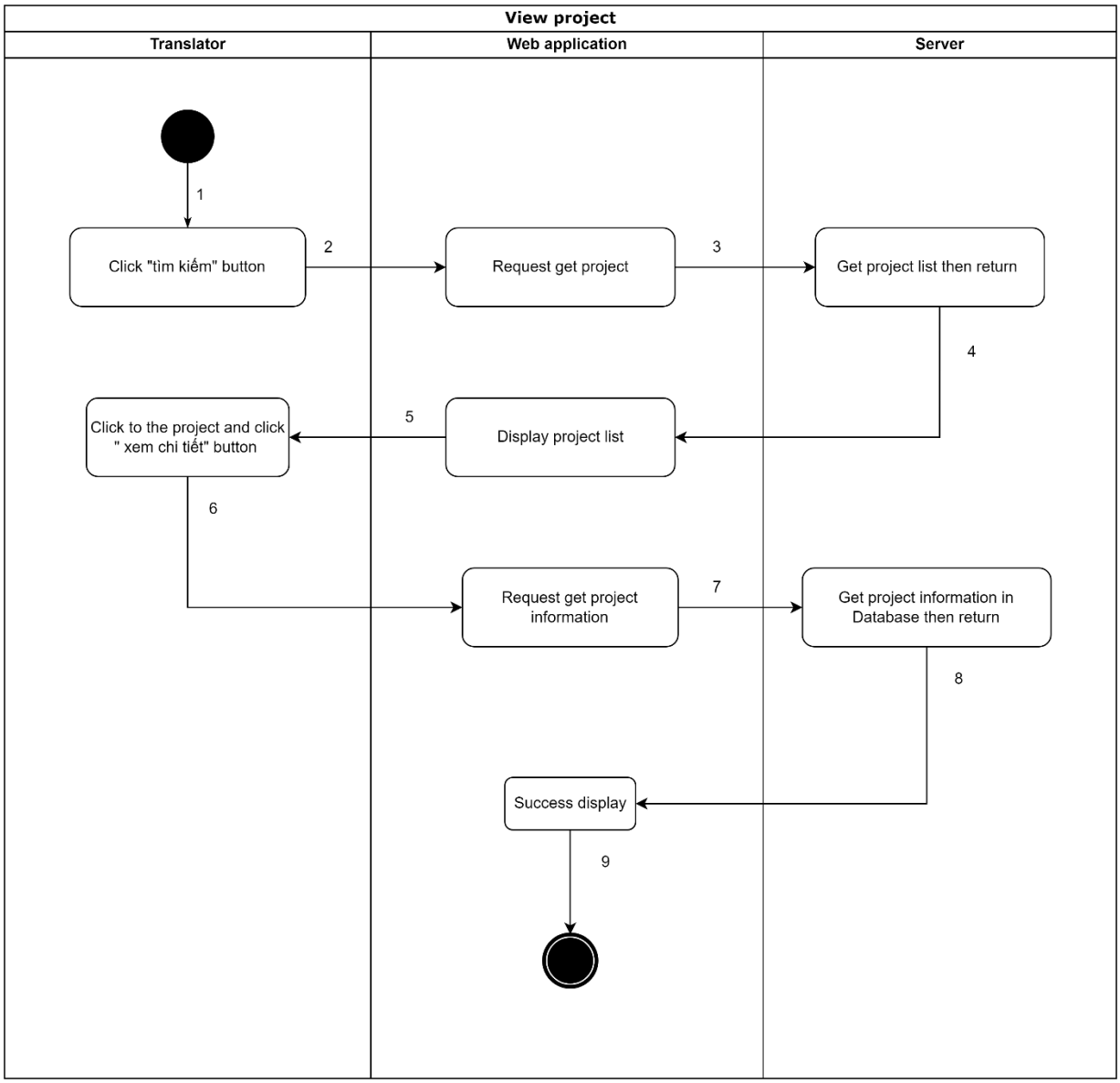


Figure 57 - <Activity Diagram> Translator view project

3.4.25 <Translator> Approve invitation

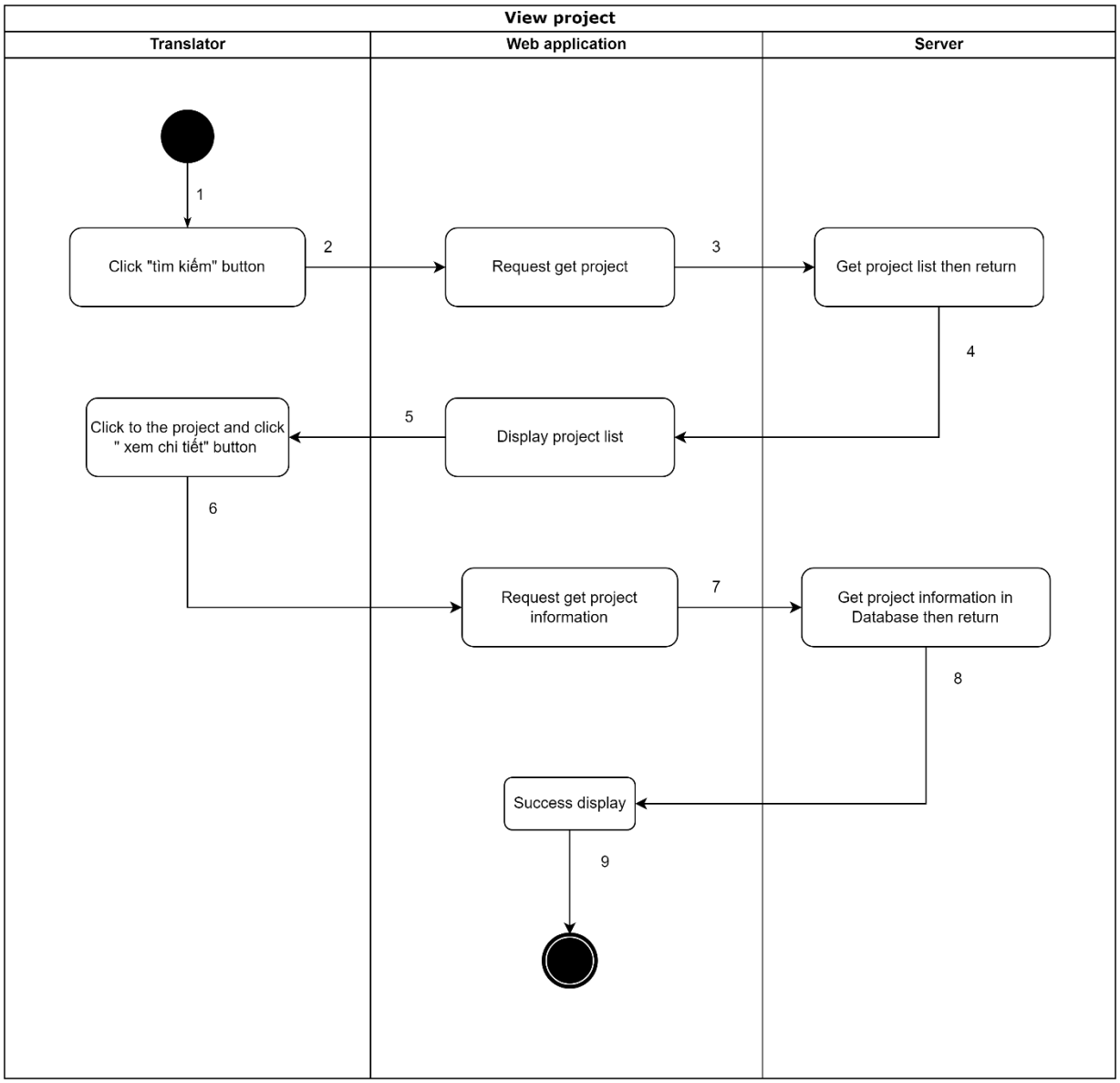


Figure 58 - <Activity Diagram> Translator approve invitation

3.4.26 <Translator> Reject invitation

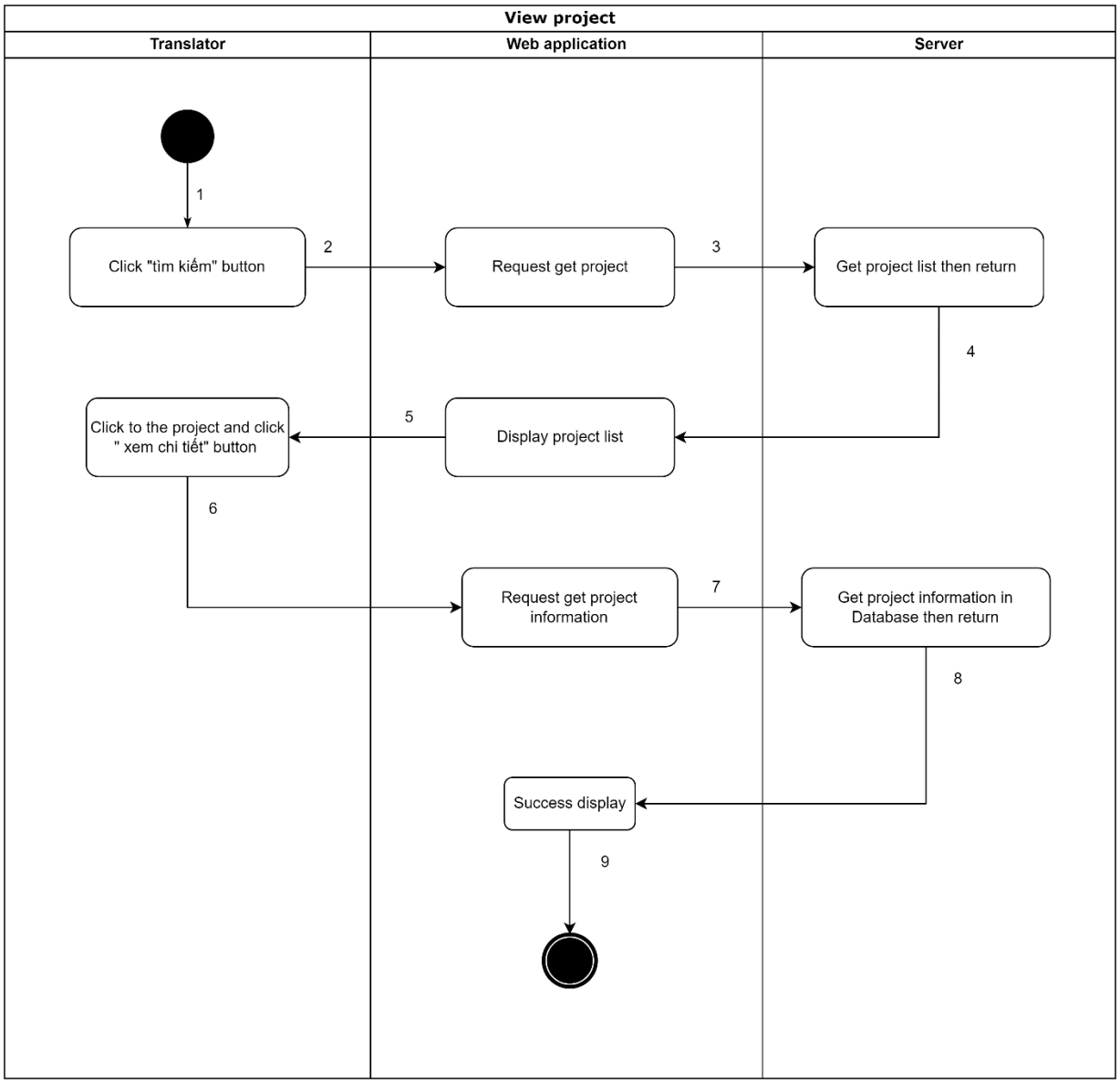


Figure 59 - <Activity Diagram> Translator reject invitation

3.4.27 <Translator> Request to join project

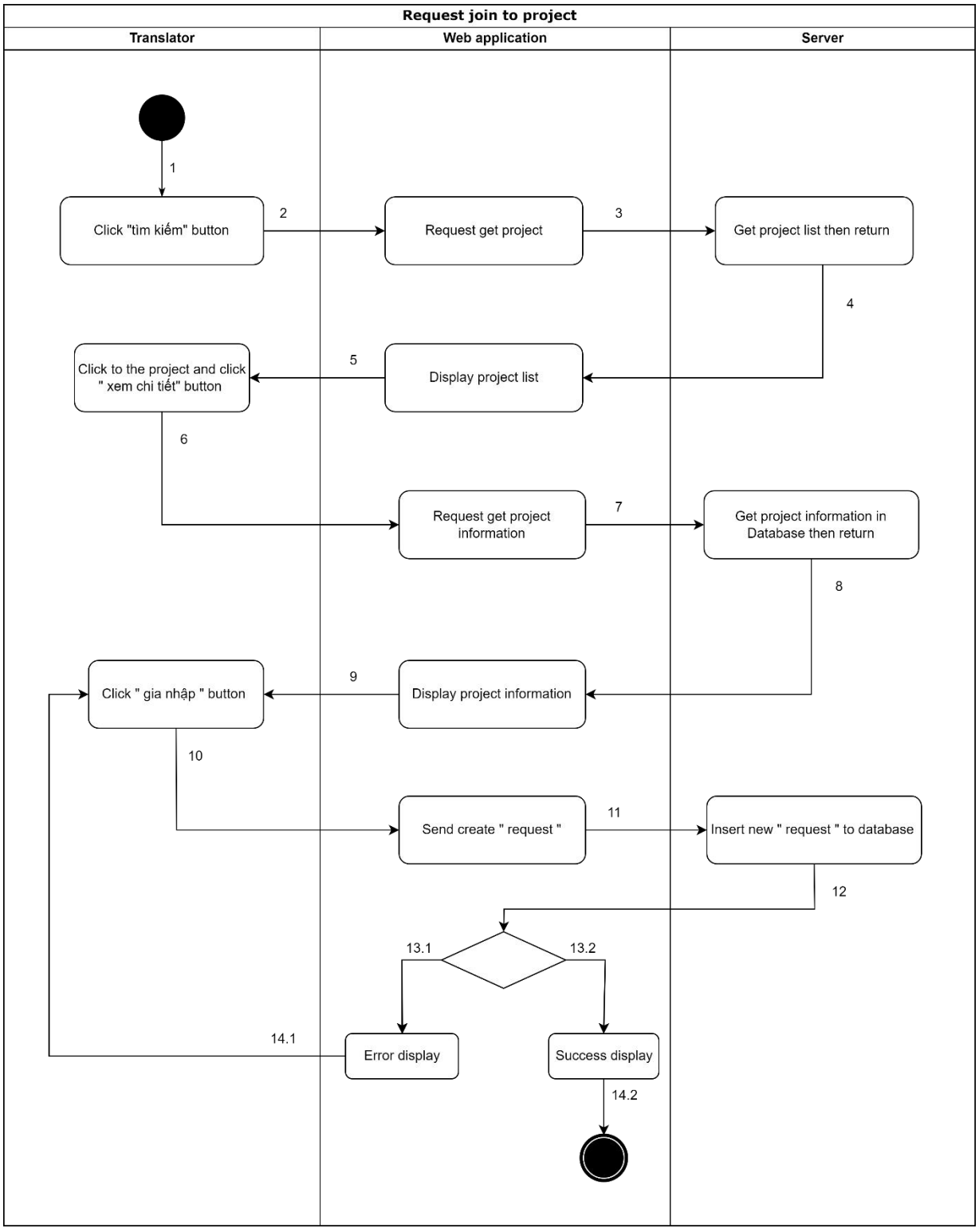


Figure 60 - <Activity Diagram> Translator request to join project

V. Software Testing Documentation

1. Overall Description

1.1 Test Scope

We plan to define test cases in main flow only.

1.2 Test Model

We use Incremental testing for some approaches:

- Each module is required a role to use clearly
- Each module could test at the early stage of the develop process

1.3 Testing Levels

- Testing Level: User Acceptance Test
- Testing Types: Functional Testing, User Interface Testing

2. Test Plan

2.1 Test Stages

Type of Test	Test Level			
	Unit	Integration	System	Acceptance
Function Testing				X
User Interface Testing				X

Table 37 - Test Stages

2.2 Resources

Worker	Role	Specific Responsibilities
Võ Nhứt Thanh Long	Leader	Planning, execute, verify test result
Võ Nhị Thanh Long	Member	Planning, execute, verify test result
Phạm Đức Anh Tuấn	Member	Planning, execute, verify test result
Lê Đức Quang	Member	Planning, execute, verify test result

Table 38 - Resources

2.3 Test Milestones

No.	Deliverables	Due Date
1	Test Document	25/04/2023

Table 39 - Test Milestones

3. Test Cases and Test Reports

Testing Level	File Name
User Acceptance Test	TestDocuments\2023.Spring-_-iTranslator-_-Capstone-Project-_-Test-Report.xlsx

Table 40 - Test Cases and Test Reports

VI. Release Package & User Guides

1. Deliverable Package

1.1 Source codes & documents

No.	Items	Sub-Items	Type	Version
Code Package				
1	Web Module	iTranslator.ManagementWebApplication.zip	New	1.0.0
2	Web API Module	iTranslator.API.zip	New	1.0.0
Database				
1	Initialise Database	iTranslator.Script.sql	New	
Documents				
1	Final Project Report	iTranslator Final Project Report.docx		

Table 41 - Source codes & documents

1.2 Known Issues, Limitations & Restrictions

- Installers for each projects haven't been packaged
- Database must be initialised by using database scripts

2. Installation Guides

2.1 System Requirements

2.1.1 Web application

PC	Minimum	Recommend
Internet connection	LAN, internet access capability	LAN, Wi-fi (16 Mbps)
Processor	Intel Core i5 1.5Ghz	Intel Core i7 2.6Ghz
Memory	4GB RAM	8GB RAM up
Storage	4GB	6GB up
Web Browser	Chrome (v69)	Chrome latest stable version

Table 42 - <System Requirements> Web application

2.1.2 Software requirement

Component	Name and Version	Description
Operating System	Window 10/11, other open sources (Linux, Ubuntu)	

Database	SqlServer 19	
.NET Core Runtime	5.0	Used to hosting API server

Table 43 - <System Requirements> Software requirement

2.2 Setup Files

- iTranslator.Script.sql: Database script for system
- iTranslator.API.zip: API for services
- iTranslator.ManagementWebApplication.zip

2.3 Installation Instruction

2.3.1 Setup Database

Install Sql Server 2019 by following guidelines link:
https://www.thegioididong.com/game-app/huong-dan-cai-dat-sql-server-2019-cuc-don-gian-chi-1312926

2.3.2 Setup Backend API

+ Install Visual Studio by following guidelines link:
https://thuthuatphanmem.vn/huong-dan-tai-va-cai-dat-visual-studio/

+ Extracting: iTranslator.API.zip

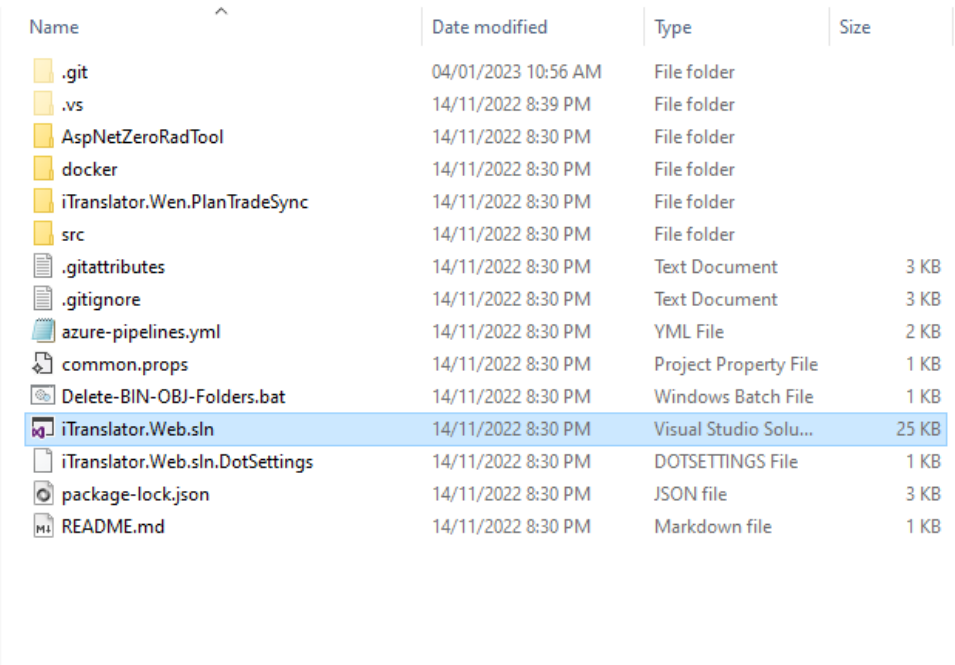


Figure 61 - Setup Backend Api

+ Open iTranslator.Web.sln file with Visual Studio

+ Config your database connection

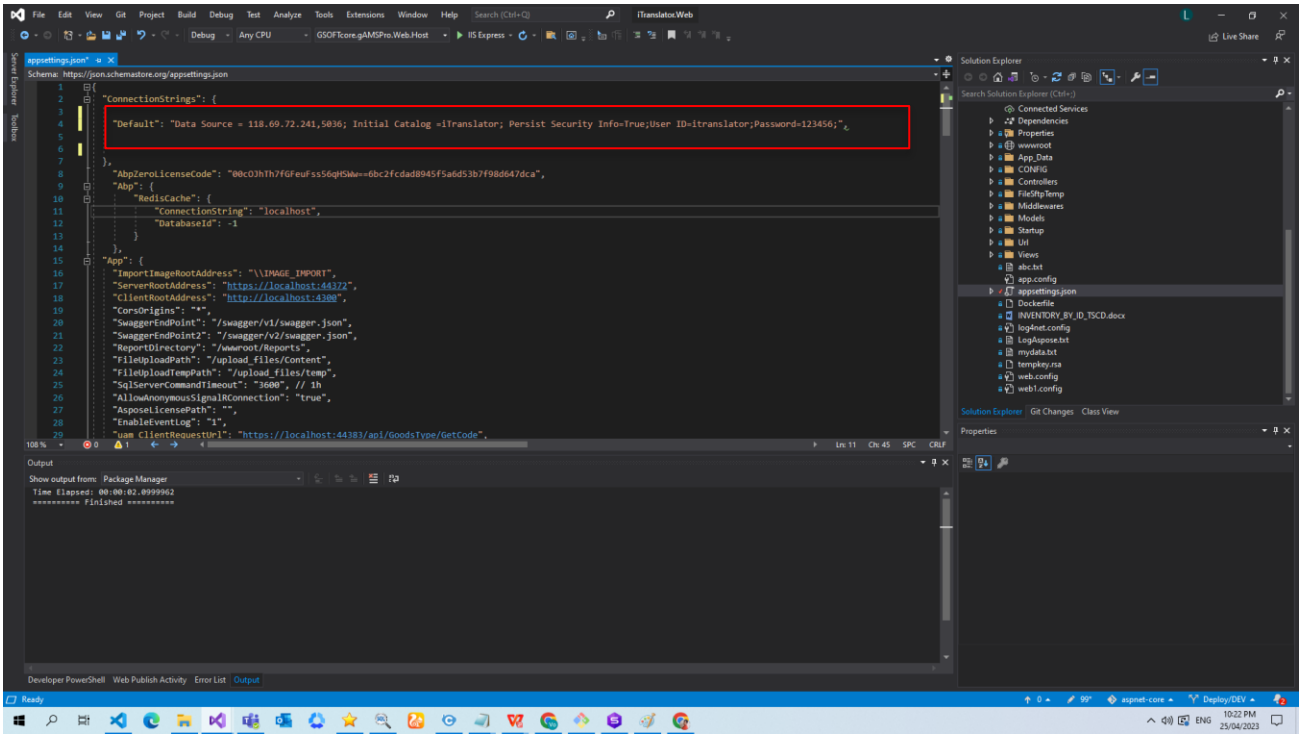


Figure 62 - Setup Backend Api

+ Run project

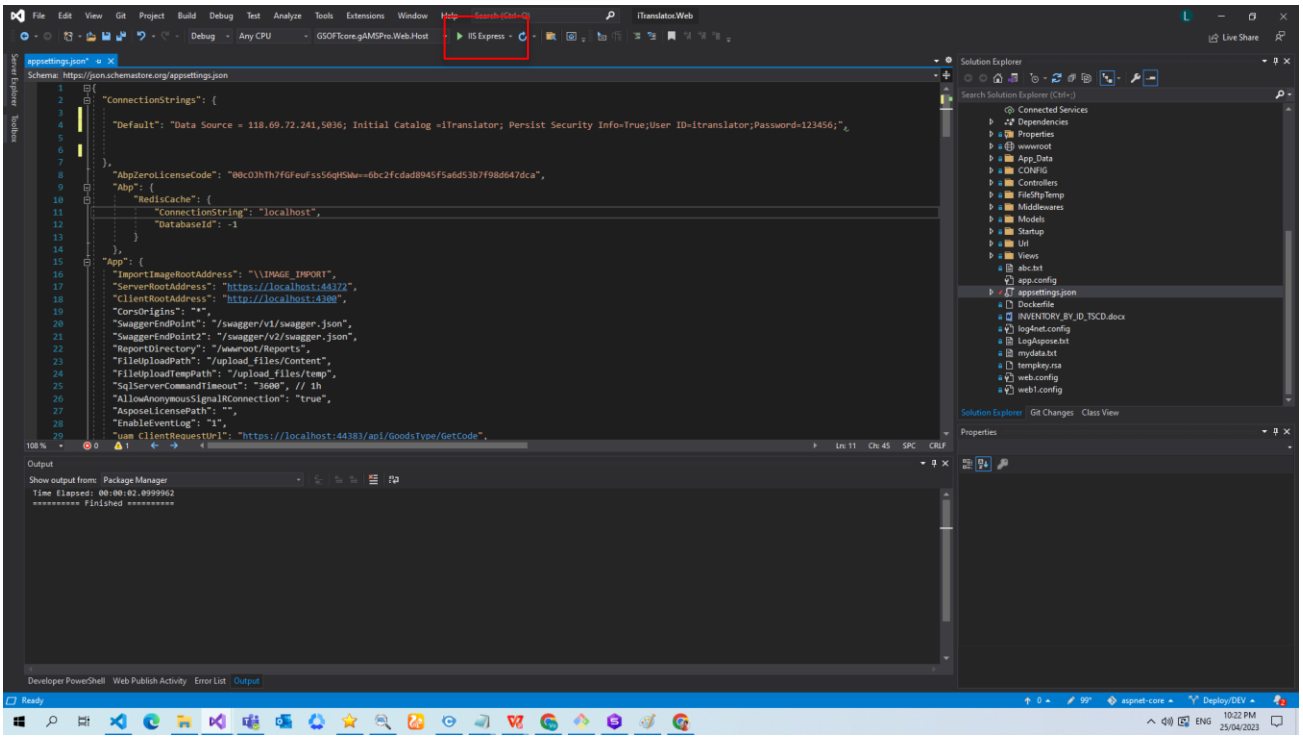


Figure 63 - Setup Backend Api

+ Test project in browser by go to url: https://localhost:44372/Ui ;

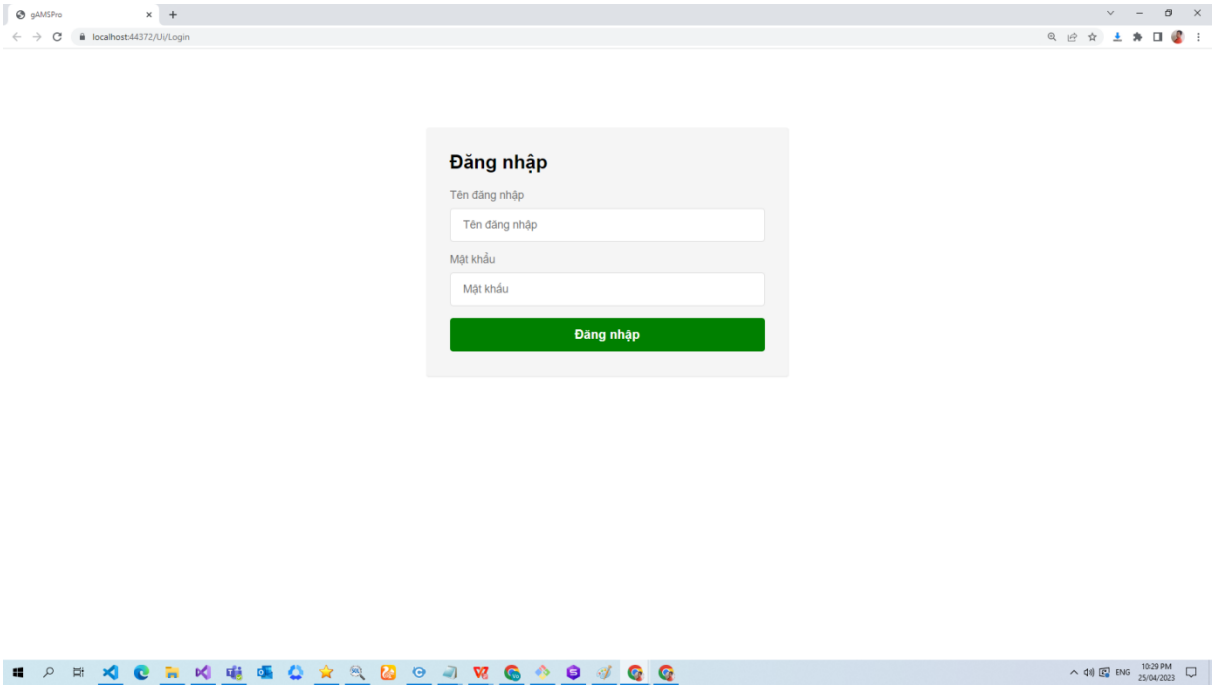


Figure 64 - Setup Backend Api

+ Username: admin1 pass 123456

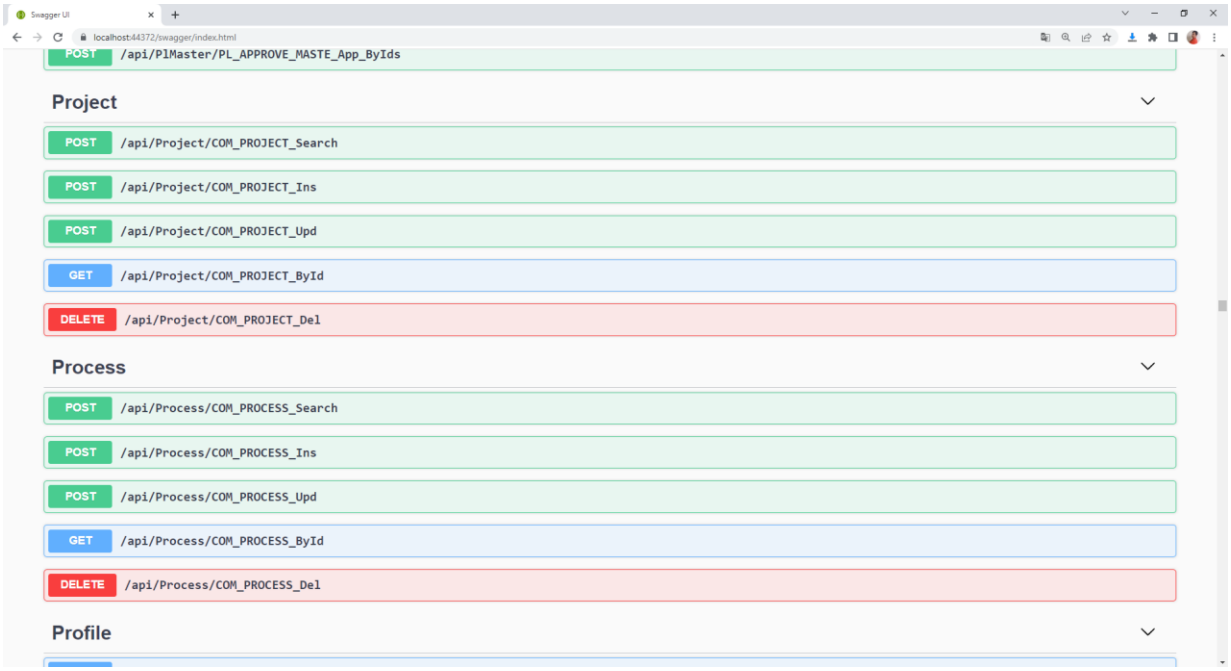


Figure 65 - Setup Backend Api

2.3.3 Setup Management Web

+ Install Visual Studio Code by following guidelines link: https://thaynhuom.edu.vn/cai-visual-studio-code-lap-trinh-c-c/

+ Install npm through Visual Studio Code or you can install npm via NodeJS https://nodejs.org/en/

+ Extract iTranslator.ManagementWebApplication.zip and open folder with Visual Studio Code

+ Open terminal and enter npm I:



Figure 66 - Setup Management Web

+ Change baseUrl based on your backend configuration

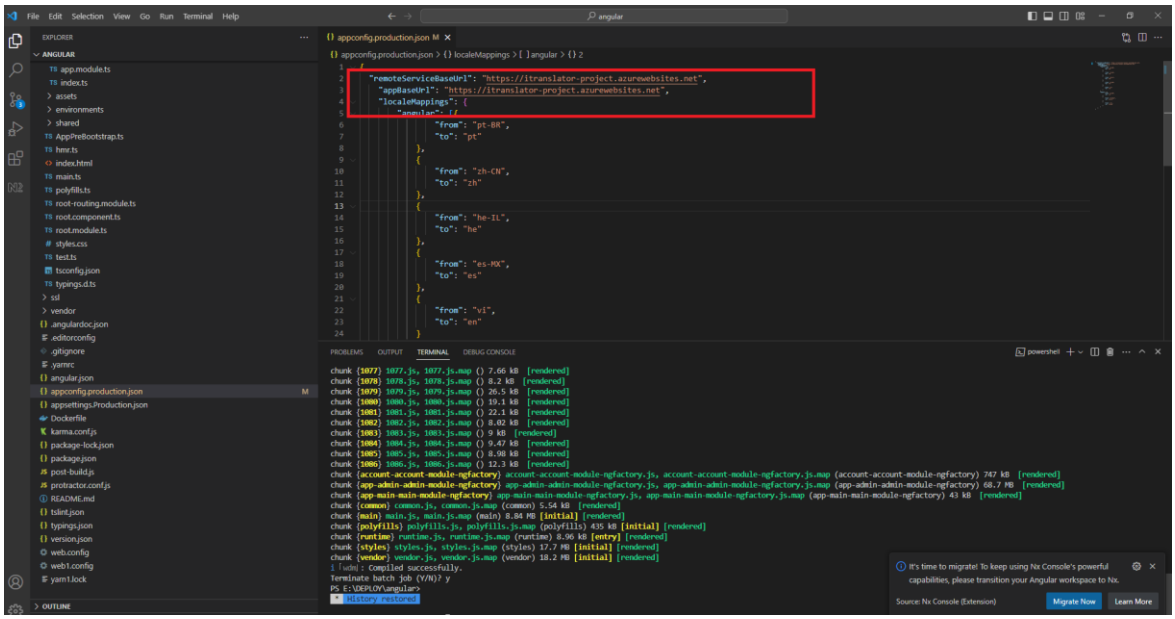


Figure 67 - Setup Management Web

+ In terminal, enter npm start



Figure 68 - Setup Management Web

+ Project start with http://localhost:4200/



3 User Manual

3.1 Overview

- Admin manage project, rating, category and type
- Moderator manage projects, categories, topic, problem report, languages, achievements also they can approve payment, chat, view rating
- Customer manage projects, payments and they can chat, view history rating
- Translator manage invitation, they also can search project, view project, request to join project, view translated experience, update translation, send project completion notice and withdraw moneyf

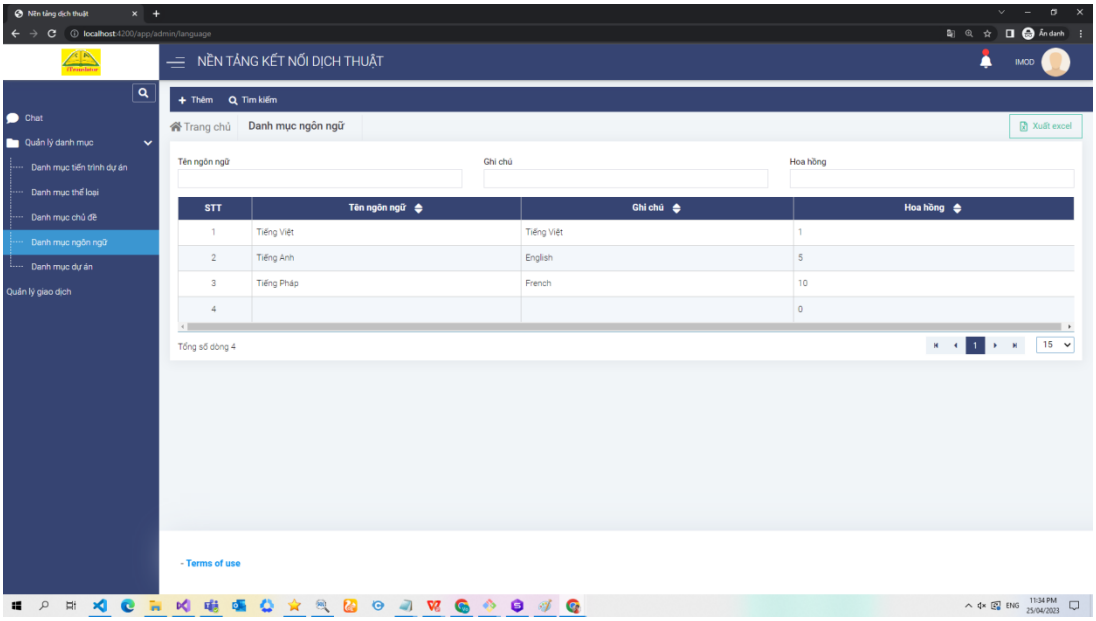
3.2 Application Usage

3.2.1 Moderator login into system



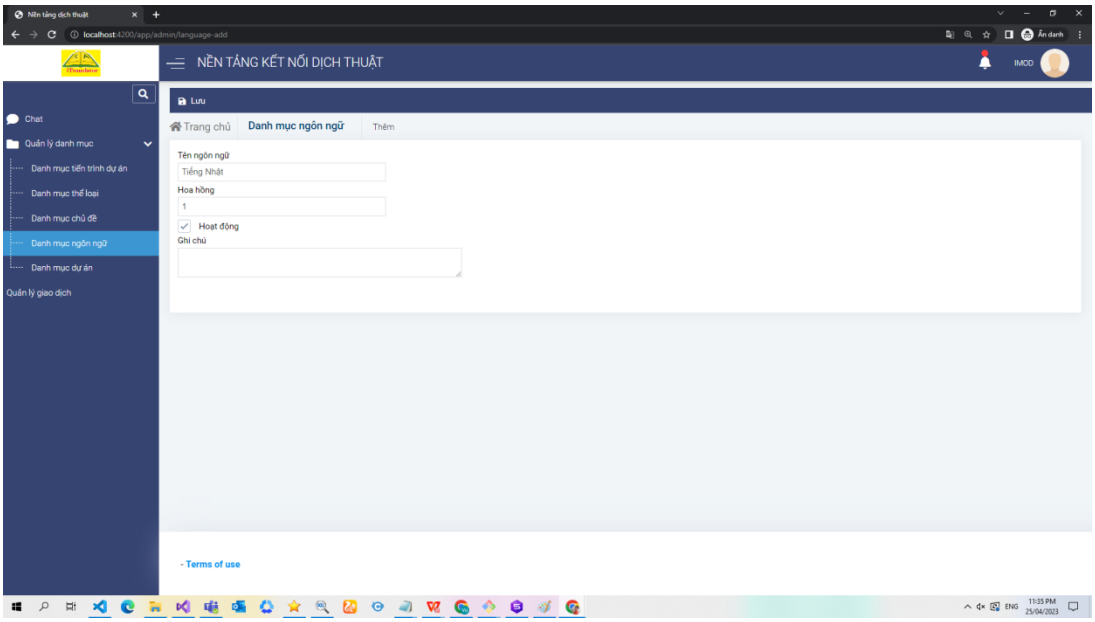
1. Enter username: imod
2. Enter password: 123456
3. Click “đăng nhập”

3.2.2 Moderator search language



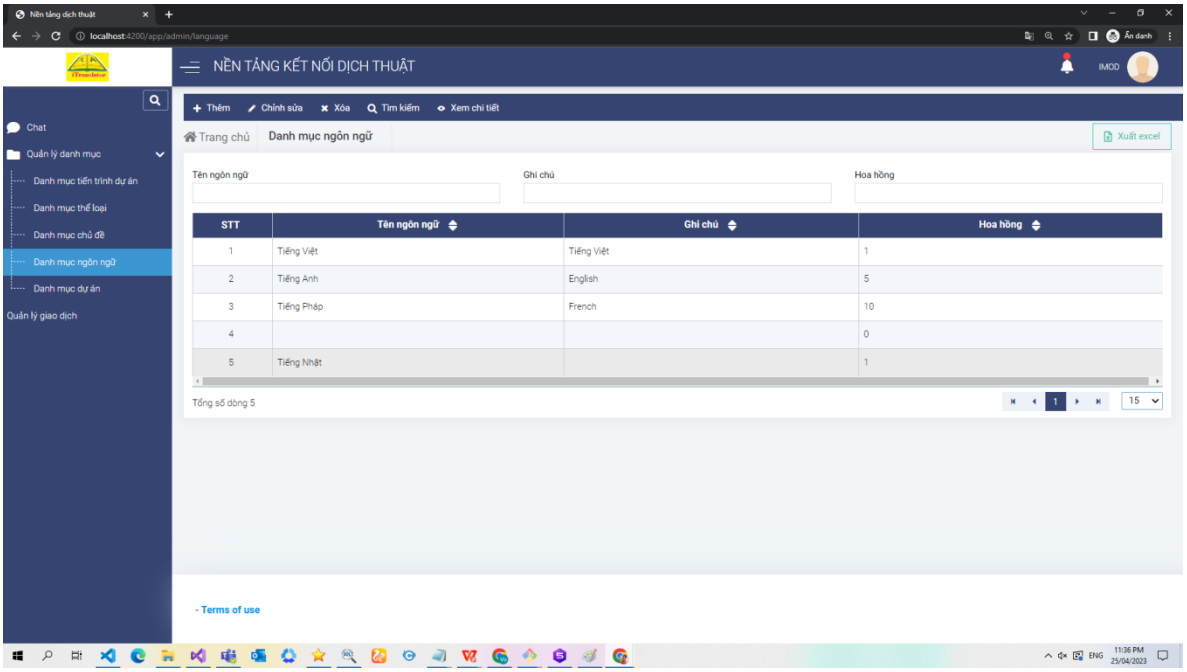
1. Click “Quản lý danh mục”
2. Click “Danh mục ngôn ngữ”
3. Click “Tìm kiếm”

3.2.3 Moderator create language



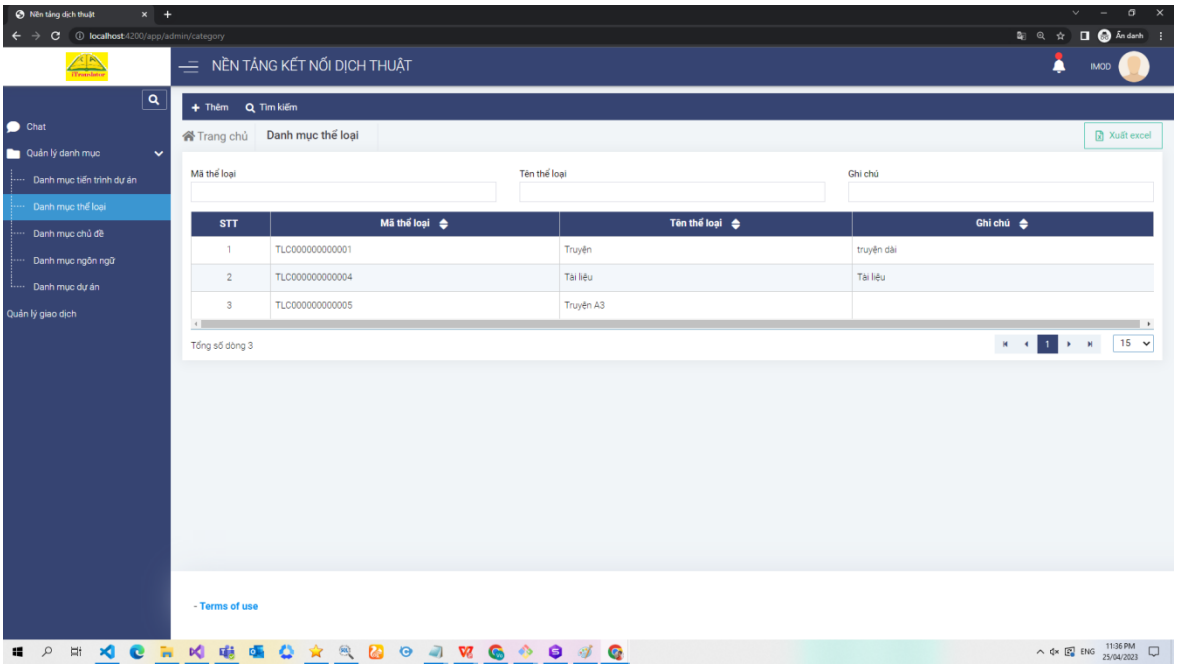
1. Click “Thêm”
2. Fill all information
3. Click “Lưu”

3.2.4 Moderator update, view information, delete language



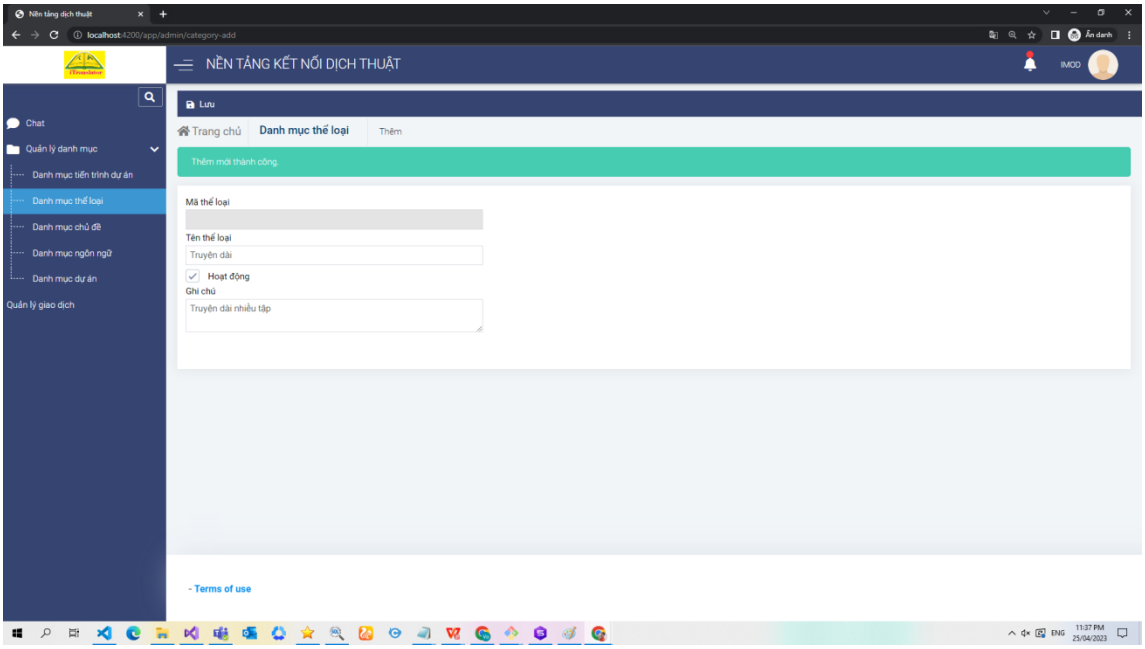
1. Click icon on toolbar to update, view information or delete

3.2.5 Moderator search category



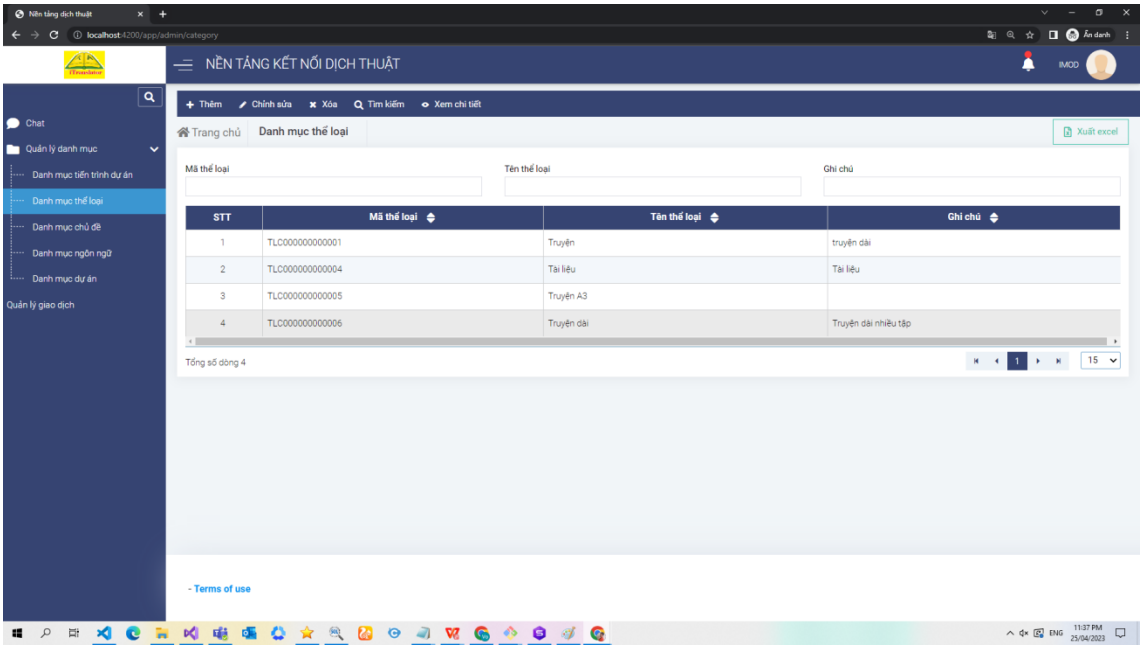
1. Click “Quản lý danh mục”
2. Click “Danh mục thể loại ”
3. Click “Tìm kiếm”

3.2.6 Moderator create category



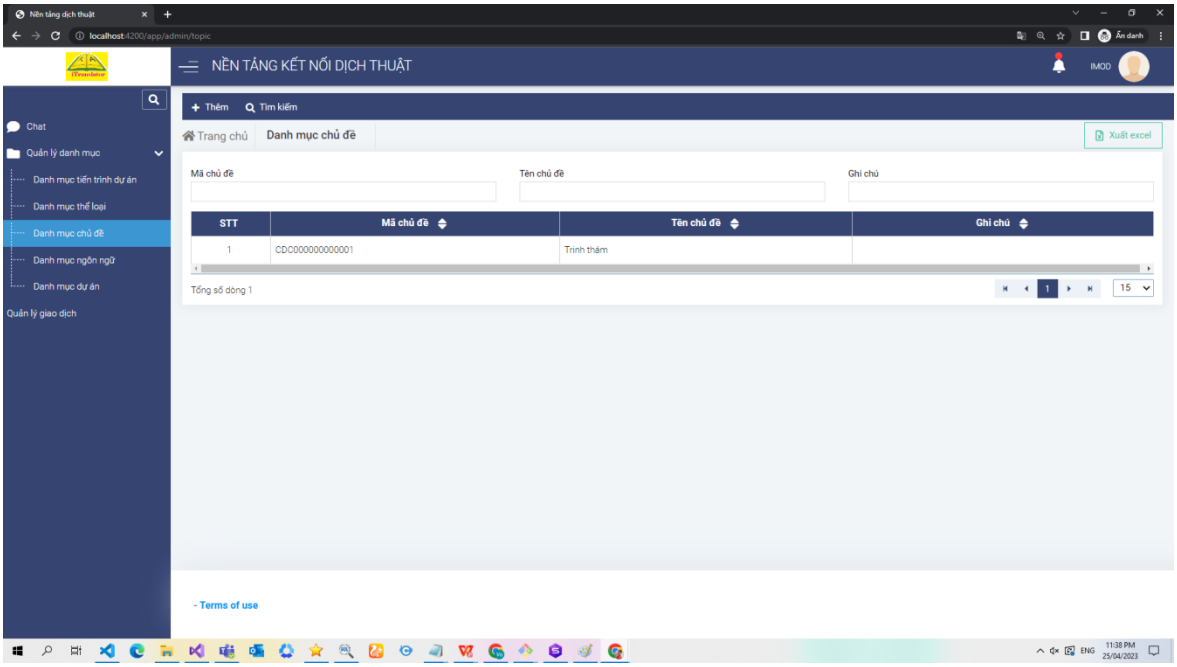
1. Click “Thêm”
2. Fill all information
3. Click “Lưu”

3.2.7 Moderator update, view information, delete category



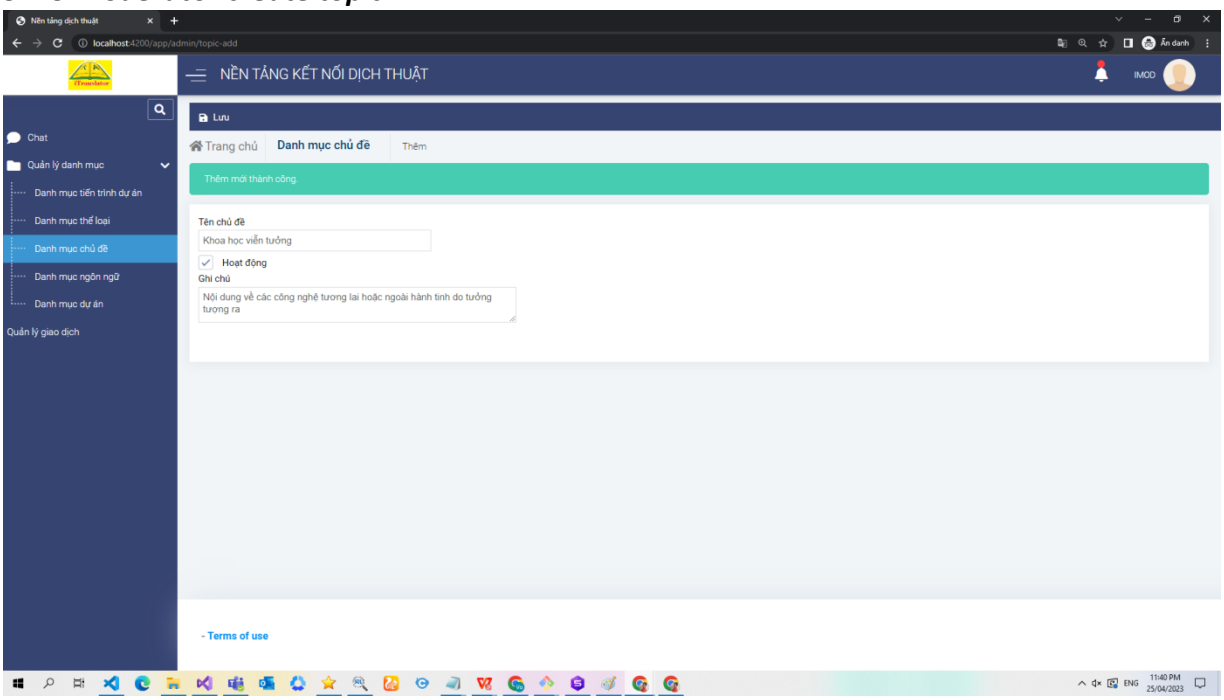
1. Click icon on toolbar to update, view information or delete

3.2.8 Moderator search topic



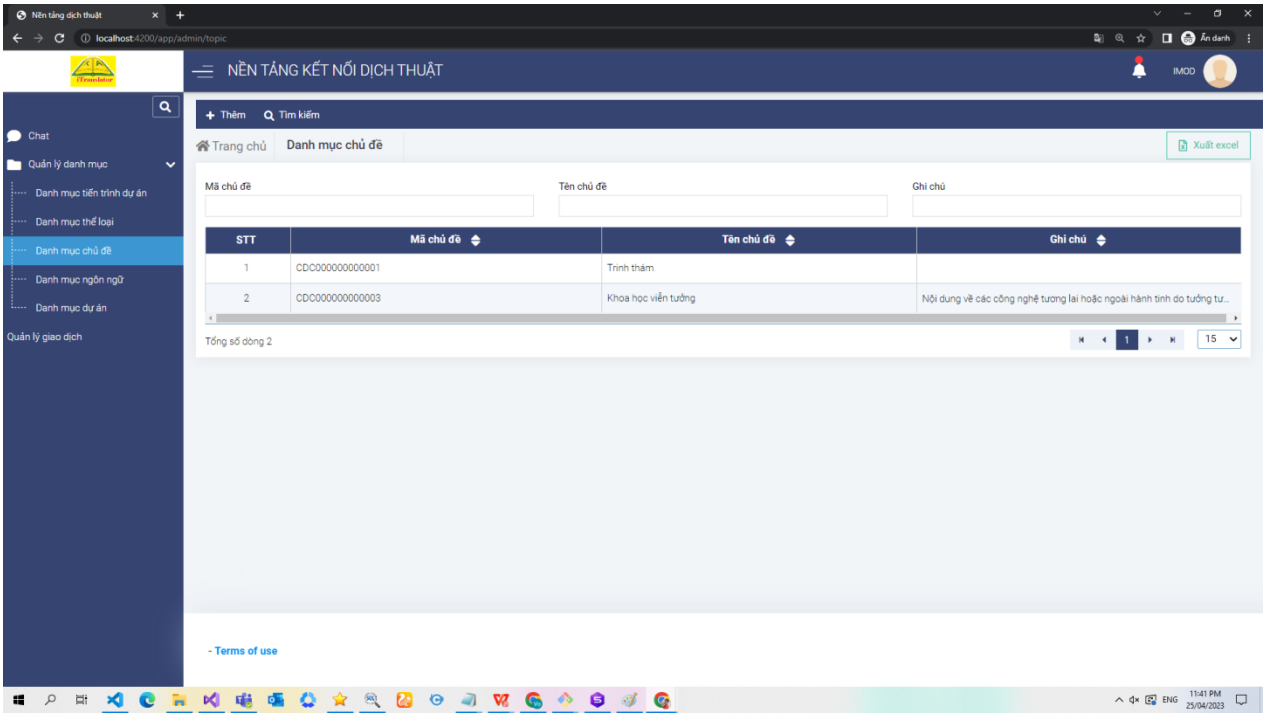
1. Click “Quản lý danh mục”
2. Click “Danh mục thể loại ”
3. Click “Tìm kiếm”

3.2.9 Moderator create topic



1. Click “Thêm”
2. Fill all information
3. Click “Lưu”

3.2.10 Moderator update, view information, delete topic



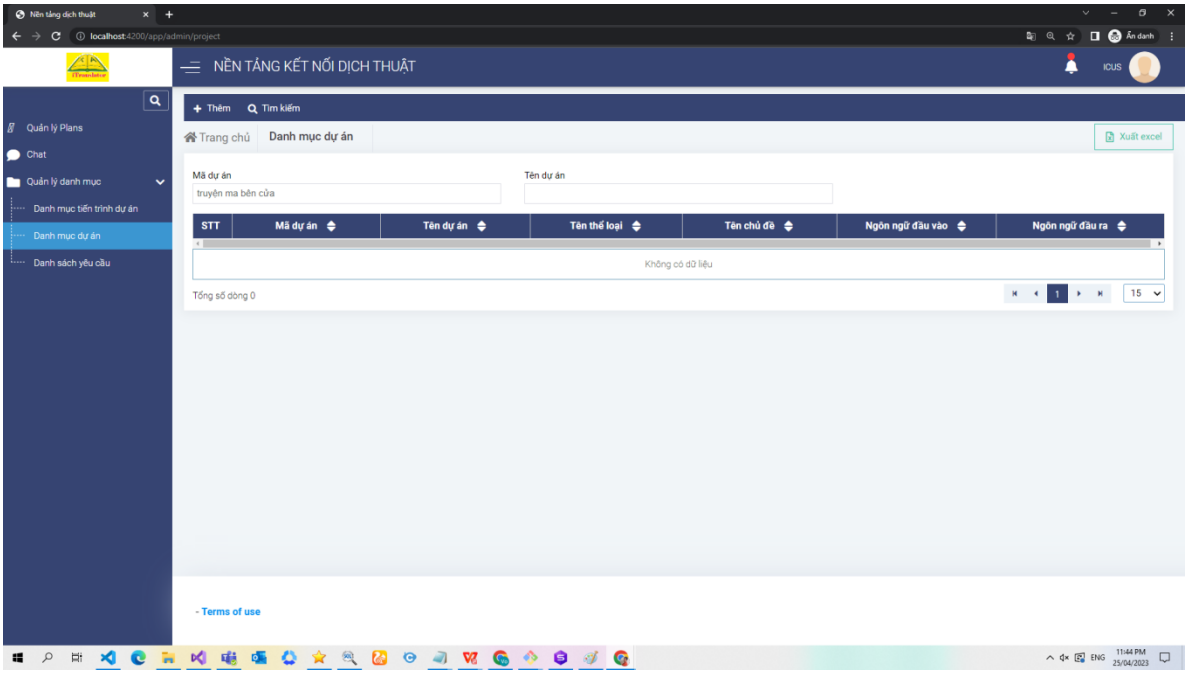
1. Click one topic
2. Click icon on toolbar to update, view information or delete

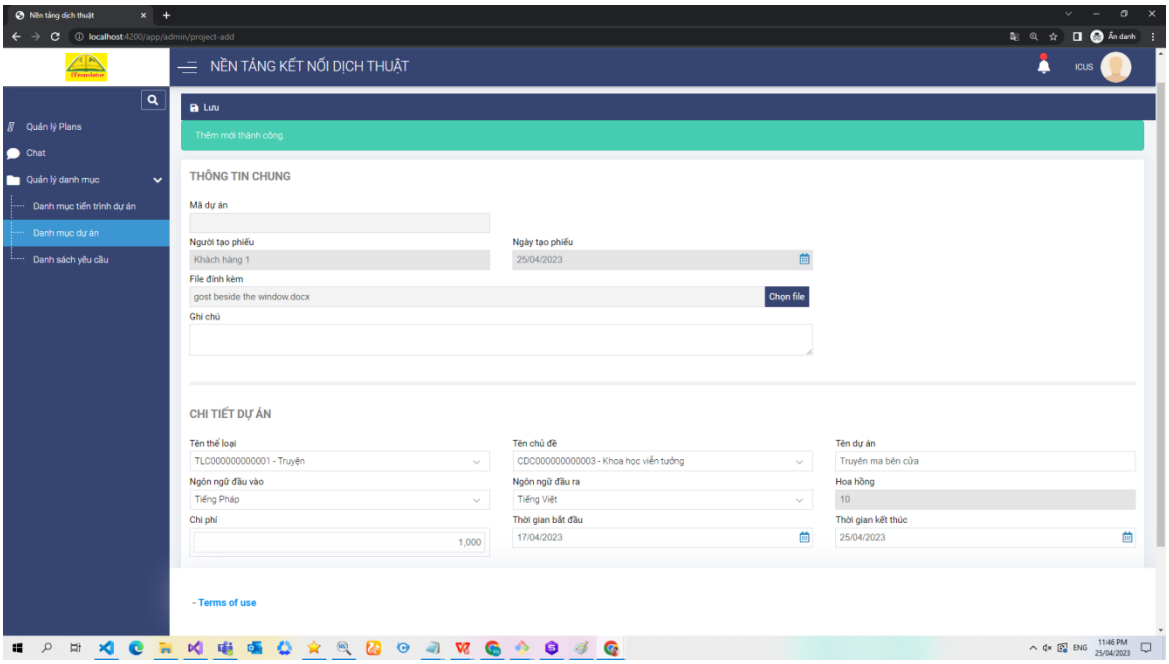
3.2.11 Customer login into system

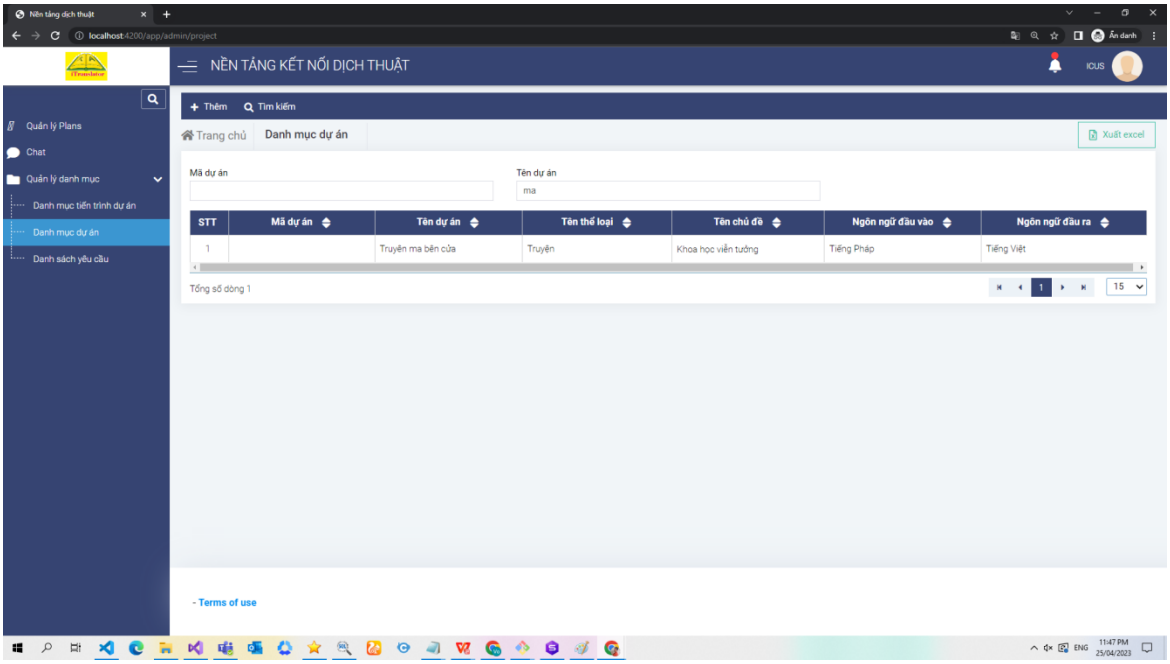


1. Enter username: icus
2. Enter password: 123456
3. Click “đăng nhập”

3.2.12 Customer create project

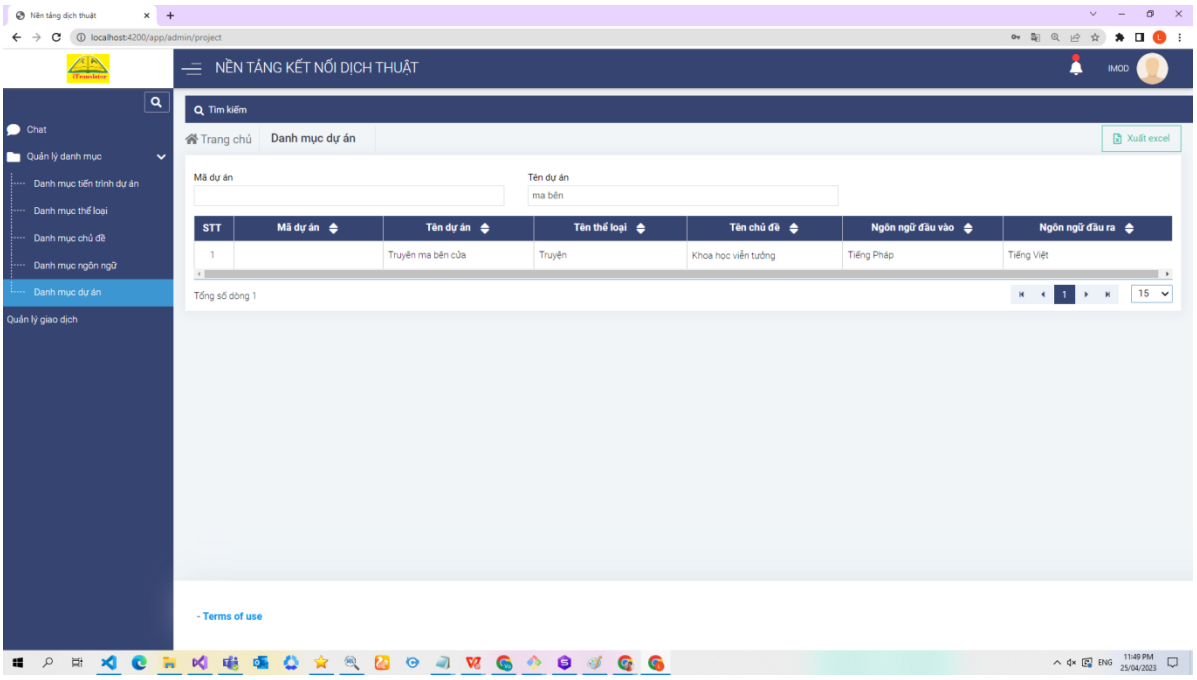


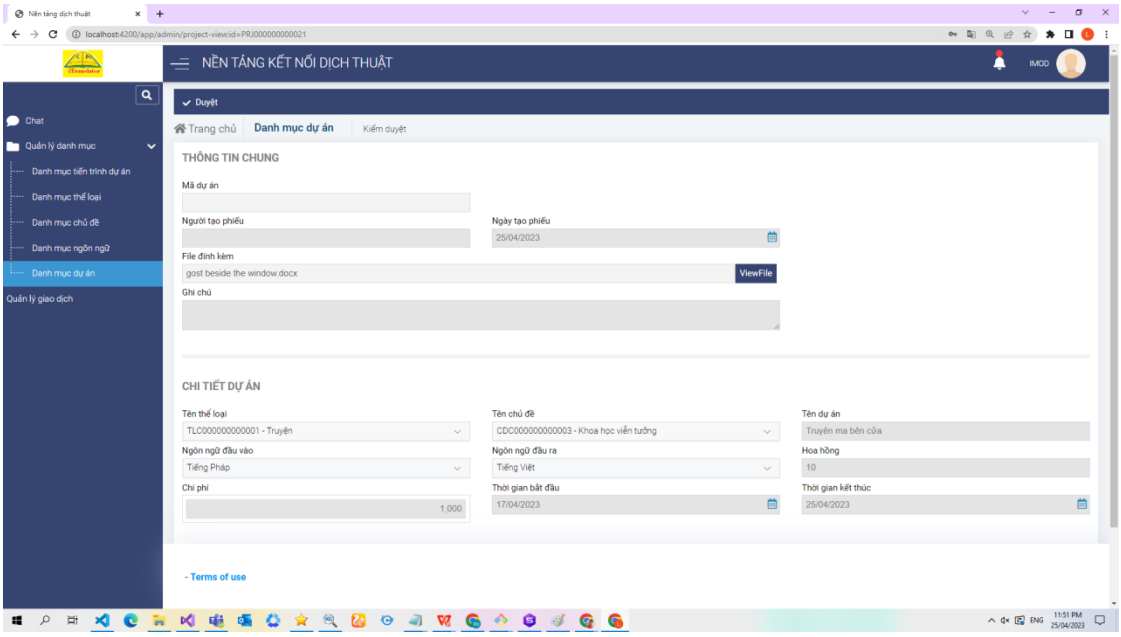


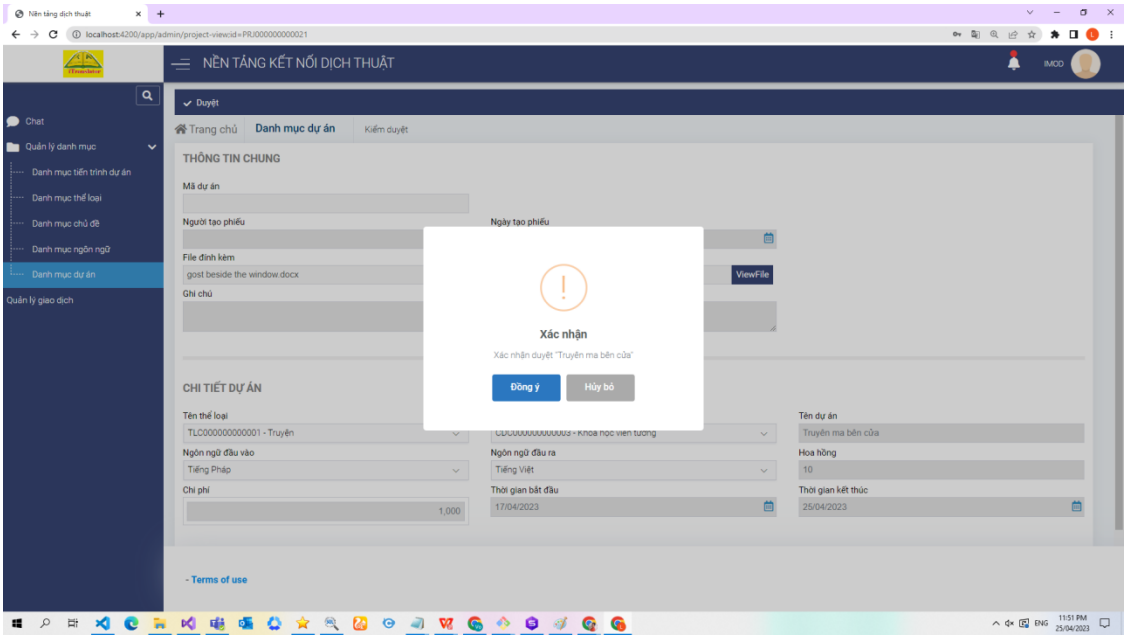


1. Click “Thêm”
2. Fill all information
3. Click “Lưu”

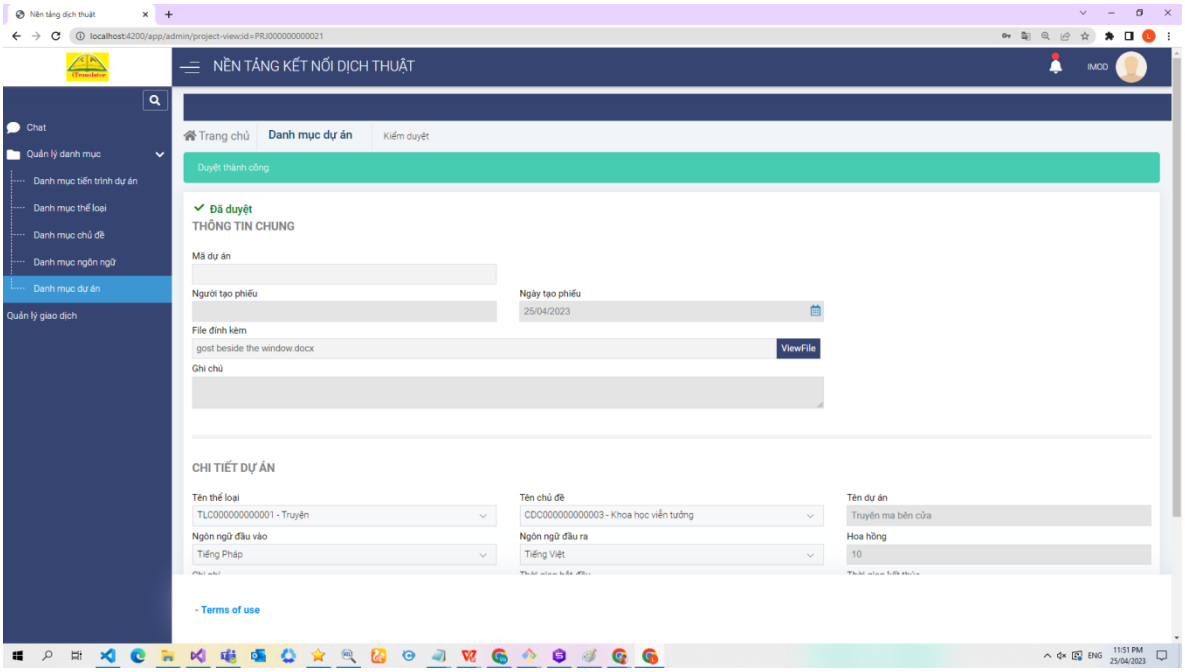
3.2.13 Moderator approved project





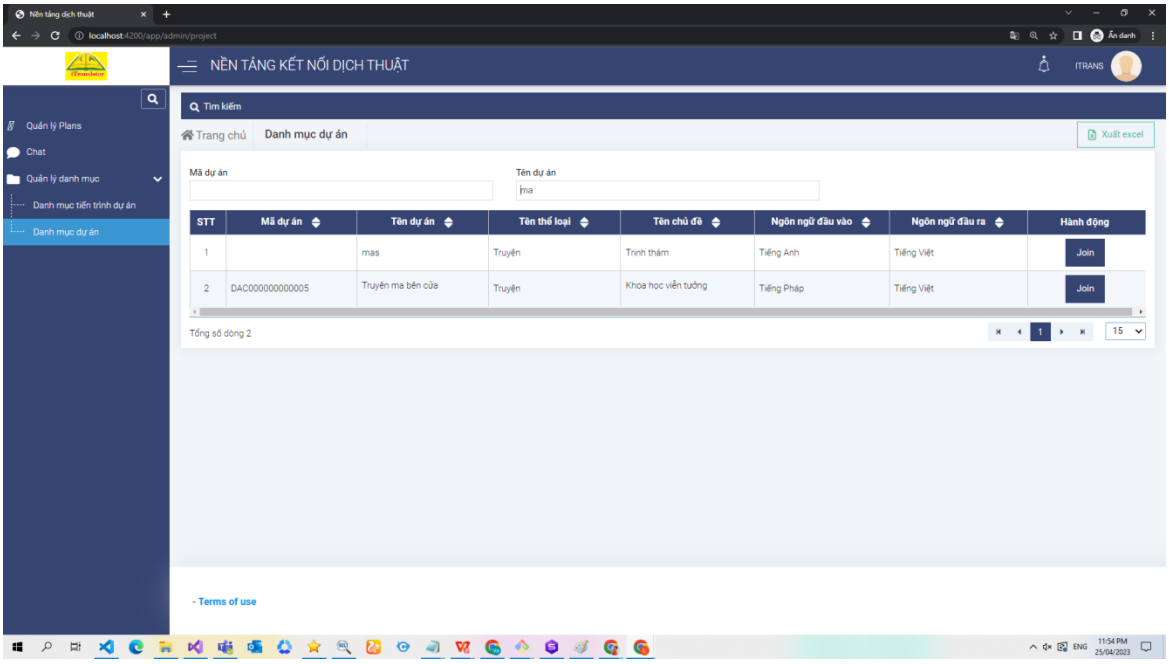


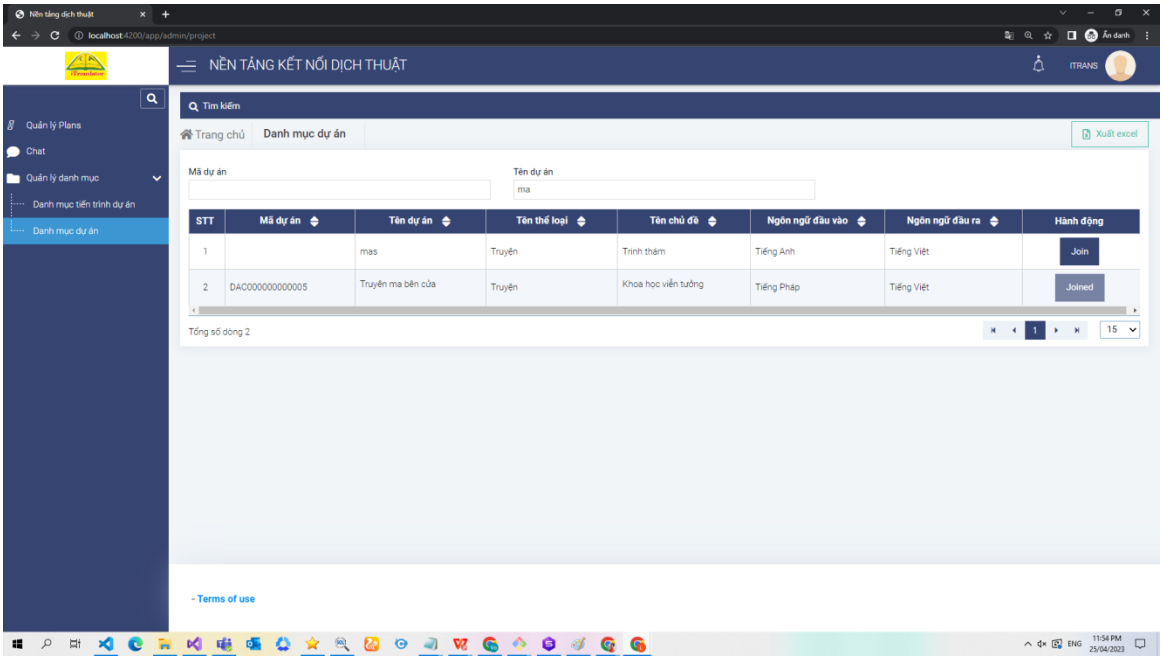
3.2.14 Moderator approved project



1. Click “Tìm kiếm” project
2. Check information of project
3. Click “Duyệt”
4. Click “Đồng ý”

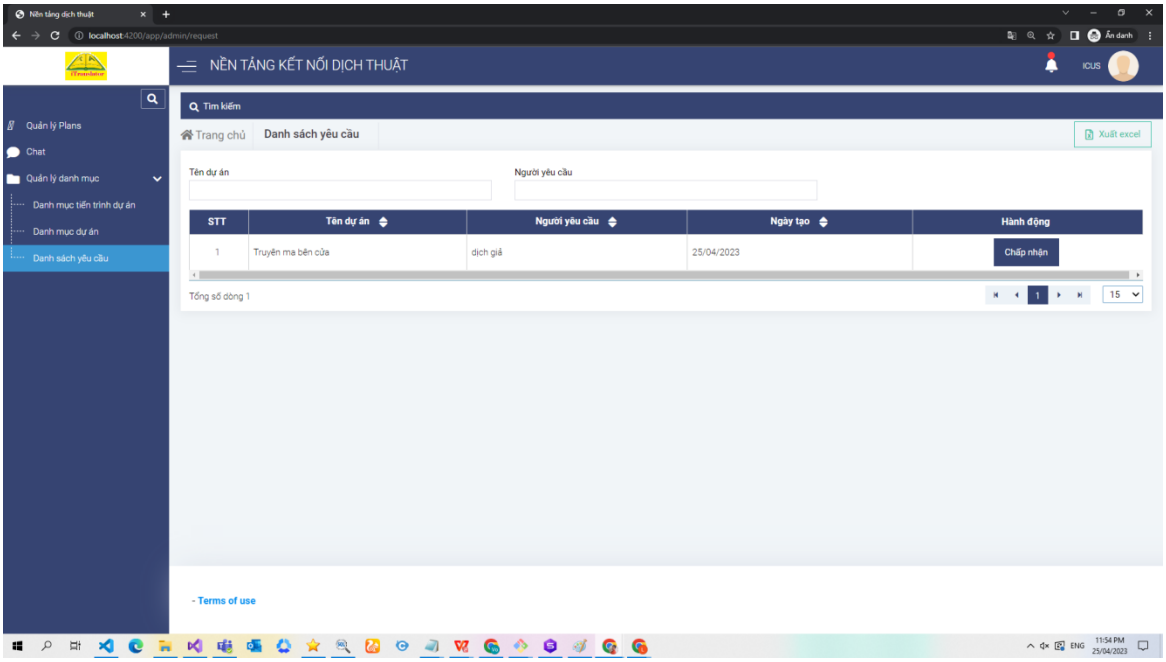
3.2.15 Translator request join project

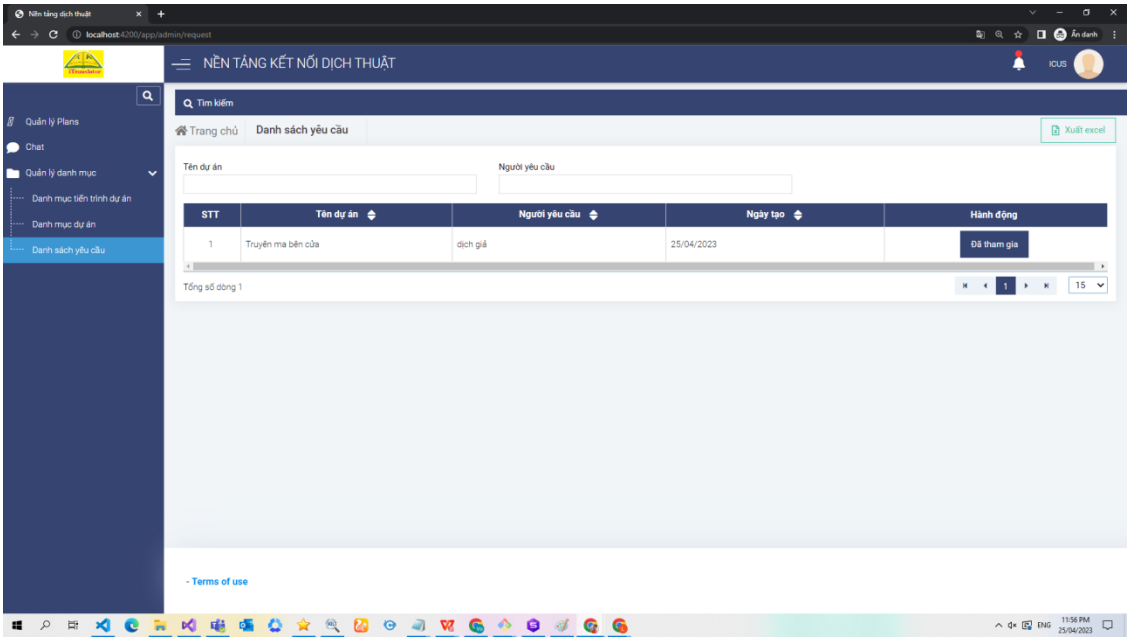




1. Click “Tìm kiếm”
2. Click “Join”

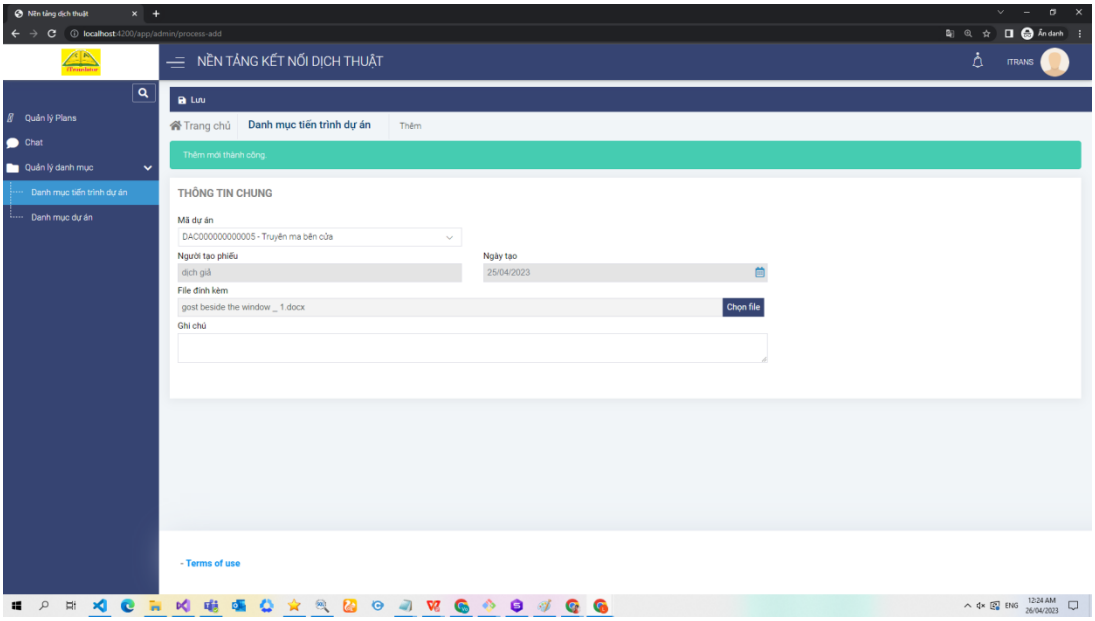
3.2.16 Customer accept translator join into project



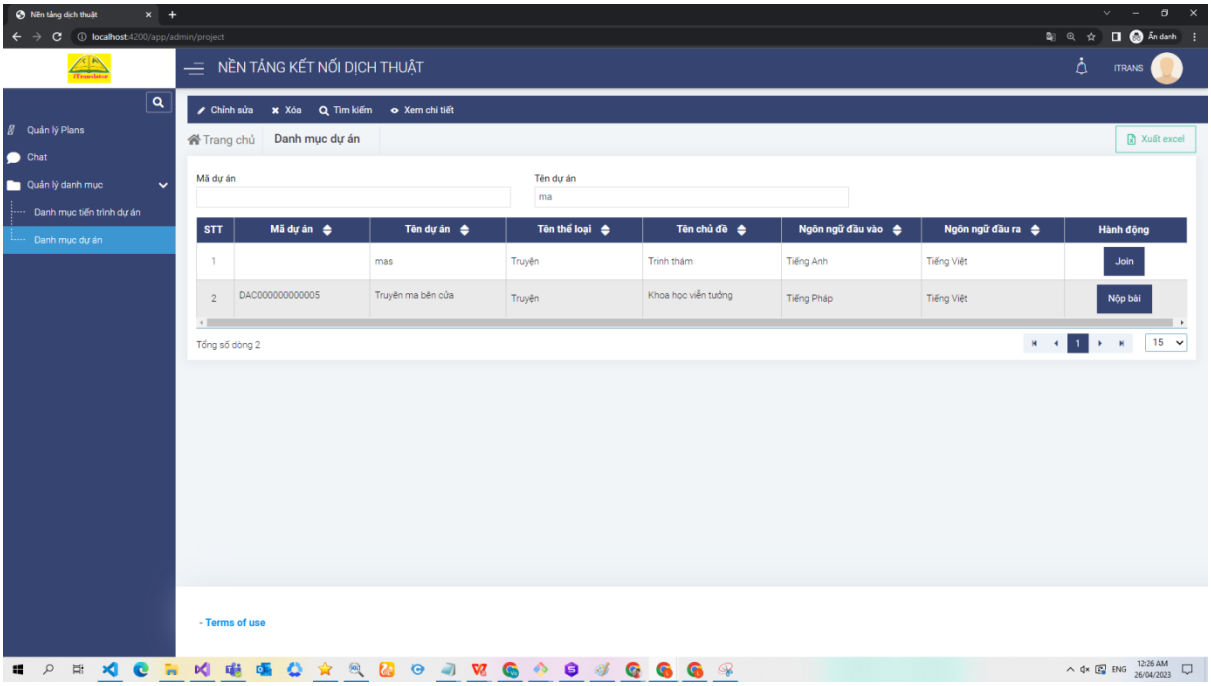


1. Click “Tìm kiếm”
2. Click “Chấp nhận”

3.2.17 Customer translate and complete project

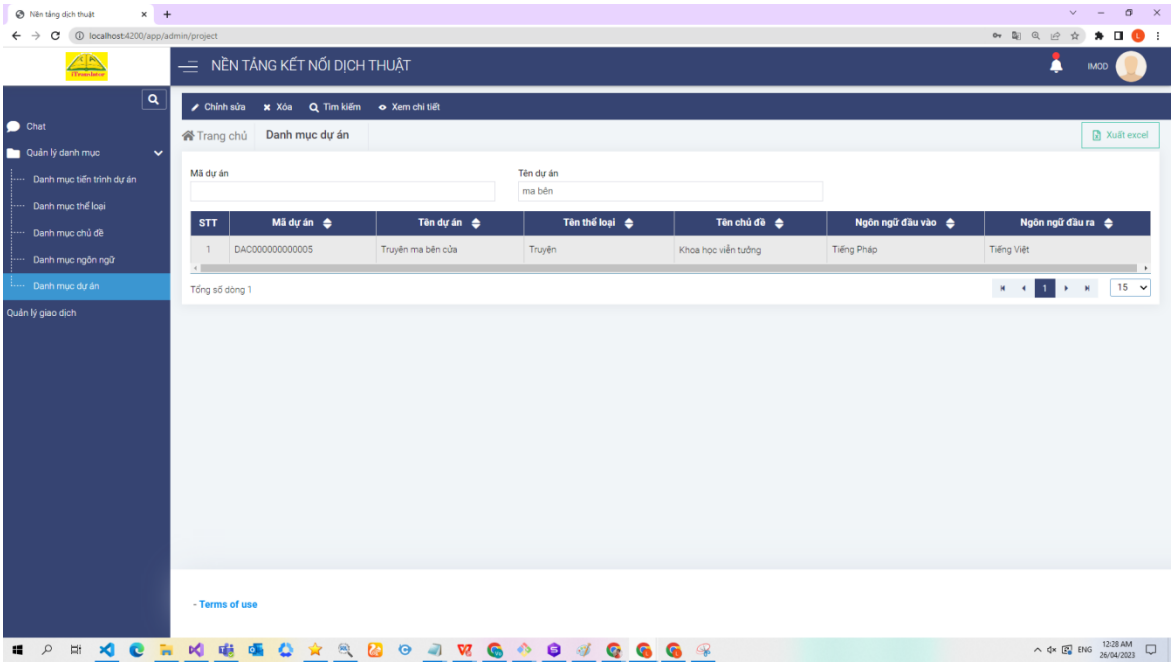


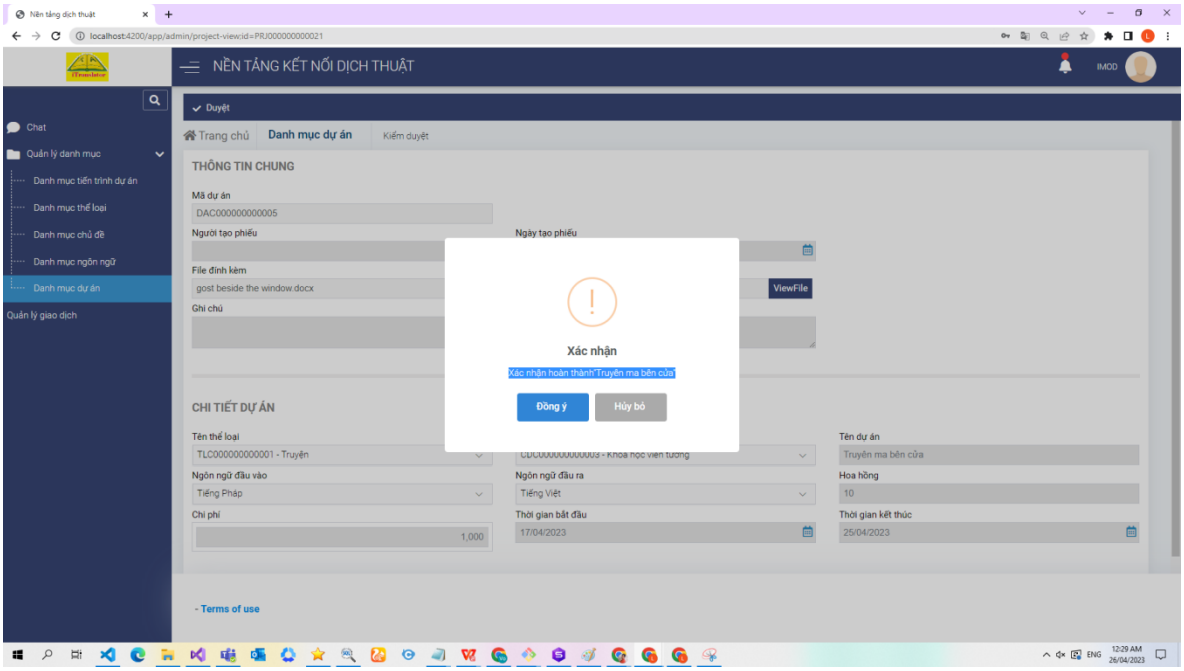
1. Click “Danh mục tiến trình”
2. Click “Thêm”
3. Fill all information
4. Click “Lưu”

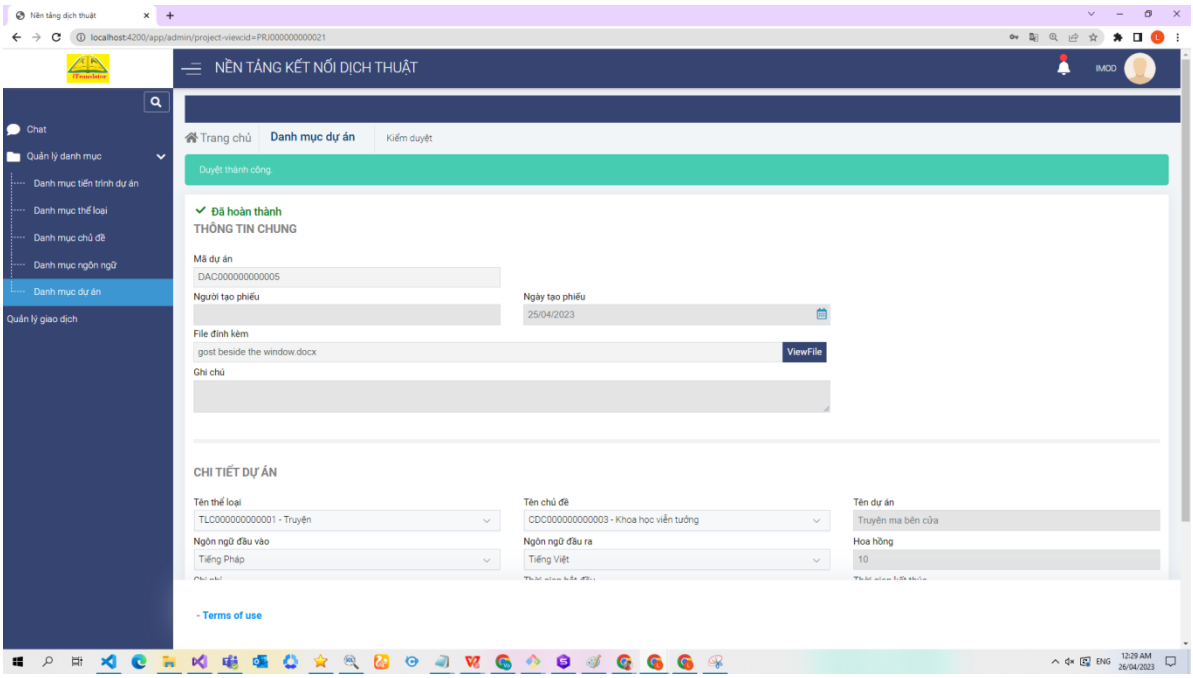


5. Click “Nộp bài”

3.2.18 Moderator approved completed project

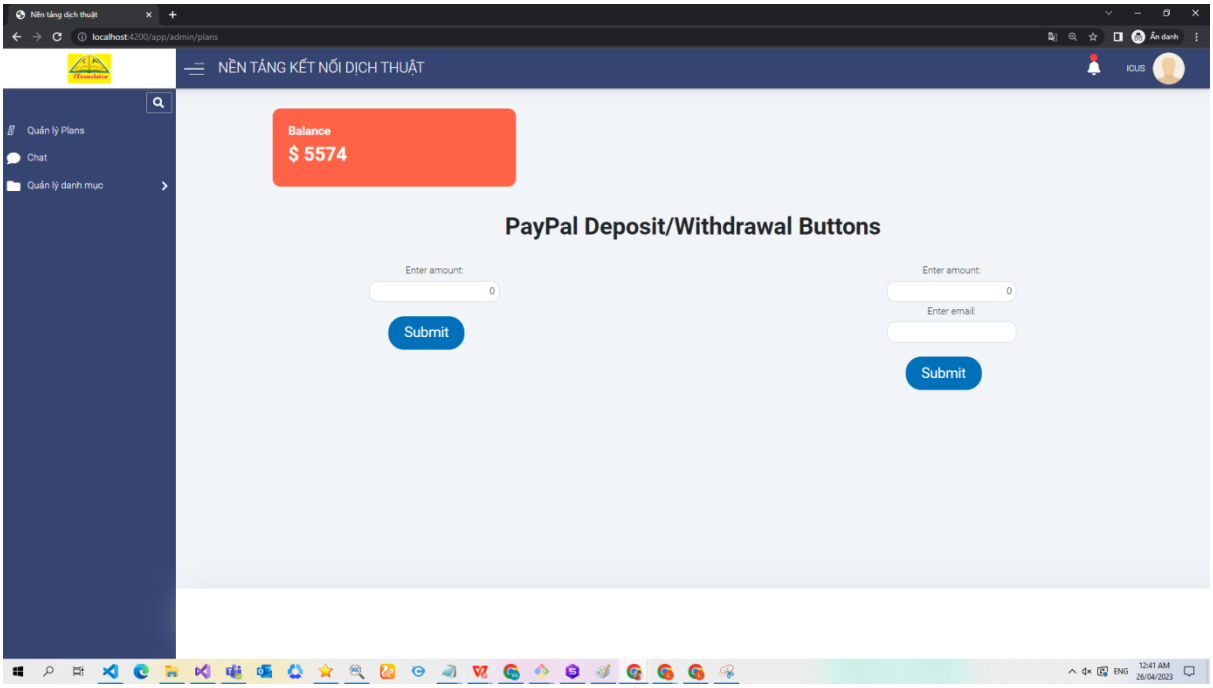


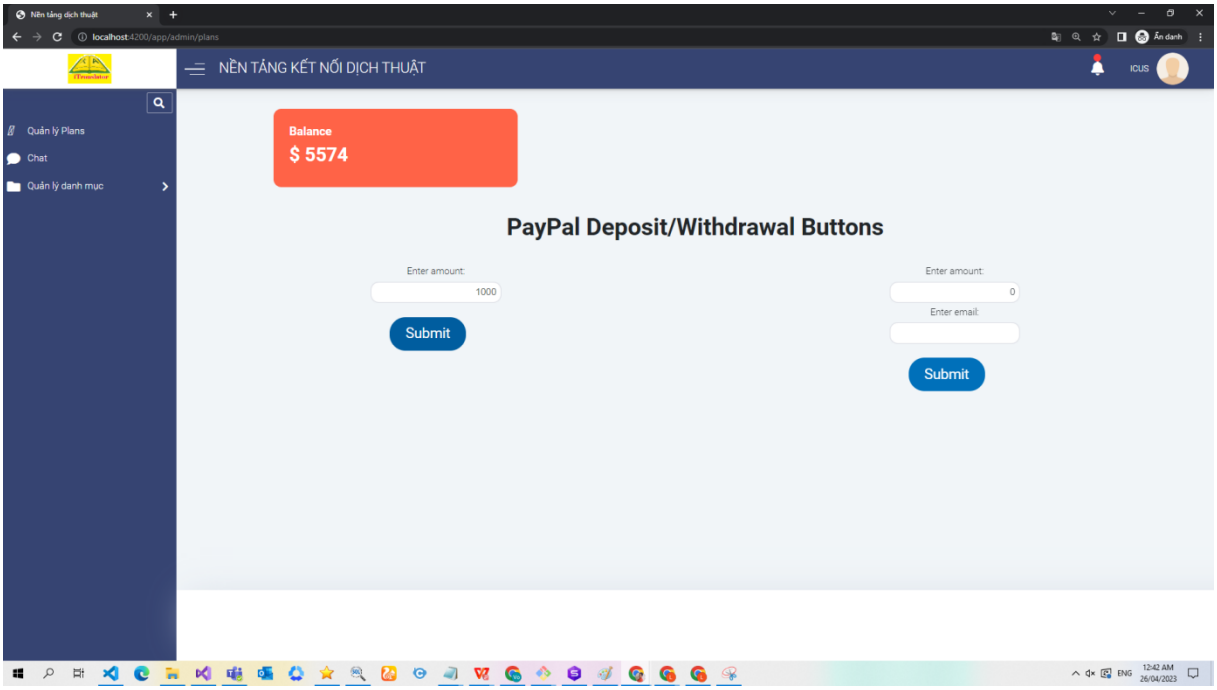


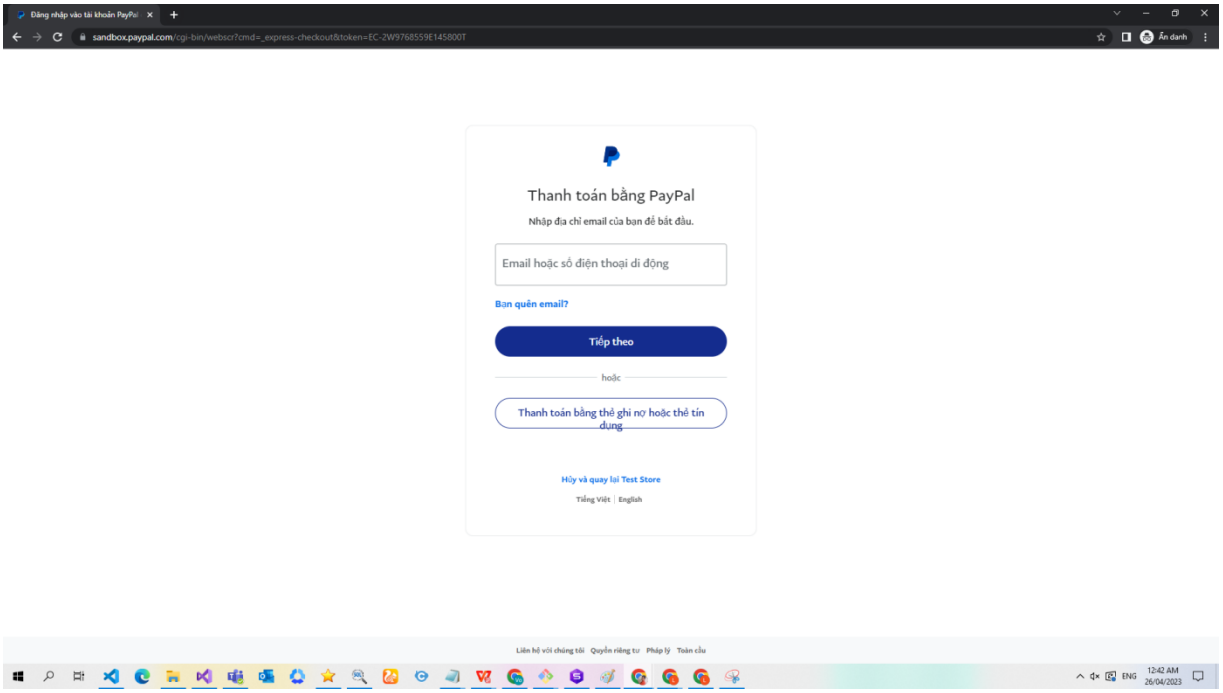


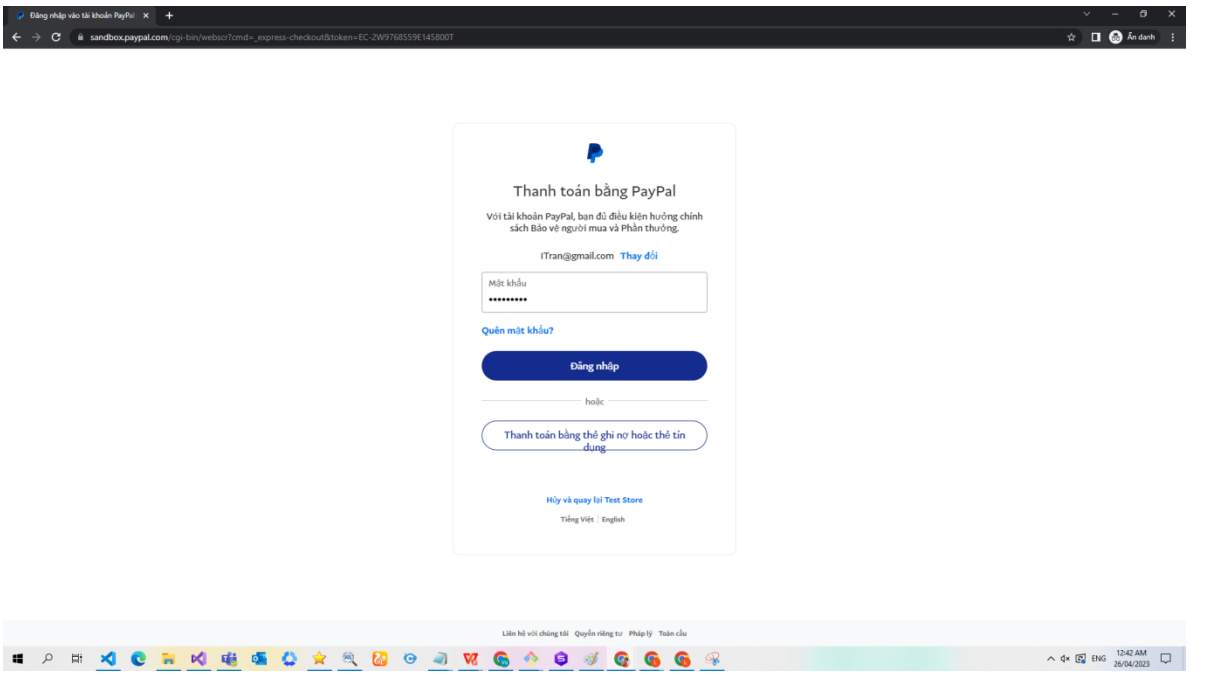
1. Click “Tìm kiếm” project
2. Check information of project
3. Click “Duyệt”
4. Click “Đồng ý”

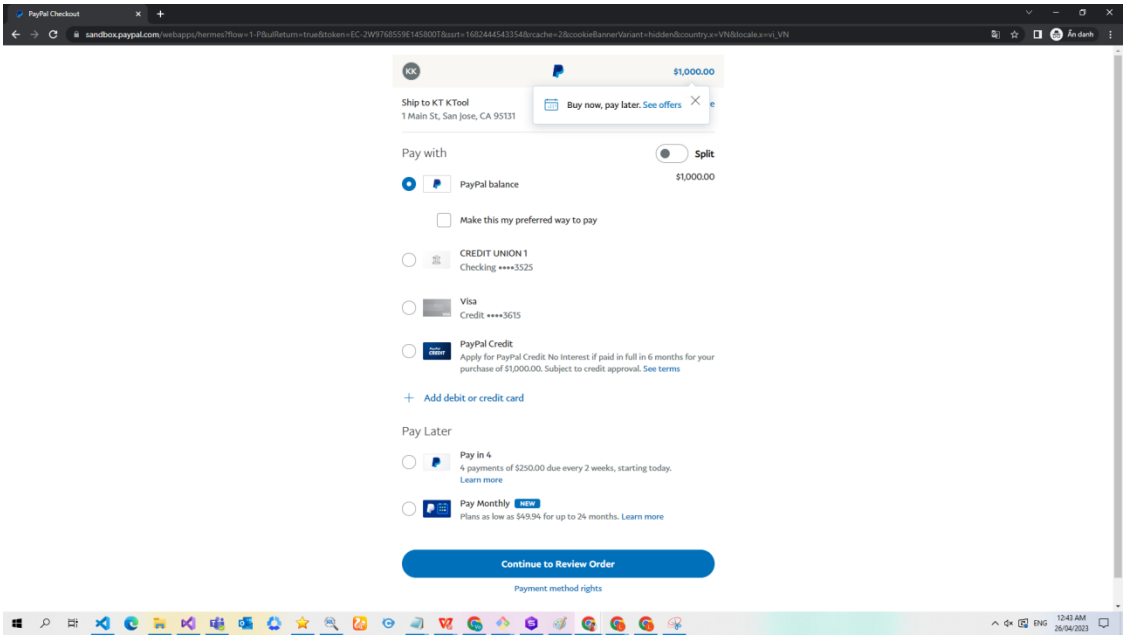
3.2.19 Customer and Translator deposit

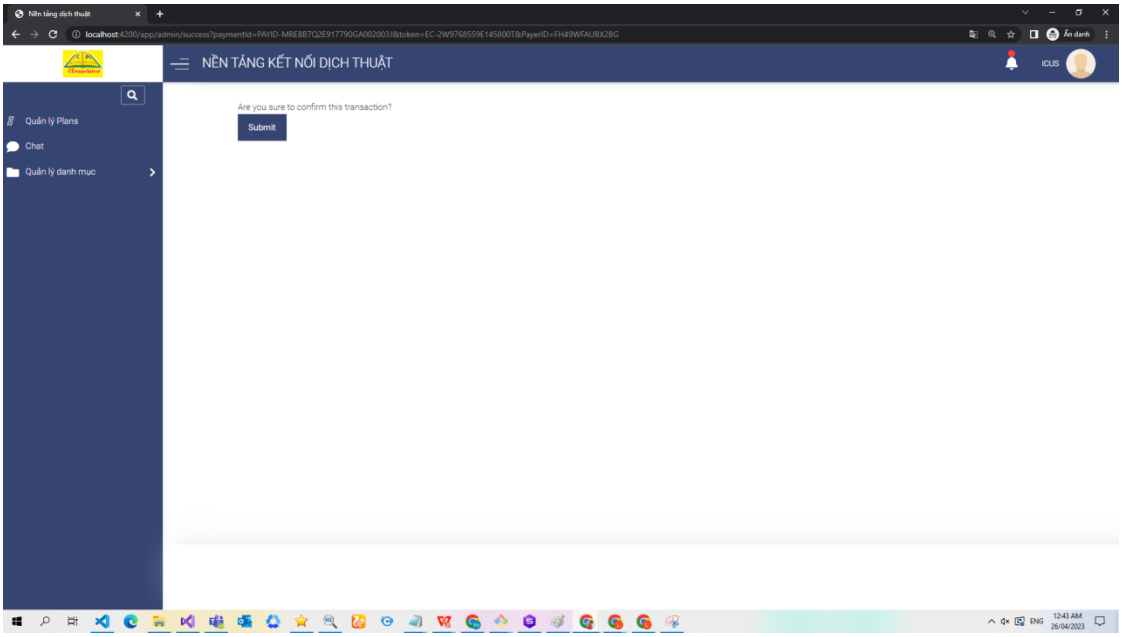





Đăng nhập vào tài khoản PayPal
sandbox.paypal.com/cgi-bin/webscr?cmd=_express-checkout&token=EC-2W9768559E145800T
Ẩn danh
Thanh toán bằng PayPal
Nhập địa chỉ email của bạn để bắt đầu.
Email hoặc số điện thoại di động
Bạn quên email?
Tiếp theo
hoặc
Thanh toán bằng thẻ ghi nợ hoặc thẻ tín dụng
Hủy và quay lại Test Store
Tiếng Việt | English
Liên hệ với chúng tôi Quyền riêng tư Pháp lý Toàn cầu
ENG
12:42 AM
26/04/2023


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sandbox.paypal.com/cgi-bin/webscr?cmd=_express-checkout&token=EC-2W9768559E145800T
Ẩn danh
Thanh toán bằng PayPal
Với tài khoản PayPal, bạn đủ điều kiện hưởng chính sách Bảo vệ người mua và Phần thưởng.
ITran@gmail.com Thay đổi
Mật khẩu
Quên mật khẩu?
Đăng nhập
hoặc
Thanh toán bằng thẻ ghi nợ hoặc thẻ tín dụng
Hủy và quay lại Test Store
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Liên hệ với chúng tôi Quyền riêng tư Pháp lý Toàn cầu
ENG
12:42 AM
26/04/2023





1. Click “nạp/ rút tiền”
2. Enter amount
3. Enter email or phone
4. Enter password
5. Choose card to payment
6. Click “Nộp”